

Maths and Statistics for the InfoEdge Online Test

40 questions in 40 minutes edition

Taught from zero. 352 exam calibrated MCQs.
Every question timed, every solution with a fast route and full working.

Syllabus covered (exactly as listed in the InfoEdge Pre-Placement Talk 2025-26, Maths and Statistics block, 33.33% weightage): Probability, Probability Distributions, Sampling, Correlations, Hypothesis Testing, Statistical Significance, Central Limit Theorem, Paired Means Tests, Bayes' Theorem, Linear Algebra (SVD, Eigenvalue Decomposition, Matrix Operations).

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Every page carries a "Stuck on this page?" link in the footer. Every section starts with a companion resource.
All links are clickable.

How to use this book

The two halves of every chapter

- **The teaching half** assumes you know nothing. Every idea starts from an everyday picture, then a tiny example you can do in your head, then the formula, then a “Try it now” box, then where it shows up in ML. If a formula appears, every symbol in it has been explained before it appears.
- **The question bank** is calibrated to the InfoEdge online test: single answer MCQs, a mix of instant recall questions and short calculations, spread across the whole syllabus.
- Chapter 0 is a toolkit (sums, powers, logs, e , areas under curves, mental arithmetic). If any symbol in a later chapter looks unfamiliar, it is explained there.

How to read a question

- Every question carries one time tag, which tells you how long it should take *on exam day*:
 - **RECALL · about 15 s** You either know the fact or you do not. Pure memory, zero calculation. If you are thinking for more than 20 seconds, you have not memorised it yet.
 - **REASON · about 45 s** One idea plus one small step (a sign, a ratio, a single formula).
 - **CALC · 1.5 to 2 min** A short calculation: two to five arithmetic steps, numbers chosen so that no calculator is needed.
- Questions marked ★ **PYQ (...)** are faithful reconstructions of questions candidates reported from InfoEdge papers. They always come first in a bank.
- Every solution has three parts:
 - **Fast route**: the exam day thinking, usually one or two lines.
 - **Full working**: every step, slowly, so that you could teach it.
 - **Why the other options fail**: because with negative marking, knowing why a wrong option is wrong is as valuable as knowing the right one.
- Practise honestly: cover the solution, start a timer, answer, then check both the answer and your time.

What we know about the InfoEdge online test

- **Latest information (told to candidates, 2026 drive)**: 40 questions in 40 minutes, covering the **breadth** of the syllabus. Some questions can be answered in about 15 seconds if you remember the fact; others need 1.5 to 2 minutes because of a small calculation. This book is built around exactly that mix.
- **Selection process (Pre-Placement Talk 2025-26)**: CV shortlisting plus online test, then two technical interview rounds (Maths and Statistics 33.33%, Classical ML and Deep Learning 66.66%), a programming round, and a case study round.
- **Earlier paper structure (IIT Roorkee drive, reported on Scribd)**: Maths and Statistics 10 questions (matrices, probability distributions), ML 5, DL 7, Python 2. Source: [Infoedge test in IITR \(Scribd\)](#).
- **Platform and marking (Glassdoor, Nov 2025)**: HirePro portal, **negative marking**, all MCQ (Python, aptitude, statistics, data science, maths). Source: [Glassdoor](#), [Info Edge Data Scientist](#).

- **40 MCQ paper recalled by an InfoEdge DS hire (IIT BHU):** variance of a continuous function over an interval; alpha and beta for Type I and II errors; kurtosis; homoscedasticity vs heteroscedasticity; Dickey-Fuller test; meaning of eigenvectors and eigenvalues in PCA; sum of two independent normals; entropy formula; what a false negative means. Source: [Placement guidance repo](#).
- **Interview topics (same sources):** Bayes and its assumptions, Naive Bayes, Poisson for customers, deriving the geometric distribution, linearity of expectation, p-values, chi-square test, rank and null space, PCA via SVD, heteroscedasticity, multicollinearity and VIF. These can also appear as OA questions, so they are in the banks.

Honesty note

- InfoEdge does not publish papers. Every PYQ here keeps the reported concept and form; the numbers are mine.
- We do not know how the 40 questions split between Maths, ML, DL and Python. If the split follows the 33.33% interview weightage, expect roughly 13 maths questions; the 2023 paper had 10 maths out of 24. Either way, maths questions compete for the same 40 minutes, so speed on them buys time for ML and DL.
- Sampling, CLT and Paired tests have no reported InfoEdge question; those banks are modelled on the reported style.

The 40 in 40 time plan

- **The arithmetic of the paper.** One minute per question on average. Suppose 24 questions are recall or quick reasoning and 16 need a calculation:
 - 24 fast questions at about 20 seconds each: 8 minutes.
 - 16 calculations at about 1.75 minutes each: 28 minutes.
 - Total 36 minutes, leaving about 4 minutes of buffer.
- The plan only works if recall questions really take 15 to 20 seconds. That is why every chapter ends its teaching half with a **15 second recall card**. Those cards are the highest value pages in this book.
- **Two pass strategy.**
 - Pass 1 (about 12 to 15 minutes): go through all 40. Answer every recall and quick question immediately. For a calculation, answer it only if you can see the route at once; otherwise mark it and move on.
 - Pass 2: return to marked calculations, easiest first. Hard stop at 2.5 minutes on any one question.
 - Last 2 minutes: decide on guesses using the rule below. Never leave the paper with time unused and questions untouched that you could have eliminated.
- **Speed tricks that recur throughout the book:** estimate before computing and use the options (they are often far apart); check units and ranges (a probability cannot exceed 1, a variance cannot be negative, a correlation lies in $[-1, 1]$); know the standard numbers by heart (tables on the next page).

Negative marking: when to guess

- Suppose a right answer earns $+1$, a wrong answer costs $-m$, and you have narrowed the options down to k equally plausible ones.
- Expected score of guessing $= \frac{1}{k} \cdot 1 - \frac{k-1}{k} \cdot m$.

•

Penalty m	guess among 4	among 3	among 2
1/4	+0.06	+0.17	+0.38
1/3	0	+0.11	+0.33
1/2	-0.13	0	+0.25
1	-0.5	-0.33	0

- Rule: check the penalty in the instructions. With the common $-1/4$ or $-1/3$, guess as soon as you have eliminated **one** option. With a full -1 penalty, only answer when you are down to one option.

Quick reference tables (memorise)

Standard normal CDF $\Phi(z) = P(Z \leq z)$

z	$\Phi(z)$
0.00	0.5000
0.50	0.6915
1.00	0.8413
1.28	0.8997
1.50	0.9332
1.645	0.9500
1.96	0.9750
2.00	0.9772
2.326	0.9900
2.50	0.9938
2.576	0.9950
3.00	0.99865

Symmetry: $\Phi(-z) = 1 - \Phi(z)$.

Empirical rule: 68.27%, 95.45%, 99.73% within 1, 2, 3 SDs.

Tail areas: $P(Z > 1) = 0.1587$, $P(Z > 2) = 0.0228$, $P(Z > 3) = 0.00135$.

Two sided critical t (5%), one sided 5% in brackets

df	$t_{0.025}$	$(t_{0.05})$
4	2.776	(2.132)
5	2.571	(2.015)
8	2.306	(1.860)
9	2.262	(1.833)
15	2.131	(1.753)
18	2.101	(1.734)
20	2.086	(1.725)
25	2.060	(1.708)
∞	1.960	(1.645)

Chi-square upper 5% points: df 1: 3.841, df 2: 5.991, df 3: 7.815, df 4: 9.488, df 5: 11.070, df 6: 12.592, df 20: 31.410.

Numbers worth knowing by heart

- $e^{-1} = 0.3679$, $e^{-2} = 0.1353$, $e^{-3} = 0.0498$, $e^{-4} = 0.0183$, $e^{-5} = 0.0067$, $e^{-0.5} = 0.6065$, $e^{-1.5} = 0.2231$.
- $\ln 2 = 0.693$, $\ln 3 = 1.099$, $\ln 10 = 2.303$, $\log_2 3 = 1.585$.
- $\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\sqrt{5} = 2.236$, $\sqrt{10} = 3.162$.
- $0.9^{10} = 0.349$, $0.95^{20} = 0.358$, $(5/6)^4 = 0.482$, $(1 - 1/n)^n \approx e^{-1}$ for large n .
- Factorials: $4! = 24$, $5! = 120$, $6! = 720$, $7! = 5040$. Useful counts: $\binom{5}{2} = 10$, $\binom{6}{3} = 20$, $\binom{10}{2} = 45$, $\binom{10}{3} = 120$, $\binom{52}{2} = 1326$.
- Powers of 2: $2^{10} = 1024$. Powers of 3: $3^4 = 81$, $3^5 = 243$.

Contents

How to use this book	1
0 The Toolkit: Every Symbol You Will Meet	10
0.1 Sets: collections of possible outcomes	10
0.2 Sigma and Pi: shorthand for long sums and products	10
0.3 Factorials and counting	11
0.4 Powers, roots, logarithms and the number e	11
0.5 Series you must recognise	12
0.6 Derivatives: slopes (only what we need)	12
0.7 Integrals: areas under curves (only what we need)	12
0.8 Mental arithmetic for the 2 minute questions	13
1 Probability	14
1.1 What probability is	14
1.1.1 The vocabulary, one word at a time	14
1.1.2 The equally likely rule	14
1.2 The three rules everything else follows from	15
1.2.1 Complement rule: the most useful trick in the chapter	15
1.2.2 The addition rule for events that can overlap	15
1.3 Counting: when outcomes are equally likely, probability is counting	15
1.3.1 The multiplication principle	15
1.3.2 Order matters: permutations	16
1.3.3 Order does not matter: combinations	16
1.4 Conditional probability: shrinking the world	16
1.4.1 A famous surprise: the second card	17
1.5 Independence	17
1.6 The law of total probability (a preview of Bayes)	17
1.7 Random variables: attaching numbers to outcomes	17
1.8 Expectation: the long run average	18
1.8.1 Linearity of expectation (a reported InfoEdge interview topic)	18
1.8.2 First step analysis: expectations that refer to themselves	19
1.9 Variance and standard deviation: how spread out	19
1.10 Continuous random variables: probability as area	20
1.10.1 Worked example (the reported InfoEdge OA type)	20
1.10.2 Two continuous shapes you meet immediately	20
1.11 Two inequalities for when you know almost nothing	21
1.12 Entropy: measuring surprise (a reported InfoEdge OA topic)	21
1.13 Question bank: Probability	22

2	Bayes' Theorem	41
2.1	The question Bayes answers	41
2.1.1	Natural frequencies: the method that never fails	41
2.2	The formula, built from what you already know	42
2.3	The odds form: Bayes in five seconds	42
2.4	Vocabulary of diagnostic tests and classifiers	43
2.5	Worked classics you should recognise instantly	43
2.6	Naive Bayes: Bayes' theorem as a classifier	43
2.6.1	The anemia table	44
2.7	Priors, MLE and MAP	44
2.8	Question bank: Bayes' theorem	45
3	Probability Distributions	55
3.1	What a distribution is, and why we give them names	55
3.2	Discrete distributions	55
3.2.1	Bernoulli(p): one yes or no	55
3.2.2	Binomial(n, p): counting successes in n tries	55
3.2.3	Geometric(p): waiting for the first success (a reported InfoEdge derivation)	56
3.2.4	Negative Binomial(r, p): waiting for the r -th success	56
3.2.5	Hypergeometric: drawing without replacement	56
3.2.6	Poisson(λ): counts of events at a steady rate (a reported InfoEdge question)	57
3.3	Continuous distributions	57
3.3.1	Uniform(a, b)	57
3.3.2	Exponential(λ): waiting times	57
3.3.3	Normal (Gaussian) $N(\mu, \sigma^2)$	57
3.3.4	Log normal	58
3.3.5	Beta and Gamma, one line each	58
3.4	The three test distributions: χ^2 , t and F	58
3.5	Shape: skewness and kurtosis (a reported InfoEdge OA topic)	59
3.6	Mixtures and the law of total variance	59
3.7	Summary table	59
3.8	Question bank: Probability distributions	60
4	Sampling	75
4.1	Why we sample at all	75
4.2	Population, sample, parameter, statistic	75
4.2.1	A sampling distribution you can see completely	75
4.3	Probability sampling: every unit has a known chance	76
4.3.1	Simple random sampling (SRS)	76
4.3.2	Systematic sampling	76
4.3.3	Stratified sampling	76
4.3.4	Cluster sampling	76
4.4	Non probability sampling and the biases it brings	77

4.5	Estimating the mean and the variance, and why $n - 1$	77
4.6	Standard error, margin of error and sample size	77
4.7	Resampling: the bootstrap and cross validation	78
4.8	Sampling inside algorithms	78
4.9	Question bank: Sampling	79
5	Central Limit Theorem	89
5.1	The surprise: averages forget where they came from	89
5.1.1	Seeing it with two dice, by hand	89
5.2	The precise statement	90
5.3	When it works, and when it fails	90
5.4	Law of Large Numbers vs Central Limit Theorem	90
5.5	Using the CLT: a four step recipe	91
5.6	What the CLT powers	91
5.7	Question bank: Central Limit Theorem	92
6	Correlations	101
6.1	Do two things move together?	101
6.2	Covariance: built from scratch	101
6.3	Pearson correlation r : covariance without units	101
6.3.1	What r cannot see	102
6.4	Correlation and the regression line	102
6.5	Rank correlations and other relatives	102
6.6	Correlation is not causation, and other traps	103
6.7	Heteroscedasticity (a reported InfoEdge OA and interview topic)	103
6.8	Multicollinearity and VIF (a reported InfoEdge interview topic)	103
6.9	Two more diagnostics	104
6.10	Question bank: Correlations	105
7	Hypothesis Testing	114
7.1	The logic, as a courtroom	114
7.2	The standard steps (a reported InfoEdge interview question)	114
7.3	The menu of tests: which test when	115
7.3.1	Worked: one sample t	115
7.3.2	Worked: A/B test with two proportions	115
7.4	Chi-square tests (a reported InfoEdge interview topic)	116
7.5	ANOVA in one picture	116
7.6	Tests about variances; confidence intervals as tests	116
7.7	Non parametric and resampling tests	116
7.8	Stationarity and the Dickey Fuller test (a reported InfoEdge OA topic)	117
7.9	Question bank: Hypothesis testing	118
8	Statistical Significance, Errors and Power	128

8.1	What “statistically significant” really means	128
8.2	The two ways to be wrong (a reported InfoEdge OA topic)	128
8.3	Power and sample size	129
8.4	Effect size: significant is not the same as important	129
8.5	Multiple testing: the more you look, the more you find	129
8.6	How often is a significant result real?	130
8.7	Question bank: Statistical significance	131
9	Paired Means Tests	140
9.1	What “paired” means	140
9.2	Why pairing is so powerful	140
9.3	The paired t test, step by step	141
9.4	Paired tests without normality, and for yes/no outcomes	141
9.5	Design traps in before and after studies	142
9.6	Question bank: Paired means tests	143
10	Linear Algebra I: Matrix Operations, Rank and Null Space	152
10.1	Vectors: lists of numbers that are also arrows	152
10.2	Matrices: two ways to look at a grid of numbers	152
10.3	Multiplying matrices	153
10.3.1	NumPy conventions you will be asked about	153
10.4	Determinant: how much a matrix stretches area	154
10.5	The inverse: undoing a matrix	154
10.6	Independence, rank, column space and null space (a reported InfoEdge interview topic)	154
10.6.1	How to find the rank and the null space by hand	155
10.7	Solving $Ax = b$ and least squares	155
10.8	Positive definite matrices and matrix gradients	156
10.9	Question bank: Matrix operations	157
11	Linear Algebra II: Eigenvalues and Eigendecomposition	167
11.1	The idea: directions a matrix does not turn	167
11.2	How to find them	168
11.3	Eigendecomposition: the matrix in its own coordinates	168
11.4	Symmetric matrices: the well behaved family	169
11.5	PCA: what eigenvectors and eigenvalues mean (a reported InfoEdge OA and inter- view topic)	169
11.6	Other places eigenvalues rule	170
11.7	Question bank: Eigenvalues and eigendecomposition	171
12	Linear Algebra III: Singular Value Decomposition and PCA	181
12.1	Why we need one more decomposition	181
12.2	Connecting SVD to eigenvalues, and computing it by hand	182
12.2.1	What the singular values tell you instantly	182

12.3 Shapes, thin SVD, and building a matrix from rank 1 layers	182
12.4 PCA via SVD (a reported InfoEdge interview question)	183
12.5 Pseudoinverse and least squares	183
12.6 Question bank: SVD	184
Timed mock papers (40 in 40)	194
Mock paper 1	194
Mock paper 2	195
Mock paper 3	196
A 7 day revision plan	197

Chapter 0

The Toolkit: Every Symbol You Will Meet

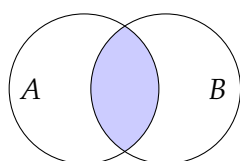
Companion resource for this section: [Maths Is Fun](#) (short, friendly pages on sigma notation, exponents, logarithms and integration)

Why this chapter exists

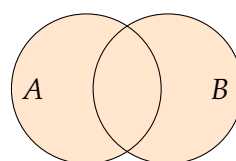
- Statistics looks hard mostly because of its notation. The ideas themselves are everyday ideas: counting, averaging, measuring spread, being surprised.
- This chapter explains every symbol used later, once, slowly, with a tiny example. If you already know a section, skim its “Try it now” box; if you get it right, move on.
- Nothing here is tested directly, but every calculation question in the book uses it.

0.1 Sets: collections of possible outcomes

- A **set** is just a collection of distinct things, written inside curly brackets. The outcomes of a die: $\{1, 2, 3, 4, 5, 6\}$.
- $x \in A$ reads “ x is in A ”. $3 \in \{1, 2, 3\}$ is true; $7 \in \{1, 2, 3\}$ is false.
- **Union** $A \cup B$: everything in A **or** B (or both). $\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$. Think of the symbol $\cup$ as a cup that collects everything.
- **Intersection** $A \cap B$: only what is in **both**. $\{1, 2\} \cap \{2, 3\} = \{2\}$. Think of $\cap$ as a bridge standing on the common part.
- **Complement** A^c (also written A' or $\bar{A}$): everything **not** in A . On a die, if $A = \{6\}$ then $A^c = \{1, 2, 3, 4, 5\}$.
- **Empty set** $\emptyset$: the set with nothing in it. Two sets with $A \cap B = \emptyset$ are **disjoint** (they cannot happen together).
- $|A|$ means the number of elements of A . $|\{1, 2, 3\}| = 3$.



shaded: $A \cap B$



shaded: $A \cup B$

Try it now (answer right below, cover it first)

$A = \{2, 4, 6\}$ (even numbers on a die), $B = \{4, 5, 6\}$ (numbers at least 4). Find $A \cup B$, $A \cap B$ and A^c .

Answer: $A \cup B = \{2, 4, 5, 6\}$; $A \cap B = \{4, 6\}$; $A^c = \{1, 3, 5\}$.

0.2 Sigma and Pi: shorthand for long sums and products

- $\sum_{i=1}^n x_i$ means “add up $x_1 + x_2 + \dots + x_n$ ”. The letter i is a counter that runs from the number below to the number on top.

- Example: data $x_1 = 2, x_2 = 5, x_3 = 8$. $\sum_{i=1}^3 x_i = 2 + 5 + 8 = 15$.
- $\sum x_i^2$ means square first, then add: $4 + 25 + 64 = 93$. This is **not** the same as $(\sum x_i)^2 = 15^2 = 225$. This difference is the heart of the variance formula later.
- Two rules you will use constantly:
 - Constants come out: $\sum c x_i = c \sum x_i$. (Doubling every number doubles the total.)
 - Sums split: $\sum (x_i + y_i) = \sum x_i + \sum y_i$.
- The **mean** (average) is written $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. For 2, 5, 8: $\bar{x} = 15/3 = 5$.
- $\prod_{i=1}^n x_i$ is the same idea with multiplication: $\prod_{i=1}^3 x_i = 2 \times 5 \times 8 = 80$. It appears when we multiply probabilities of independent events (likelihoods, Naive Bayes).

Try it now (answer right below, cover it first)

For $x = 1, 2, 3, 4$: compute $\sum x_i$, $\sum x_i^2$, $(\sum x_i)^2$ and $\bar{x}$.

Answer: 10; $1 + 4 + 9 + 16 = 30$; 100; 2.5.

0.3 Factorials and counting

- $n!$ (“ n factorial”) = $n \times (n-1) \times \cdots \times 2 \times 1$. $4! = 24$. By agreement $0! = 1$ (there is exactly one way to arrange nothing).
- Why it matters: $n!$ is the number of ways to put n different things in a row. Three books A, B, C: ABC, ACB, BAC, BCA, CAB, CBA, which is $3! = 6$.
- $\binom{n}{k}$ (“ n choose k ”) counts the ways to **choose** k things out of n when order does not matter:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$
- Fast way to compute: $\binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$. Take k numbers counting down from n on top, $k!$ below.
- Symmetry: $\binom{n}{k} = \binom{n}{n-k}$. Choosing 3 people to invite is the same as choosing 7 people to leave out: $\binom{10}{3} = \binom{10}{7}$.

0.4 Powers, roots, logarithms and the number e

- a^n means a multiplied by itself n times: $2^3 = 8$. Rules: $a^m a^n = a^{m+n}$; $(a^m)^n = a^{mn}$; $a^{-n} = 1/a^n$; $a^0 = 1$; $a^{1/2} = \sqrt{a}$.
- A **logarithm** undoes a power. $\log_2 8 = 3$ because $2^3 = 8$. Read $\log_b y$ as “what power of b gives y ?”.
- Log rules (they turn multiplication into addition, which is why likelihoods are logged):
 - $\log(ab) = \log a + \log b$; $\log(a/b) = \log a - \log b$; $\log(a^n) = n \log a$; $\log 1 = 0$.
- The **number** $e \approx 2.718$ is the natural base for growth and decay. $\ln$ means $\log_e$. Key facts: $\ln e = 1$, $e^{\ln x} = x$.
- Where e comes from: $(1 + \frac{1}{n})^n \rightarrow e$ as n grows, and $(1 - \frac{1}{n})^n \rightarrow e^{-1} \approx 0.368$. The second fact is behind the bootstrap’s 36.8% and many “rare event” calculations.
- Handy approximation for small x : $e^{-x} \approx 1 - x$. For instance $e^{-0.05} \approx 0.95$.
- Bits vs nats: entropy with $\log_2$ is measured in bits, with $\ln$ in nats. $\log_2 x = \ln x / \ln 2$.

Try it now (answer right below, cover it first)

Compute $\log_2 16$, $\log_{10} 1000$, $\ln(e^3)$, and simplify $\ln(2) + \ln(3)$.

Answer: 4; 3; 3; $\ln 6$.

0.5 Series you must recognise

- **Arithmetic:** $1 + 2 + \dots + n = \frac{n(n+1)}{2}$. So $1 + \dots + 6 = 21$ (useful for a die: mean $21/6 = 3.5$).
- Sum of squares: $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$. For $n = 6$: 91.
- **Geometric:** $1 + r + r^2 + \dots = \frac{1}{1-r}$ when $|r| < 1$. For $r = \frac{1}{2}$: $1 + \frac{1}{2} + \frac{1}{4} + \dots = 2$. This one sum powers the geometric distribution.
- Finite geometric: $1 + r + \dots + r^{n-1} = \frac{1-r^n}{1-r}$.
- Exponential series: $e^\lambda = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots$. This is why Poisson probabilities add up to 1.

0.6 Derivatives: slopes (only what we need)

- The derivative $f'(x)$ is the **slope** of the curve at x : how fast f changes when x nudges up.
- Rules: $(x^n)' = nx^{n-1}$; $(e^x)' = e^x$; $(\ln x)' = 1/x$; constants disappear; sums split.
- Example: $f(x) = x^2$ has slope $2x$; at $x = 3$ the slope is 6.
- Why we care: maximum or minimum points have slope zero. Maximum likelihood estimation, least squares and gradient descent all rely on “set the slope to zero” or “walk against the slope”.

0.7 Integrals: areas under curves (only what we need)

- $\int_a^b f(x) dx$ is the **area under the curve** f between $x = a$ and $x = b$. For continuous random variables, probability is area, so this symbol appears whenever a variable can take any value in an interval.
- It is the continuous version of $\sum$: slice the area into thin strips of width dx and height $f(x)$, then add them up.
- The one rule you need: $\int_a^b x^n dx = \left[\frac{x^{n+1}}{n+1} \right]_a^b = \frac{b^{n+1} - a^{n+1}}{n+1}$ (for $n \neq -1$).
- Example: $\int_0^1 x dx = \frac{1}{2}$: the area of a triangle with base 1 and height 1. The formula and the geometry agree.
- Exponential: $\int_0^\infty \lambda e^{-\lambda x} dx = 1$ and $\int_t^\infty \lambda e^{-\lambda x} dx = e^{-\lambda t}$.
- Geometry shortcut: when the curve is flat (uniform) or a straight line, skip the integral and compute a rectangle or triangle area. This saves a full minute on many OA questions.

Try it now (answer right below, cover it first)

Compute $\int_0^2 3x^2 dx$ and $\int_1^3 1 dx$.

Answer: $[x^3]_0^2 = 8$; the rectangle of width 2 and height 1 has area 2.

0.8 Mental arithmetic for the 2 minute questions

- **Squares near 50 and 100:** $48^2 = (50 - 2)^2 = 2500 - 200 + 4 = 2304$. $103^2 = 10609$.
- **Square roots:** $\sqrt{45}$ lies between 6 and 7, closer to 6.7 (since $6.7^2 = 44.89$). For an MCQ this is always enough.
- **Percent of:** $0.95 \times 0.01 = 0.0095$; move decimals, then multiply the small numbers.
- **Fractions to decimals:** $1/6 = 0.1667$, $1/7 = 0.1429$, $1/8 = 0.125$, $1/9 = 0.111$, $1/11 = 0.0909$, $1/12 = 0.0833$.
- **Divide by estimation:** $0.0099/0.0594$: both about 0.01 and 0.06, ratio about $1/6 = 0.167$. Then check which option is near.
- **Use the options:** if the options are 0.02, 0.16, 0.50, 0.84, a rough estimate decides it. Only compute precisely when two options are close.
- **Power estimates:** $(1 - p)^n \approx e^{-np}$ when p is small. $0.99^{70} \approx e^{-0.7} \approx 0.50$.

15 second recall card for Chapter 0

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$; $\binom{10}{3} = 120$, $\binom{52}{5} = 2,598,960$.
- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$; $1 + r + r^2 + \dots = \frac{1}{1-r}$; $e^\lambda = \sum \frac{\lambda^k}{k!}$.
- $\log(ab) = \log a + \log b$; $\log_2 x = \ln x / \ln 2$; $(1 - \frac{1}{n})^n \rightarrow e^{-1} = 0.368$.
- $\int_a^b x^n dx = \frac{b^{n+1} - a^{n+1}}{n+1}$; probability for continuous variables is area.

References for this chapter

- [Maths Is Fun: Sigma notation; Logarithms; Integration.](#)
- [3Blue1Brown: Essence of calculus.](#)

Chapter 1

Probability

Companion resource for this section: [Seeing Theory, Basic Probability](#) (interactive: toss coins and roll dice to watch probability emerge)

1.1 What probability is

Start here: the everyday picture

- Toss a coin 10 times: you might see 7 heads. Toss it 10,000 times: you will see very close to 5,000. The fraction of heads settles down to a fixed number as you keep going. That settling value is what we call the **probability** of heads: 0.5.
- So a probability is a number between 0 and 1 that answers “in the long run, what fraction of the time does this happen?”. 0 means never, 1 means always, 0.5 means half the time.
- A second, equally valid reading: probability is a **degree of belief**. “There is a 70% chance this candidate accepts the offer” is a belief, not something we can repeat 10,000 times. The same rules apply to both readings.

1.1.1 The vocabulary, one word at a time

- An **experiment** is anything with an uncertain result: rolling a die, showing a job to a user, drawing a card.
- An **outcome** is one possible result: “the die shows 4”.
- The **sample space** S (also written Ω) is the set of *all* outcomes. Die: $S = \{1, 2, 3, 4, 5, 6\}$. Two coins: $S = \{HH, HT, TH, TT\}$.
- An **event** is any collection of outcomes you care about. “Even number” = $\{2, 4, 6\}$. An event *happens* if the outcome that actually occurs is inside it.
- $P(A)$ is the probability of event A .

1.1.2 The equally likely rule

- If every outcome in S is equally likely (fair die, fair coin, well shuffled cards), then

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S} = \frac{|A|}{|S|}.$$

- Example: $P(\text{even}) = \frac{|\{2, 4, 6\}|}{6} = \frac{3}{6} = \frac{1}{2}$.
- Example with two dice: there are $6 \times 6 = 36$ equally likely ordered pairs (first, second). Sum 7 happens for $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$: 6 pairs, so $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$. Seven is the most likely sum of two dice.
- Warning: the rule only works when outcomes are truly equally likely. “Either it rains or it does not” has two outcomes, but rain is not automatically 50%.

1.2 The three rules everything else follows from

The axioms

1. $P(A) \geq 0$ for every event: probabilities are never negative.
2. $P(S) = 1$: something in the sample space certainly happens.
3. If A and B cannot happen together (**mutually exclusive**, $A \cap B = \emptyset$), then $P(A \cup B) = P(A) + P(B)$.

Everything below is a consequence of these three lines.

1.2.1 Complement rule: the most useful trick in the chapter

- A and “not A ” (A^c) are mutually exclusive and together make up S . So $P(A) + P(A^c) = 1$, i.e. $P(A^c) = 1 - P(A)$.
- Why it matters: “at least one” is hard to count directly (exactly one, exactly two, ...), but its complement “none” is easy.
- Example: roll a die 4 times. $P(\text{no six in any roll}) = \left(\frac{5}{6}\right)^4 = 0.482$. So $P(\text{at least one six}) = 1 - 0.482 = 0.518$.

1.2.2 The addition rule for events that can overlap

- If A and B can both happen, adding $P(A) + P(B)$ counts the overlap twice. Remove it once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- Example: pick a number from 1 to 100. Divisible by 2: 50 numbers; by 5: 20 numbers; by both (that is, by 10): 10 numbers. $P(2 \text{ or } 5) = 0.50 + 0.20 - 0.10 = 0.60$.
- Three events (**inclusion exclusion**): add the singles, subtract the pairs, add back the triple:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

- Picture it with a Venn diagram: each pairwise overlap was counted twice, so subtract once; but then the centre (all three) has been added 3 times and subtracted 3 times, so add it back once.

Try it now (answer right below, cover it first)

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$ and $P(\text{neither})$.

Answer: $0.5 + 0.4 - 0.1 = 0.8$; $\text{neither} = 1 - 0.8 = 0.2$.

1.3 Counting: when outcomes are equally likely, probability is counting

1.3.1 The multiplication principle

- If a task has step 1 with a ways and step 2 with b ways (whatever happened in step 1), the whole task has $a \times b$ ways.
- Example: a shirt (3 choices) and trousers (4 choices) give $3 \times 4 = 12$ outfits. Two dice give $6 \times 6 = 36$ outcomes. A 4 digit PIN gives $10^4 = 10,000$.

1.3.2 Order matters: permutations

- Arrange all n distinct objects in a row: $n!$ ways (first spot n choices, second $n - 1$, and so on).
- Arrange only k of them: $n \times (n - 1) \times \cdots \times (n - k + 1) = \frac{n!}{(n-k)!}$ ways. Gold, silver, bronze among 10 runners: $10 \times 9 \times 8 = 720$.
- **Repeated letters:** arrangements of a word = $\frac{n!}{(\text{repeat counts})!}$. "BOOK": $\frac{4!}{2!} = 12$ (swapping the two O's gives the same word, so divide by 2!). "STATISTICS" (S 3 times, T 3, I 2, A 1, C 1): $\frac{10!}{3!3!2!} = 50,400$.

1.3.3 Order does not matter: combinations

- Choosing a committee, a hand of cards, a set of test users: only *which* items, not their order. Count: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.
- Why divide by $k!$: each group of k can be ordered in $k!$ ways, and we want to count each group once.
- Example: choose 3 from 9 people: $\binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{6} = 84$.
- "At least one" with combinations: committees of 3 from 5 men and 4 women with at least one woman = all committees – all male committees = $84 - \binom{5}{3} = 84 - 10 = 74$.
- Card example: 5 cards from 52. Exactly 2 aces: choose the 2 aces ($\binom{4}{2} = 6$ ways) and the other 3 cards from the 48 non aces ($\binom{48}{3} = 17,296$). Probability = $\frac{6 \times 17,296}{\binom{52}{5}} = \frac{103,776}{2,598,960} \approx 0.040$.

Decision in 5 seconds

- Does rearranging the chosen items give a different result? Yes: permutations (passwords, rankings, seatings). No: combinations (committees, hands, samples).
- Can an item be used again? Yes: powers (n^k). No: factorial style products.

1.4 Conditional probability: shrinking the world

Start here: the everyday picture

- 100 users visited a job page. 40 came from mobile, and of those 10 applied. 60 came from desktop, and of those 30 applied.
- Question: a user applied. What is the chance they were on mobile? We only look at the 40 users who applied: 10 of them were on mobile. Answer $\frac{10}{40} = 0.25$.
- That is all conditional probability is: **restrict attention to the cases where the given information is true, then count again.**

- Formula: $P(A | B) = \frac{P(A \cap B)}{P(B)}$, read "probability of A given B ". Dividing by $P(B)$ rescales the shrunken world so that it has total probability 1.
- Check on the story: $P(\text{mobile} \cap \text{applied}) = \frac{10}{100}$, $P(\text{applied}) = \frac{40}{100}$, ratio 0.25.
- Order matters: $P(\text{applied} | \text{mobile}) = \frac{10}{40} = 0.25$ here by coincidence of numbers, but in general $P(A | B) \neq P(B | A)$. "Most offers go to people with GitHub profiles" is $P(\text{GitHub} | \text{offer})$, not $P(\text{offer} | \text{GitHub})$.
- **Multiplication rule** (rearranged formula): $P(A \cap B) = P(B) P(A | B)$. This is how we handle drawing **without replacement**.
- Example: an urn has 5 red and 3 blue balls. Draw 2 without replacement. $P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14} \approx 0.357$. The second factor is conditional: after one red is gone, 4 red remain

among 7 balls.

- With replacement the draws do not affect each other: $\left(\frac{5}{8}\right)^2 = 0.391$.

1.4.1 A famous surprise: the second card

- Two cards are drawn and you do not look at the first. $P(\text{second is an ace})$? It is $\frac{4}{52} = \frac{1}{13}$, exactly the same as for the first card.
- Reason (symmetry): before anything is revealed, every position in a shuffled deck is equally likely to hold any card. Not seeing the first card gives you no information.

1.5 Independence

- A and B are **independent** if knowing one happened does not change the chance of the other: $P(A | B) = P(A)$. Equivalent and easier to test: $P(A \cap B) = P(A)P(B)$.
- Example: two fair coin tosses. $P(\text{both heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. The coin has no memory.
- Test with numbers: $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Then $P(A \cap B) = 0.6 + 0.5 - 0.8 = 0.3 = 0.6 \times 0.5$: independent.
- For independent events, $P(A \cup B) = P(A) + P(B) - P(A)P(B)$. With 0.3 and 0.4: $0.3 + 0.4 - 0.12 = 0.58$.

Independent is NOT the same as mutually exclusive

- Mutually exclusive means they cannot happen together: $P(A \cap B) = 0$.
- If two events with positive probability are mutually exclusive, they are strongly *dependent*: learning that A happened tells you B certainly did not.
- Quick check in the exam: exclusive $\Rightarrow P(A \cup B) = P(A) + P(B)$; independent $\Rightarrow P(A \cap B) = P(A)P(B)$.
- **Pairwise vs mutual independence**: toss two coins; A : first is H; B : second is H; C : both show the same face. Each pair is independent (for example $P(A \cap C) = P(HH) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$). But all three together: $P(A \cap B \cap C) = P(HH) = \frac{1}{4} \neq \frac{1}{8}$. Knowing A and B decides C completely. So they are pairwise independent but not mutually independent.

1.6 The law of total probability (a preview of Bayes)

- If the world splits into non overlapping cases $B_1, B_2, \dots$ that together cover everything, then

$$P(A) = \sum_i P(A | B_i)P(B_i).$$

Read it as: “the overall rate is a weighted average of the rates in each case, weighted by how common each case is”.

- Example: machine A makes 60% of items with a 2% defect rate, machine B makes 40% with 5%. $P(\text{defective}) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.020 = 0.032$.
- Chapter 2 turns this around to ask “given a defective item, which machine made it?”.

1.7 Random variables: attaching numbers to outcomes

- A **random variable** (RV) is a rule that gives a number to every outcome. Toss 3 coins; let X = number of heads. The outcome HTH gets $X = 2$.

- Capital X is the variable (before the experiment); lower case x is a particular value it can take.
- **Discrete** RVs take separate values you can list (0, 1, 2, 3). **Continuous** RVs take any value in an interval (a waiting time of 3.27 minutes).
- The **probability mass function** (PMF) lists $P(X = x)$ for each value. For 3 fair coins:

x	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

The 3/8 for $x = 1$ comes from the three outcomes HTT, THT, TTH. The probabilities sum to 1.

- The **cumulative distribution function** (CDF) is $F(x) = P(X \leq x)$: a running total. Here $F(1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$.

1.8 Expectation: the long run average

Start here: the everyday picture

- A game: roll a die and win that many rupees. Play it 6,000 times. Each face comes up about 1,000 times, so you win about $1000(1 + 2 + 3 + 4 + 5 + 6) = 21,000$ rupees, which is 3.5 per game.
- That 3.5 is the **expected value**. You never actually win 3.5 in one game; it is the average you approach over many games.

- Definition (discrete): $E[X] = \sum_x x P(X = x)$: each value weighted by how often it happens.
- Die: $E[X] = \frac{1+2+3+4+5+6}{6} = 3.5$. Three coins: $E[X] = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = 1.5$.
- **Expectation of a function** (called LOTUS): $E[g(X)] = \sum_x g(x)P(X = x)$. You do not need the distribution of $g(X)$; just apply g to each value. Die: $E[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx 15.17$.
- Caution: $E[g(X)] \neq g(E[X])$ in general. $E[X^2] = 15.17$ but $(E[X])^2 = 12.25$.

1.8.1 Linearity of expectation (a reported InfoEdge interview topic)

The most powerful shortcut in probability

- $E[aX + b] = aE[X] + b$ and $E[X + Y] = E[X] + E[Y]$, **always**, whether or not X and Y are independent.
- Why: expectation is a weighted sum, and sums can be split and constants pulled out (Chapter 0).
- Contrast: $E[XY] = E[X]E[Y]$ needs independence (or at least zero correlation).
- **The indicator trick.** An **indicator** I_A is 1 if event A happens and 0 otherwise. Its expectation is $E[I_A] = 1 \cdot P(A) + 0 = P(A)$. To find an expected *count*, write the count as a sum of indicators and add their probabilities.
- Example 1 (hat check): n people get their hats back at random. Let $I_k = 1$ if person k gets their own hat; $P = \frac{1}{n}$. Expected number of correct hats = $n \times \frac{1}{n} = 1$, for every n . The indicators are dependent, but linearity does not care.
- Example 2 (birthday pairs): 30 people. Each of the $\binom{30}{2} = 435$ pairs shares a birthday with probability $\frac{1}{365}$. Expected number of sharing pairs = $\frac{435}{365} \approx 1.19$.
- Example 3 (patterns): 10 coin tosses; how many times does HH appear (overlaps allowed)? There are 9 adjacent positions, each HH with probability $\frac{1}{4}$: expected $9 \times \frac{1}{4} = 2.25$.

1.8.2 First step analysis: expectations that refer to themselves

- Toss a coin until the first head. Let E be the expected number of tosses. The first toss is always used. With probability $\frac{1}{2}$ it is a head and we stop; with probability $\frac{1}{2}$ it is a tail and we are back at the start. So $E = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}E$, giving $E = 2$.
- In general, waiting for an event of probability p takes $\frac{1}{p}$ trials on average. A die needs 6 rolls on average to show a six.
- Two heads in a row (HH) takes 6 tosses on average; HT takes only 4. (Worked in the question bank.)
- **Coupon collector:** expected rolls to see all six faces $= \frac{6}{6} + \frac{6}{5} + \frac{6}{4} + \frac{6}{3} + \frac{6}{2} + \frac{6}{1} = 14.7$. Each term is the wait for a new face when some are already collected.

1.9 Variance and standard deviation: how spread out

Start here: the everyday picture

- Two job portals both average 50 applications per post. On portal A every post gets between 45 and 55. On portal B some get 5 and some get 200. Same mean, very different experience. We need a number for spread.

- **Variance:** the average squared distance from the mean: $\text{Var}(X) = E[(X - \mu)^2]$ where $\mu = E[X]$.
- Why square? Distances above and below the mean would cancel if we just averaged them (they always sum to zero). Squaring makes all distances positive and punishes big deviations more.
- **Standard deviation** $\sigma = \sqrt{\text{Var}(X)}$ brings the units back (rupees instead of rupees squared).
- **Shortcut formula** (use this in exams): $\text{Var}(X) = E[X^2] - (E[X])^2$. "Mean of the squares minus square of the mean".
- Die: $E[X^2] = \frac{91}{6}$, $(E[X])^2 = \frac{49}{4}$. $\text{Var} = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12} \approx 2.917$. $\text{SD} \approx 1.708$.
- Rules:
 - $\text{Var}(X + b) = \text{Var}(X)$: shifting everything does not change spread.
 - $\text{Var}(aX) = a^2\text{Var}(X)$: scaling by a scales distances by a , squared distances by a^2 . Note $\text{Var}(-X) = \text{Var}(X)$.
 - If X, Y are independent: $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$. Yes, plus even for a difference: subtracting an uncertain quantity adds uncertainty.
 - In general: $\text{Var}(X \pm Y) = \text{Var}X + \text{Var}Y \pm 2\text{Cov}(X, Y)$.
- Example: X, Y independent, $\text{Var}X = 1$, $\text{Var}Y = 2$. $\text{Var}(2X - 3Y + 4) = 4(1) + 9(2) = 22$.
- **Covariance** $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ measures whether X and Y tend to be high together (positive) or move oppositely (negative). Chapter 6 develops it fully.
- Example of negative covariance: roll a die n times; X = number of 1s, Y = number of 6s. More 1s leave fewer rolls for 6s, so $\text{Cov}(X, Y) = -\frac{n}{36}$.

1.10 Continuous random variables: probability as area

Start here: the everyday picture

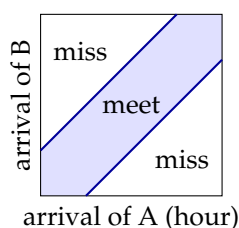
- A delivery time can be 12.3 minutes, 12.31, 12.3107... There are infinitely many possible values, so the chance of hitting any *exact* value is 0. What makes sense is the chance of landing in a *range*, like 12 to 13 minutes.
- Picture sand spread along a line. The **density** $f(x)$ is how thickly sand lies at x . The probability of an interval is the amount of sand over it: the **area under** f .
- Rules for a density: $f(x) \geq 0$ and total area $\int_{-\infty}^{\infty} f(x) dx = 1$.
- $P(a \leq X \leq b) = \int_a^b f(x) dx$. And $P(X = a) = 0$, so $<$ and $\leq$ give the same answer for continuous variables.
- A density is **not** a probability. It can exceed 1: the uniform density on $[0, 0.5]$ is $f(x) = 2$ (so that the area $0.5 \times 2 = 1$).
- CDF $F(x) = P(X \leq x)$ = area to the left of x . It rises from 0 to 1 and never decreases. The density is the slope of the CDF.
- Mean and variance: replace sums by integrals. $E[X] = \int x f(x) dx$, $E[X^2] = \int x^2 f(x) dx$, then $\text{Var} = E[X^2] - (E[X])^2$.

1.10.1 Worked example (the reported InfoEdge OA type)

- $f(x) = 3x^2$ on $[0, 1]$. Check the area: $\int_0^1 3x^2 dx = [x^3]_0^1 = 1$. Valid.
- $E[X] = \int_0^1 x \cdot 3x^2 dx = 3 \cdot \frac{x^4}{4} \Big|_0^1 = \frac{3}{4}$.
- $E[X^2] = \int_0^1 x^2 \cdot 3x^2 dx = 3 \cdot \frac{x^5}{5} \Big|_0^1 = \frac{3}{5}$.
- $\text{Var} = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{5} - \frac{9}{16} = \frac{48-45}{80} = \frac{3}{80} = 0.0375$.
- Pattern worth memorising: for $f(x) = (k+1)x^k$ on $[0, 1]$, $E[X^m] = \frac{k+1}{k+1+m}$. Here $k = 2$: $E[X] = \frac{3}{4}$, $E[X^2] = \frac{3}{5}$. This turns a 2 minute question into a 30 second one.

1.10.2 Two continuous shapes you meet immediately

- **Uniform on $[a, b]$:** flat density $\frac{1}{b-a}$. Probability of a sub interval is just its length divided by $(b-a)$. Mean $\frac{a+b}{2}$, variance $\frac{(b-a)^2}{12}$. For $[0, 1]$: mean $\frac{1}{2}$, variance $\frac{1}{12}$, and $E[X^n] = \frac{1}{n+1}$.
- **Exponential with rate λ :** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Tail $P(X > t) = e^{-\lambda t}$. With $\lambda = 2$: $P(X > 1) = e^{-2} = 0.135$.
- **Geometry replaces integrals** whenever the variables are uniform: probability becomes area in a square. Two people arriving uniformly in an hour and waiting 15 minutes meet with probability $1 - \left(\frac{3}{4}\right)^2 = \frac{7}{16}$ (the two corner triangles are where they miss).



Meeting problem: shaded band $|A - B| \leq \frac{1}{4}$ has area $1 - 2 \cdot \frac{1}{2} \left(\frac{3}{4}\right)^2 = \frac{7}{16}$.

1.11 Two inequalities for when you know almost nothing

- **Markov:** for a variable that is never negative, $P(X \geq a) \leq \frac{E[X]}{a}$. If orders per day average 10, at most $10/50 = 20\%$ of days can have 50 or more orders. (If more than 20% did, the average would already exceed 10.)
- **Chebyshev:** for any variable with mean μ and SD σ , $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$. At least 75% of any distribution lies within 2 SDs, at least 89% within 3.
- Example: mean 50, SD 5. $P(|X - 50| \geq 10)$: $k = 2$, bound $\frac{1}{4}$. For a normal the true value is only 0.046; Chebyshev is a worst case guarantee.

1.12 Entropy: measuring surprise (a reported InfoEdge OA topic)

Start here: the everyday picture

- If a friend who is always late is late again, you are not surprised. If they arrive early, you are very surprised. Rare events carry more surprise.
- Define the surprise of an outcome with probability p as $\log_2 \frac{1}{p}$. Probability 1: surprise 0. Probability $\frac{1}{2}$: 1 bit. Probability $\frac{1}{4}$: 2 bits.
- **Entropy** is the *average* surprise, i.e. how uncertain the variable is.

- Formula: $H(X) = -\sum_i p_i \log_2 p_i = \sum_i p_i \log_2 \frac{1}{p_i}$ (bits). With $\ln$ instead of $\log_2$ the unit is nats.
- Fair coin: $H = \frac{1}{2}(1) + \frac{1}{2}(1) = 1$ bit. Probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$: $H = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{4}(2) = 1.5$ bits.
- Facts: $H \geq 0$; $H = 0$ only when one outcome is certain; with k outcomes the maximum is $\log_2 k$, reached by the uniform distribution (fair die: $\log_2 6 = 2.585$ bits).
- ML uses: decision trees split to reduce entropy (information gain); cross entropy is the classification loss; entropy of a classifier's output measures its uncertainty.

Where this shows up in ML and at a company like Naukri

- Click, apply, churn: every binary label is an event with a probability. Models output estimates of $P(\text{event} \mid \text{features})$, which is conditional probability.
- Expected values drive decisions: expected revenue = (probability of conversion) $\times$ (value). Linearity lets you add expected clicks across 50 recommendations without modelling their dependence.
- Variance is uncertainty: ensembles average models to cut variance; the variance of a sum explains why correlated errors do not cancel.
- Entropy and cross entropy appear in decision tree splits and in the loss of every classifier you will train.

Formula sheet

$P(A^c) = 1 - P(A)$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ independent:
 $P(A \cap B) = P(A)P(B)$ $P(A) = \sum P(A \mid B_i)P(B_i)$ $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ arrangements with repeats
 $\frac{n!}{n_1!n_2!\dots}$ $E[X] = \sum xp(x)$ or $\int xf(x)dx$ $E[aX + b] = aE[X] + b$ $E[X + Y] = E[X] + E[Y]$
 always $\text{Var} = E[X^2] - (E[X])^2$ $\text{Var}(aX + b) = a^2\text{Var}X$ $\text{Var}(X \pm Y) = \text{Var}X + \text{Var}Y \pm 2\text{Cov}$
 Markov $P(X \geq a) \leq E[X]/a$ Chebyshev $P(|X - \mu| \geq k\sigma) \leq 1/k^2$ $H = -\sum p \log_2 p$.

15 second recall card: memorise every line

- Two dice: 36 outcomes; $P(\text{sum} = 7) = \frac{1}{6}$; die mean 3.5, variance $\frac{35}{12} \approx 2.92$.
- $P(\text{at least one six in 4 rolls}) = 1 - (5/6)^4 = 0.518$. Birthday: 23 people give just over 50%.
- Expected tosses: first head 2; HH 6; HT 4; all six die faces 14.7. Waiting for probability p : $1/p$.
- Random permutation: expected fixed points 1. Second card unseen first: same as first card.
- Monty Hall: switch, win $\frac{2}{3}$. Two children, at least one boy: both boys $\frac{1}{3}$.
- Uniform $[0, 1]$: mean $\frac{1}{2}$, variance $\frac{1}{12}$, $E[X^n] = \frac{1}{n+1}$; min of two has mean $\frac{1}{3}$, max $\frac{2}{3}$.
- Stick broken at two random points forms a triangle with probability $\frac{1}{4}$. Point in unit square inside quarter circle: $\frac{\pi}{4}$.
- Density can exceed 1; $P(X = a) = 0$ for continuous X .
- Entropy: fair coin 1 bit; $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ gives 1.5 bits; maximum $\log_2 k$ at uniform.
- Exclusive events with positive probability are dependent. Pairwise independence does not imply mutual independence.

1.13 Question bank: Probability**Q1.1 CALC · 1.5 to 2 min**

★ **PYQ (OA: variance of a continuous function over an interval)** A random variable X has density $f(x) = 3x^2$ for $0 \leq x \leq 1$ (and 0 elsewhere). $\text{Var}(X)$ is

- (A) $3/80$
- (B) $3/5$
- (C) $3/4$
- (D) $1/16$

Solution

Fast route: For $f = (k+1)x^k$ on $[0, 1]$, $E[X^m] = \frac{k+1}{k+1+m}$. With $k = 2$: $E[X] = \frac{3}{4}$, $E[X^2] = \frac{3}{5}$. Variance $\frac{3}{5} - \frac{9}{16} = \frac{3}{80}$.

Full working:

- $E[X] = \int_0^1 3x^3 dx = \frac{3}{4}$.
- $E[X^2] = \int_0^1 3x^4 dx = \frac{3}{5}$.
- $\text{Var} = \frac{3}{5} - \frac{9}{16} = \frac{48-45}{80} = \frac{3}{80} = 0.0375$.

Why the other options fail: (B) is $E[X^2]$, not the variance. (C) is the mean. (D) squares $1 - E[X] = \frac{1}{4}$, which is not a variance.

Answer: A.

Q1.2 CALC · 1.5 to 2 min

★ **PYQ (OA: variance of a function over an interval)** X is uniform on $[0, 1]$. $\text{Var}(X^2)$ equals

- (A) $1/5$
- (B) $1/45$
- (C) $4/45$
- (D) $1/12$

Solution

Fast route: $E[X^n] = \frac{1}{n+1} \cdot E[X^4] - (E[X^2])^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}$.

Full working:

- Let $Y = X^2$. We need $E[Y^2] - (E[Y])^2 = E[X^4] - (E[X^2])^2$.
- $E[X^2] = \int_0^1 x^2 dx = \frac{1}{3}$; $E[X^4] = \int_0^1 x^4 dx = \frac{1}{5}$.
- $\text{Var}(Y) = \frac{1}{5} - \frac{1}{9} = \frac{9-5}{45} = \frac{4}{45} \approx 0.089$.

Why the other options fail: (D) is $\text{Var}(X)$, not $\text{Var}(X^2)$. (A) is $E[X^4]$ alone. (B) drops a factor.

Answer: C.

Q1.3 RECALL · about 15 s

★ **PYQ (INT and OA: linearity of expectation)** Which statement is always true for any two random variables X and Y with finite means?

- (A) $E[X/Y] = E[X]/E[Y]$
- (B) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- (C) $E[XY] = E[X]E[Y]$
- (D) $E[X + Y] = E[X] + E[Y]$

Solution

Fast route: Linearity of expectation needs no independence. Everything else does.

Full working:

- $E[X + Y] = \sum_{x,y} (x + y)p(x, y)$ splits into $\sum x p_X(x) + \sum y p_Y(y)$: always true.

Why the other options fail: (C) needs zero correlation. (B) needs zero covariance. (A) is false even for independent variables (Jensen: $E[1/Y] \neq 1/E[Y]$).

Answer: D.

Q1.4 REASON · about 45 s

★ **PYQ (OA: entropy formula)** A variable takes three values with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. Its entropy in bits is

- (A) 2
- (B) 1
- (C) 1.5
- (D) 1.585

Solution

Fast route: Surprises are 1, 2, 2 bits. Weighted: $\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{4}(2) = 1.5$.

Full working:

- $H = -\sum p \log_2 p = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{1}{4} \log_2 4 = 0.5 + 0.5 + 0.5 = 1.5$ bits.

Why the other options fail: (D) $\log_2 3$ is the maximum for 3 outcomes (uniform), which this is not. (B) is a fair coin. (A) is uniform over 4 outcomes.

Answer: C.

Q1.5 RECALL · about 15 s

Two fair dice are rolled. $P(\text{sum} = 7)$ is

- (A) $1/11$
- (B) $7/36$
- (C) $1/6$
- (D) $1/12$

Solution

Fast route: 6 favourable ordered pairs out of 36.

Full working:

- $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$: 6 out of 36 = $\frac{1}{6}$.

Why the other options fail: (A) treats the 11 possible sums as equally likely; they are not.

Answer: C.

Q1.6 REASON · about 45 s

A fair die is rolled 4 times. The probability of at least one six is closest to

- (A) 0.482
- (B) 0.333
- (C) 0.667
- (D) 0.518

Solution

Fast route: Complement: $1 - (5/6)^4 = 1 - 0.482$.

Full working:

- $P(\text{no six}) = (5/6)^4 = 625/1296 = 0.482$. At least one: 0.518.

Why the other options fail: (C) adds $\frac{1}{6}$ four times, which double counts rolls with several sixes. (A) is the complement.

Answer: D.

Q1.7 REASON · about 45 s

$P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Then A and B are

- (A) such that $A \subset B$
- (B) neither exclusive nor independent
- (C) mutually exclusive
- (D) independent

Solution

Fast route: $P(A \cap B) = 0.6 + 0.5 - 0.8 = 0.3 = 0.6 \times 0.5$.

Full working:

- Addition rule gives the overlap 0.3. Product of marginals is also 0.3: independent.

Why the other options fail: (C) would need overlap 0. (A) would need $P(A) \leq P(B)$, but $0.6 > 0.5$.

Answer: D.

Q1.8 REASON · about 45 s

$P(A) = 0.3$ and $P(B) = 0.4$, and A, B are independent. $P(A \cup B)$ is

- (A) 0.88
- (B) 0.12
- (C) 0.58
- (D) 0.70

Solution

Fast route: $0.3 + 0.4 - 0.12$.

Full working:

- Independent: $P(A \cap B) = 0.12$. Union = 0.58. Equivalently $1 - P(\text{neither}) = 1 - 0.7 \times 0.6 = 0.58$.

Why the other options fail: (D) is the answer only if they were mutually exclusive. (B) is the intersection.

Answer: C.

Q1.9 CALC - 1.5 to 2 min

A committee of 3 is chosen from 5 men and 4 women. How many committees contain at least one woman?

- (A) 60
- (B) 74
- (C) 80
- (D) 84

Solution

Fast route: All committees minus all male: $\binom{9}{3} - \binom{5}{3} = 84 - 10$.

Full working:

- Total $\binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{6} = 84$. All male $\binom{5}{3} = 10$. Answer 74.
- Check by cases: 1 woman $4 \times \binom{5}{2} = 40$; 2 women $\binom{4}{2} \times 5 = 30$; 3 women 4. Total 74.

Why the other options fail: (D) is all committees, including the 10 all male ones. (C) and (A) come from miscounting the cases.

Answer: B.

Q1.10 CALC - 1.5 to 2 min

The number of distinct arrangements of the letters of STATISTICS is

- (A) 50,400
- (B) 3,628,800
- (C) 25,200
- (D) 100,800

Solution

Fast route: $\frac{10!}{3!3!2!} = \frac{3628800}{72}$.

Full working:

- Letters: S 3, T 3, I 2, A 1, C 1 (total 10).
- $\frac{10!}{3!3!2!} = \frac{3,628,800}{6 \cdot 6 \cdot 2} = 50,400$.

Why the other options fail: (B) ignores repeated letters. (C) and (D) divide by a wrong factor (144 or 36).

Answer: A.

Q1.11 **RECALL** · about 15 s

In a room of 23 people (365 equally likely birthdays), the probability that at least two share a birthday is closest to

- (A) 0.25
- (B) 0.94
- (C) 0.51
- (D) 0.06

Solution

Fast route: The classic birthday result: 23 people cross 50%.

Full working:

- $P(\text{all different}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{343}{365} \approx 0.493$, so at least one pair ≈ 0.507 .
- Intuition: 23 people form $\binom{23}{2} = 253$ pairs, each a small chance.

Why the other options fail: (D) is $23/365$, which thinks about one person only.

Answer: C.

Q1.12 **CALC** · 1.5 to 2 min

A 5 card hand is dealt from a standard 52 card deck. The probability it contains exactly 2 aces is closest to

- (A) 0.10
- (B) 0.20
- (C) 0.04
- (D) 0.004

Solution

Fast route: $\frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}} = \frac{6 \times 17296}{2598960} \approx 0.04$.

Full working:

- Choose 2 aces: 6. Choose 3 non aces: $\binom{48}{3} = \frac{48 \cdot 47 \cdot 46}{6} = 17,296$.
- Favourable = 103,776. Total $\binom{52}{5} = 2,598,960$. Ratio 0.0399.
- Estimate check: treat draws as independent with $P(\text{ace}) = \frac{1}{13}$: $\binom{5}{2} \left(\frac{1}{13}\right)^2 \left(\frac{12}{13}\right)^3 \approx 0.047$. Close to 0.04, far from the other options.

Answer: C.

Q1.13 **REASON** · about 45 s

An urn has 5 red and 3 blue balls. Two are drawn **without** replacement. $P(\text{both red})$ is

- (A) $25/64$
- (B) $5/8$
- (C) $15/56$
- (D) $5/14$

Solution**Fast route:** $\frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56}$.**Full working:**

- First red $\frac{5}{8}$; given that, 4 red of 7 remain: $\frac{4}{7}$. Product $\frac{5}{8} \cdot \frac{4}{7} \approx 0.357$.
- Counting check: $\binom{5}{2} / \binom{8}{2} = 10/28 = 5/14$.

Why the other options fail: (A) is the with replacement answer. (C) is P(one red then one blue).**Answer:** D.**Q1.14 REASON · about 45 s**

Same urn (5 red, 3 blue), but now the two draws are **with** replacement. Compared with drawing without replacement, $P(\text{both red})$ is

- (A) smaller, equal to $5/14$
- (B) the same
- (C) larger, equal to $25/64$
- (D) larger, equal to $5/8$

Solution**Fast route:** With replacement: $(5/8)^2 = 0.391 > 0.357$.**Full working:**

- Replacing the red ball keeps the red share at $\frac{5}{8}$ for the second draw, so both red is more likely.

Answer: C.**Q1.15 REASON · about 45 s**

A family has two children, each independently a boy or a girl with probability $\frac{1}{2}$. Given that at least one is a boy, the probability that both are boys is

- (A) $2/3$
- (B) $1/3$
- (C) $1/2$
- (D) $1/4$

Solution**Fast route:** Condition removes GG: remaining BB, BG, GB; one of three.**Full working:**

- $P(BB \mid \text{at least one B}) = \frac{1/4}{3/4} = \frac{1}{3}$.

Why the other options fail: (C) is the answer to a different question: “the older child is a boy”. Wording decides the conditioning event.**Answer:** B.**Q1.16 REASON · about 45 s**

Two fair coins are tossed. A: first is H. B: second is H. C: both show the same face. Which is correct?

- (A) A and C are dependent
- (B) $P(A \cap B \cap C) = 1/8$

- (C) A, B, C are mutually independent
 (D) They are pairwise independent but not mutually independent

Solution

Fast route: Any two of them are independent, but A and B together decide C .

Full working:

- Each has probability $\frac{1}{2}$. $P(A \cap C) = P(HH) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$; same for $B \cap C$ and $A \cap B$.
- $P(A \cap B \cap C) = P(HH) = \frac{1}{4} \neq \frac{1}{8}$; not mutually independent.

Answer: D.

Q1.17 **RECALL** · about 15 s

$P(A \cap B) = 0.2$ and $P(B) = 0.5$. $P(A | B)$ is

- (A) 0.7
 (B) 0.4
 (C) 0.25
 (D) 0.1

Solution

Fast route: $0.2/0.5$.

Full working:

- Definition: $P(A | B) = \frac{P(A \cap B)}{P(B)} = 0.4$.

Answer: B.

Q1.18 **REASON** · about 45 s

Machine A makes 60% of items with a 2% defect rate; machine B makes 40% with a 5% defect rate. The probability a random item is defective is

- (A) 0.020
 (B) 0.032
 (C) 0.070
 (D) 0.035

Solution

Fast route: Weighted average: $0.6(0.02) + 0.4(0.05)$.

Full working:

- Law of total probability: $0.012 + 0.020 = 0.032$.

Why the other options fail: (D) is the simple average of 2% and 5%, ignoring the 60/40 mix. (C) adds the rates.

Answer: B.

Q1.19 **RECALL** · about 15 s

For one roll of a fair die, $E[X]$ and $\text{Var}(X)$ are

- (A) 3 and $35/12$
 (B) 3.5 and $\sqrt{35/12}$

- (C) 3.5 and $35/12$
 (D) 3.5 and 3.5

Solution

Fast route: Memorise: 3.5 and $\frac{35}{12} \approx 2.92$ (discrete uniform on $1..n$ has variance $\frac{n^2-1}{12}$).

Full working:

- $E[X^2] = \frac{91}{6}$, variance $\frac{91}{6} - 12.25 = \frac{35}{12}$.

Why the other options fail: (B) gives the SD, not the variance.

Answer: C.

Q1.20 RECALL · about 15 s

A fair coin is tossed until the first head appears. The expected number of tosses is

- (A) 1
 (B) 2
 (C) 1.5
 (D) 3

Solution

Fast route: Waiting for probability p takes $1/p$ trials: $1/0.5 = 2$.

Full working:

- $E = 1 + \frac{1}{2}E \Rightarrow E = 2$ (first step analysis).

Answer: B.

Q1.21 REASON · about 45 s

The expected number of fair coin tosses needed to see two heads in a row (HH) is

- (A) 4
 (B) 8
 (C) 3
 (D) 6

Solution

Fast route: Known result: HH takes 6 (HT takes 4). General p : $\frac{1+p}{p^2}$.

Full working:

- Let a = expected remaining tosses from the start, b = from “just saw one H”.
- $a = 1 + \frac{1}{2}b + \frac{1}{2}a$ and $b = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}a$.
- From the second, $b = 1 + \frac{a}{2}$. Substitute: $a = 1 + \frac{1}{2} + \frac{a}{4} + \frac{a}{2} \Rightarrow \frac{a}{4} = \frac{3}{2} \Rightarrow a = 6$.

Why the other options fail: (A) is the answer for HT: after an H, a further H does not reset your progress.

Answer: D.

Q1.22 REASON · about 45 s

The expected number of rolls of a fair die until every face has appeared at least once is closest to

- (A) 14.7

- (B) 36
(C) 6
(D) 8.5

Solution

Fast route: Coupon collector: $6 \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right) = 6 \times 2.45$.

Full working:

- With k faces seen, a new face has probability $\frac{6-k}{6}$, so the wait is $\frac{6}{6-k}$ rolls. Sum over $k = 0..5$:
 $1 + 1.2 + 1.5 + 2 + 3 + 6 = 14.7$.

Answer: A.

Q1.23 REASON · about 45 s

X, Y independent with $\text{Var}(X) = 1, \text{Var}(Y) = 2$. $\text{Var}(2X - 3Y + 4)$ is

- (A) -4
(B) 14
(C) 26
(D) 22

Solution

Fast route: Square the coefficients, drop the constant: $4(1) + 9(2)$.

Full working:

- $\text{Var}(aX + bY + c) = a^2\text{Var}X + b^2\text{Var}Y$ for independent X, Y . $= 4 + 18 = 22$.

Why the other options fail: (A) and (B) subtract variances or forget to square. (C) keeps the constant somehow.

Answer: D.

Q1.24 REASON · about 45 s

$\text{Var}(X) = 4, \text{Var}(Y) = 9, \text{Cov}(X, Y) = 2$. $\text{Var}(X + Y)$ and $\text{Var}(X - Y)$ are

- (A) 17 and 9
(B) 13 and 13
(C) 17 and 5
(D) 15 and 11

Solution

Fast route: $13 \pm 2(2)$.

Full working:

- $\text{Var}(X \pm Y) = 4 + 9 \pm 4$: 17 and 9.

Why the other options fail: (B) ignores covariance. (D) uses $\pm\text{Cov}$ instead of $\pm 2\text{Cov}$.

Answer: A.

Q1.25 CALC · 1.5 to 2 min

$f(x) = kx$ for $0 \leq x \leq 2$ is a density. The value of k and $P(X > 1)$ are

- (A) $k = 1/2, P = 1/2$

- (B) $k = 1/2, P = 3/4$
 (C) $k = 1/4, P = 1/4$
 (D) $k = 1, P = 3/4$

Solution

Fast route: Triangle of base 2 and height $2k$ has area $2k = 1$, so $k = \frac{1}{2}$. Area left of 1 is $\frac{1}{2} \cdot 1 \cdot \frac{1}{2} = \frac{1}{4}$.

Full working:

- $\int_0^2 kx \, dx = 2k = 1 \Rightarrow k = \frac{1}{2}$.
- $P(X > 1) = \int_1^2 \frac{x}{2} \, dx = \frac{4-1}{4} = \frac{3}{4}$.

Why the other options fail: (A) assumes a flat density. (D) has area 2, not a valid density.

Answer: B.

Q1.26 REASON · about 45 s

X has density $f(x) = 2e^{-2x}$ for $x \geq 0$. $P(X > 1)$ is closest to

- (A) 0.368
 (B) 0.135
 (C) 0.865
 (D) 0.5

Solution

Fast route: Exponential tail $e^{-\lambda t} = e^{-2}$.

Full working:

- $\int_1^\infty 2e^{-2x} \, dx = e^{-2} = 0.1353$.

Why the other options fail: (A) is e^{-1} (wrong rate). (C) is $P(X \leq 1)$.

Answer: B.

Q1.27 CALC · 1.5 to 2 min

X has density $f(x) = 2x$ on $[0, 1]$. The median of X is

- (A) 0.707
 (B) 0.667
 (C) 0.5
 (D) 0.75

Solution

Fast route: CDF $F(x) = x^2$; set $x^2 = \frac{1}{2}$.

Full working:

- $F(x) = \int_0^x 2t \, dt = x^2$. Median m : $m^2 = \frac{1}{2} \Rightarrow m = \frac{1}{\sqrt{2}} = 0.707$.

Why the other options fail: (B) is the mean ($\int_0^1 2x^2 \, dx = \frac{2}{3}$). Mean < median here because the density is left skewed (mass piles up near 1).

Answer: A.

Q1.28 RECALL · about 15 s

For a continuous random variable with density f and CDF F , which is true?

- (A) $\int f(x) dx$ can equal 2
 (B) F can decrease
 (C) $f(x)$ can be greater than 1
 (D) $P(X = 3) = f(3)$

Solution

Fast route: A density is not a probability; Uniform $[0, 0.5]$ has $f = 2$.

Full working:

- Only the area must be 1; the height can be anything non negative.

Why the other options fail: (D) any exact value has probability 0. (B) a CDF is non decreasing. (A) total area must be exactly 1.

Answer: C.

Q1.29 REASON · about 45 s

In a group of 30 people (365 equally likely birthdays), the expected number of pairs of people sharing a birthday is closest to

- (A) 1.19
 (B) 2.38
 (C) 0.71
 (D) 0.08

Solution

Fast route: Indicators: $\binom{30}{2} \times \frac{1}{365} = \frac{435}{365}$.

Full working:

- Each pair shares with probability $\frac{1}{365}$; linearity adds 435 such indicators: 1.19.

Why the other options fail: (D) is $30/365$ (people, not pairs). (B) doubles the pair count (30×29).

Answer: A.

Q1.30 REASON · about 45 s

A fair coin is tossed 10 times. The expected number of occurrences of HH (overlaps allowed, so HHH counts 2) is

- (A) 5
 (B) 1.25
 (C) 2.25
 (D) 2.5

Solution

Fast route: 9 adjacent positions, each HH with probability $\frac{1}{4}$.

Full working:

- Indicator per position $i = 1..9$: $E = 9 \times \frac{1}{4} = 2.25$.

Why the other options fail: (D) uses 10 positions; there are only 9 adjacent pairs.

Answer: C.

Q1.31 REASON · about 45 s

Three ants sit at the corners of a triangle. Each independently picks one of its two edges and walks along it at the same speed. The probability that no two ants collide is

- (A) $1/4$
- (B) $1/8$
- (C) $3/4$
- (D) $1/2$

Solution

Fast route: Only “all clockwise” or “all anticlockwise” avoid collisions: $2/8$.

Full working:

- $2^3 = 8$ equally likely direction choices; 2 of them are safe.

Answer: A.

Q1.32 CALC · 1.5 to 2 min

Two friends arrive independently and uniformly between 5:00 and 6:00 and each waits 15 minutes for the other. The probability they meet is

- (A) $9/16$
- (B) $7/16$
- (C) $1/2$
- (D) $1/4$

Solution

Fast route: Unit square; miss region is two triangles of total area $(\frac{3}{4})^2 = \frac{9}{16}$.

Full working:

- Scale the hour to $[0, 1]$. They meet iff $|X - Y| \leq \frac{1}{4}$.
- Complement: two corner triangles each with legs $\frac{3}{4}$, area $\frac{1}{2} \cdot \frac{9}{16}$ each, total $\frac{9}{16}$.
- $P(\text{meet}) = 1 - \frac{9}{16} = \frac{7}{16} \approx 0.44$. (Picture in the teaching section.)

Why the other options fail: (A) is the probability of missing.

Answer: B.

Q1.33 RECALL · about 15 s

A stick is broken at two points chosen independently and uniformly along its length. The probability that the three pieces can form a triangle is

- (A) $1/4$
- (B) $1/3$
- (C) $1/8$
- (D) $1/2$

Solution

Fast route: Classic result: $\frac{1}{4}$.

Full working:

- Triangle iff every piece is shorter than half the stick. In the unit square of break points, this region is 4 small triangles of area $\frac{1}{16}$ each: total $\frac{1}{4}$.

Answer: A.

Q1.34 REASON · about 45 s

A point is chosen uniformly in the unit square $[0, 1]^2$. The probability it lies inside the quarter circle $x^2 + y^2 \leq 1$ is

- (A) 0.637
- (B) 0.785
- (C) 0.866
- (D) 0.5

Solution

Fast route: Area of a quarter unit circle = $\frac{\pi}{4}$.

Full working:

- Probability = area = $\frac{\pi}{4} \approx 0.785$. This is how Monte Carlo estimates π .

Answer: B.

Q1.35 RECALL · about 15 s

A die is rolled repeatedly. The probability that the first six appears on the third roll is

- (A) 1/216
- (B) 1/6
- (C) 5/36
- (D) 25/216

Solution

Fast route: Not six, not six, six: $\frac{5}{6} \cdot \frac{5}{6} \cdot \frac{1}{6}$.

Full working:

- $\frac{25}{216} \approx 0.116$ (a geometric probability).

Answer: D.

Q1.36 RECALL · about 15 s

The probability of exactly 3 heads in 5 fair coin tosses is

- (A) 1/2
- (B) 5/16
- (C) 1/32
- (D) 3/5

Solution

Fast route: $\binom{5}{3}/2^5 = 10/32$.

Full working:

- Choose which 3 tosses are heads: 10 ways, each sequence has probability $\frac{1}{32} \cdot \frac{10}{32} = \frac{5}{16}$.

Answer: B.

Q1.37 CALC · 1.5 to 2 min

In a survey, 40% read paper A, 30% read B, 20% read C; 10% read A and B, 8% A and C, 5% B and C, and 3% read all three. The fraction reading none is

- (A) 0.13
- (B) 0.30
- (C) 0.27
- (D) 0.70

Solution

Fast route: Inclusion exclusion: $0.9 - 0.23 + 0.03 = 0.70$ read at least one.

Full working:

- Singles $0.4 + 0.3 + 0.2 = 0.9$; pairs $0.10 + 0.08 + 0.05 = 0.23$; triple 0.03.
- $P(\text{at least one}) = 0.9 - 0.23 + 0.03 = 0.70$; none = 0.30.

Why the other options fail: (C) forgets to add back the triple. (D) is “at least one”.

Answer: B.

Q1.38 REASON · about 45 s

Two cards are drawn without replacement. $P(\text{first is an ace and second is a king})$ is

- (A) $2/13$
- (B) $1/169$
- (C) $8/663$
- (D) $4/663$

Solution

Fast route: $\frac{4}{52} \cdot \frac{4}{51} = \frac{16}{2652} = \frac{4}{663}$.

Full working:

- Removing an ace does not remove any king, so 4 kings remain among 51 cards.

Why the other options fail: (B) is with replacement. (C) counts both orders (ace king or king ace).

Answer: D.

Q1.39 RECALL · about 15 s

Two cards are drawn without replacement and you do not see the first. The probability that the second is an ace is

- (A) $1/13$
- (B) $1/17$
- (C) $4/51$
- (D) $3/51$

Solution

Fast route: Symmetry: any unseen position is like the first.

Full working:

- Total probability: $\frac{4}{52} \cdot \frac{3}{51} + \frac{48}{52} \cdot \frac{4}{51} = \frac{12+192}{2652} = \frac{204}{2652} = \frac{1}{13}$.

Answer: A.

Q1.40 RECALL · about 15 s

Monty Hall: you pick door 1 of 3; the host, who knows where the car is, always opens another door showing a goat and offers a switch. Your best strategy and its win probability:

- (A) switch, $1/2$
- (B) switch, $2/3$
- (C) stay, $1/2$
- (D) stay, $2/3$

Solution

Fast route: First pick is right $\frac{1}{3}$ of the time; switching wins the other $\frac{2}{3}$.

Full working:

- The host's action cannot change the $\frac{1}{3}$ of your original door; the remaining $\frac{2}{3}$ moves to the one closed door.

Answer: B.

Q1.41 CALC · 1.5 to 2 min

Two fair dice are rolled. $E[\max]$ of the two values is

- (A) 4.47
- (B) 5
- (C) 3.5
- (D) 4

Solution

Fast route: $P(\max = k) = \frac{2k-1}{36}$; $E = \frac{\sum k(2k-1)}{36} = \frac{161}{36}$.

Full working:

- $P(\max \leq k) = \left(\frac{k}{6}\right)^2$, so $P(\max = k) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}$.
- $\sum_{k=1}^6 k(2k-1) = 1 + 6 + 15 + 28 + 45 + 66 = 161$. $E = \frac{161}{36} = 4.47$.
- Sanity: the max must exceed one die's mean 3.5, and cannot reach 6 on average.

Answer: A.

Q1.42 REASON · about 45 s

U_1, U_2 independent Uniform $[0, 1]$. $E[\min(U_1, U_2)]$ and $E[\max(U_1, U_2)]$ are

- (A) $1/3, 1/2$
- (B) $1/2, 1/2$
- (C) $1/4, 3/4$
- (D) $1/3, 2/3$

Solution

Fast route: Two points cut $[0, 1]$ into three pieces of equal expected length $\frac{1}{3}$.

Full working:

- $P(\max \leq x) = x^2$, density $2x$, mean $\int_0^1 2x^2 dx = \frac{2}{3}$.
- $\min + \max = U_1 + U_2$ has mean 1, so $E[\min] = \frac{1}{3}$.

Answer: D.

Q1.43 CALC · 1.5 to 2 min

A die is rolled n times. X = number of 1s, Y = number of 6s. $\text{Cov}(X, Y)$ equals

- (A) $n/36$
- (B) $-n/6$
- (C) $-n/36$
- (D) 0

Solution

Fast route: Negative (a 1 uses up a roll that could have been a 6); per roll $E[AB] - E[A]E[B] = 0 - \frac{1}{36}$.

Full working:

- $X = \sum A_i, Y = \sum B_j$ with indicators per roll.
- Same roll: $A_i B_i = 0$ (cannot be both), so $\text{Cov}(A_i, B_i) = 0 - \frac{1}{6} \cdot \frac{1}{6} = -\frac{1}{36}$.
- Different rolls are independent: covariance 0. Total $-\frac{n}{36}$.

Why the other options fail: (D) forgets that the counts compete for the same rolls. (A) has the wrong sign.

Answer: C.

Q1.44 RECALL · about 15 s

A variable has mean 50 and SD 5 with unknown distribution. The best general upper bound on $P(|X - 50| \geq 10)$ is

- (A) 0.05
- (B) 0.50
- (C) 0.25
- (D) 0.0455

Solution

Fast route: Chebyshev with $k = 2$: $\frac{1}{k^2} = \frac{1}{4}$.

Full working:

- 10 is 2 SDs. Without knowing the shape, only Chebyshev applies.

Why the other options fail: (D) is the exact value for a normal; the distribution is unknown.

Answer: C.

Q1.45 RECALL · about 15 s

Daily orders (never negative) average 10. The best general upper bound on $P(X \geq 50)$ is

- (A) 0.2
- (B) 0.5
- (C) 0.8
- (D) 0.02

Solution

Fast route: Markov: $E[X]/a = 10/50$.

Full working:

- If more than 20% of days had 50 or more orders, the mean would exceed $0.2 \times 50 = 10$.

Answer: A.

Q1.46 REASON · about 45 s

An integer is chosen uniformly from 1 to 100. The probability it is divisible by 2 or by 5 is

- (A) 0.50
- (B) 0.70
- (C) 0.10
- (D) 0.60

Solution

Fast route: $50 + 20 - 10 = 60$ numbers.

Full working:

- Multiples of 2: 50; of 5: 20; of both (of 10): 10. Count 60, probability 0.6.

Why the other options fail: (B) forgets to subtract the overlap.

Answer: D.

Q1.47 REASON · about 45 s

A shop gets a random number N of customers per day with $E[N] = 10$. Each spends an independent amount with mean Rs 200, independent of N . Expected daily revenue:

- (A) Rs 200
- (B) Rs 2,000
- (C) cannot be found without the distribution of N
- (D) Rs 210

Solution

Fast route: $E[\text{total}] = E[N] \times E[\text{spend}]$ (Wald's identity).

Full working:

- Condition on $N = n$: expected total $200n$. Average over N : $200E[N] = 2000$.

Why the other options fail: (C): only the variance would need more information.

Answer: B.

Q1.48 CALC · 1.5 to 2 min

You roll a die and win its face value in rupees. After seeing the first roll you may roll once more (and must then accept the second roll). With optimal play your expected win is

- (A) 4.25
- (B) 3.5
- (C) 4
- (D) 4.5

Solution

Fast route: A re roll is worth 3.5, so keep 4, 5, 6. $E = \frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 3.5$.

Full working:

- Keep if the first roll beats 3.5: faces 4, 5, 6 (average 5), probability $\frac{1}{2}$.
- Otherwise re roll and expect 3.5.

- $E = 0.5(5) + 0.5(3.5) = 4.25.$

Answer: A.

Q1.49 **RECALL** · about 15 s

A random permutation of $1, 2, \dots, n$ is formed. The expected number of fixed points (numbers left in their original position) is

- (A) 1
- (B) $n/2$
- (C) 0
- (D) $1/n$

Solution

Fast route: Indicators: $n \times \frac{1}{n} = 1$, for every n .

Full working:

- Each position keeps its number with probability $\frac{1}{n}$. Linearity adds n such indicators.

Answer: A.

Q1.50 **REASON** · about 45 s

A report says: “80% of candidates who received offers had a GitHub profile.” Which conclusion is valid?

- (A) 80% of GitHub users get offers
- (B) $P(\text{offer} \mid \text{GitHub}) = 0.8$
- (C) Having a GitHub profile causes offers
- (D) $P(\text{GitHub} \mid \text{offer}) = 0.8$

Solution

Fast route: The statement conditions on “received offer”.

Full working:

- Among offer holders, 80% had GitHub: $P(\text{GitHub} \mid \text{offer})$.

Why the other options fail: (B) and (A) swap the conditioning (the base rate fallacy); if 90% of all candidates have GitHub, the profile is not even informative. (C) confuses association with causation.

Answer: D.

Q1.51 **CALC** · 1.5 to 2 min

A deploy introduces a bug with probability $1/100$, independently. The probability of at least one bug in 70 deploys is closest to

- (A) 0.30
- (B) 0.99
- (C) 0.50
- (D) 0.70

Solution

Fast route: $1 - 0.99^{70} \approx 1 - e^{-0.7} \approx 1 - 0.50$.

Full working:

- $(1 - p)^n \approx e^{-np}$ for small p . $e^{-0.7} \approx 0.497$. Answer ≈ 0.50 .

Why the other options fail: (D) adds 70×0.01 , double counting runs with several bugs.

Answer: C.

References for this chapter

- [Seeing Theory: Basic Probability and Compound Probability](#).
- [Maths Is Fun: Combinations and permutations; Standard deviation](#).
- [Harvard Stat 110 lectures \(Joe Blitzstein\)](#).
- [StatQuest: Entropy](#).

Chapter 2

Bayes' Theorem

Companion resource for this section: [3Blue1Brown: Bayes theorem, the geometry of changing beliefs](#) (the clearest 15 minutes on this topic)

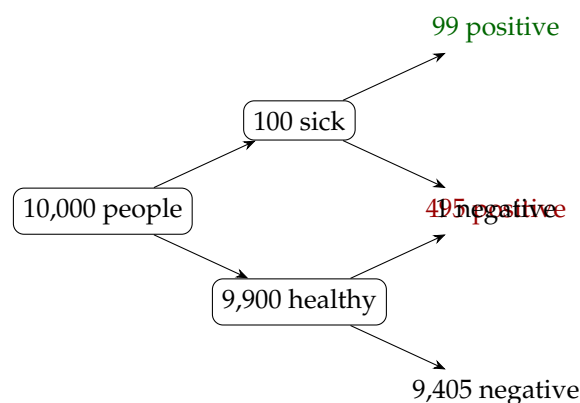
2.1 The question Bayes answers

Start here: the everyday picture

- You get a message: “Your medical test is positive.” The test is 99% accurate on sick people. Are you 99% likely to be sick?
- Almost everyone says yes. Almost everyone is wrong. If the disease is rare, most positive results come from the huge number of healthy people who get a false alarm.
- Bayes’ theorem is the tool that combines two pieces of information correctly: **how common something is to begin with** (the base rate) and **how strongly the new evidence points to it**.

2.1.1 Natural frequencies: the method that never fails

- Replace probabilities by counts of people. Imagine exactly 10,000 people.
- Suppose 1% have the disease, the test catches 99% of the sick, and it wrongly flags 5% of the healthy.
- Sick: $10,000 \times 1\% = 100$. Of these, 99 test positive.
- Healthy: 9,900. Of these, 5% = 495 test positive (false alarms).
- Total positives: $99 + 495 = 594$. Of them, only 99 are really sick.
- $P(\text{sick} \mid \text{positive}) = \frac{99}{594} = \frac{1}{6} \approx 16.7\%$.
- The positive result raised your chance from 1% to 16.7%: a big jump, but far from 99%.



Tree of natural frequencies. Positives: $99 + 495 = 594$; truly sick among them: 99.

2.2 The formula, built from what you already know

- From Chapter 1, $P(A \cap B)$ can be written two ways: $P(A | B)P(B)$ and $P(B | A)P(A)$. They are equal, so

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

- Names for each piece (worth learning, they are used everywhere in ML):
 - $P(A)$: the **prior**, what you believed before seeing the evidence.
 - $P(B | A)$: the **likelihood**, how probable the evidence is if A is true.
 - $P(B)$: the **evidence** (or marginal), how probable the evidence is overall.
 - $P(A | B)$: the **posterior**, your updated belief.
- The denominator comes from the law of total probability. If the hypotheses $A_1, \dots, A_k$ are **mutually exclusive** (no overlap) and **collectively exhaustive** (one of them must be true), then

$$P(B) = \sum_j P(B | A_j)P(A_j), \quad P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_j P(B | A_j)P(A_j)}.$$

- The assumptions, stated exactly** (a reported InfoEdge question): (1) the hypotheses form a partition, i.e. they are mutually exclusive and collectively exhaustive (MECE); (2) $P(B) > 0$. Without MECE, the sum in the denominator either double counts or misses probability. Independence is *not* an assumption of Bayes' theorem.
- Medical example in formula form: $\frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} = \frac{1}{6}$. Same as the counts.

Try it now (answer right below, cover it first)

Machines A (60% of output, 2% defective) and B (40%, 5% defective). An item is defective. $P(\text{made by B})$?

Answer: $\frac{0.4 \times 0.05}{0.6 \times 0.02 + 0.4 \times 0.05} = \frac{0.020}{0.032} = 0.625$. B makes fewer items but most of the defects.

2.3 The odds form: Bayes in five seconds

- Odds** of an event $= \frac{P}{1-P}$. Probability 0.2 is odds $\frac{0.2}{0.8} = \frac{1}{4}$, written 1 : 4. Back from odds o to probability: $P = \frac{o}{1+o}$.
- Likelihood ratio** (LR, or Bayes factor) of the evidence $= \frac{P(\text{evidence}|A)}{P(\text{evidence}|\text{not } A)}$: how many times more likely the evidence is if A is true.
- Bayes becomes one multiplication:

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}$$

- Example: prior odds of fraud 1 : 4, a signal with LR 6. Posterior odds $= \frac{1}{4} \times 6 = 1.5$, probability $\frac{1.5}{2.5} = 0.6$.
- Independent pieces of evidence simply multiply their LR. Medical test: LR of a positive $= \frac{0.99}{0.05} = 19.8$. Prior odds $\frac{1}{99}$. After one positive: odds 0.2 ($P = \frac{1}{6}$). After a second independent positive: odds $0.2 \times 19.8 = 3.96$, $P = \frac{3.96}{4.96} \approx 0.80$.

2.4 Vocabulary of diagnostic tests and classifiers

	Truly positive (sick, fraud)	Truly negative
Test says positive	True Positive (TP)	False Positive (FP)
Test says negative	False Negative (FN)	True Negative (TN)

- **Sensitivity** = recall = true positive rate (TPR) = $\frac{TP}{TP+FN} = P(+ | \text{sick})$: of the sick, how many do we catch?
- **Specificity** = true negative rate = $\frac{TN}{TN+FP} = P(- | \text{healthy})$. **False positive rate** = $1 - \text{specificity}$.
- **Precision** = positive predictive value (PPV) = $\frac{TP}{TP+FP} = P(\text{sick} | +)$: of those flagged, how many are real? This is the Bayes posterior.
- **Negative predictive value** (NPV) = $P(\text{healthy} | -)$.
- Precision depends on prevalence; sensitivity and specificity do not. The same test has PPV $\frac{1}{6}$ at 1% prevalence but about 0.69 at 10% prevalence.
- A **false negative** means the case is truly positive but the test says negative (a sick person told they are healthy). In hypothesis testing this is a Type II error (Chapter 8).

The base rate fallacy

- Mistake: treating $P(+ | \text{sick})$ as if it were $P(\text{sick} | +)$, or ignoring how rare the condition is.
- Symptom in the exam: an option that simply repeats the test's accuracy (0.99 or 0.95) as the answer to "probability of disease given positive". It is almost always wrong.

2.5 Worked classics you should recognise instantly

- **Two urns**: urn 1 has 3 red, 2 blue; urn 2 has 1 red, 4 blue. Pick an urn at random and draw red. Likelihoods $\frac{3}{5}$ and $\frac{1}{5}$, equal priors. $P(\text{urn 1} | \text{red}) = \frac{0.6}{0.6+0.2} = 0.75$.
- **Coin in a bag**: one fair coin, one double headed coin. Pick one, toss 3 times, all heads. Likelihoods $\frac{1}{8}$ and 1. $P(\text{double headed}) = \frac{1}{1+\frac{1}{8}} = \frac{8}{9}$.
- **Cab problem**: 15% of cabs are blue; a witness who is right 80% of the time says "blue". $P(\text{blue} | \text{says blue}) = \frac{0.15 \times 0.8}{0.15 \times 0.8 + 0.85 \times 0.2} = \frac{0.12}{0.29} \approx 0.41$. Most people say 0.8.
- **Monty Hall by Bayes**: you pick door 1, host opens door 3. $P(\text{host opens 3} | \text{car at 1}) = \frac{1}{2}$, | car at 2 = 1, | car at 3 = 0. Posterior for door 2 = $\frac{1 \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{2}{3}$.

2.6 Naive Bayes: Bayes' theorem as a classifier

Start here: the everyday picture

- A doctor sees a patient with low appetite, low weight and low blood pressure, and asks: anemia or not? She has 7 past records. Naive Bayes turns those records into probabilities and applies Bayes' theorem to decide.
- Goal: $P(\text{class} | x_1, \dots, x_d) \propto P(\text{class}) P(x_1, \dots, x_d | \text{class})$. The symbol $\propto$ means "proportional to": we can skip the denominator because it is the same for every class; we only compare classes.
- Problem: estimating $P(x_1, \dots, x_d | \text{class})$ for every combination needs enormous data.

- The **naive assumption**: features are **independent given the class** (conditionally independent). Then the joint likelihood becomes a product:

$$P(\text{class} \mid x) \propto P(\text{class}) \prod_{j=1}^d P(x_j \mid \text{class}).$$

- It is called “naive” because the assumption is usually false (low weight and low appetite are related). It still classifies well, because we only need the ranking of classes to be right, not the exact probabilities.

2.6.1 The anemia table

Sample	Appetite	Weight	BP	Class
1	Low	Normal	Low	No anemia
2	Low	Low	Low	Anemia
3	Normal	Low	Low	Anemia
4	Low	Low	Normal	No anemia
5	Normal	Low	Normal	Anemia
6	Normal	Normal	Low	Anemia
7	Normal	Normal	Normal	No anemia

- Priors: Anemia in rows 2, 3, 5, 6: $P(A) = \frac{4}{7}$. No anemia in rows 1, 4, 7: $P(N) = \frac{3}{7}$.
- Each conditional probability is a count inside the class. Example: among the 4 anemia rows, Appetite = Normal in rows 3, 5, 6, so $P(\text{App} = \text{Normal} \mid A) = \frac{3}{4}$.
- **Worked example**: new patient (Appetite Normal, Weight Low, BP Low).
 - Anemia score: $\frac{4}{7} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{4}{7} \times \frac{27}{64} = 0.241$.
 - No anemia score: Appetite Normal in class N: row 7 only, $\frac{1}{3}$; Weight Low: row 4, $\frac{1}{3}$; BP Low: row 1, $\frac{1}{3}$. Score $\frac{3}{7} \times \frac{1}{27} = 0.0159$.
 - Normalise: $P(A \mid x) = \frac{0.241}{0.241 + 0.0159} \approx 0.94$. Predict anemia.
- **Zero counts**: if a feature value never appears in a class, its conditional probability is 0 and wipes out the whole product. Fix: **Laplace (add one) smoothing**: $P(x_j = v \mid c) = \frac{\text{count}+1}{n_c+K}$, where K is the number of possible values of that feature.
- **Underflow**: multiplying hundreds of small probabilities (one per word in a resume) rounds to 0 on a computer. Fix: add log probabilities instead of multiplying probabilities.
- Uses: spam filtering, classifying resumes or job posts into categories, sentiment. Fast, needs little data, a strong baseline for text.

2.7 Priors, MLE and MAP

- **Maximum likelihood (MLE)**: choose the parameter that makes the observed data most probable. 7 clicks in 10 impressions: $\hat{p} = 0.7$.
- **Maximum a posteriori (MAP)**: choose the parameter with the highest posterior $\propto$ likelihood $\times$ prior. The prior pulls the estimate towards what you believed before.
- With a flat (uniform) prior, MAP = MLE: the prior adds nothing.
- With lots of data, the likelihood dominates and MAP approaches MLE; with little data, the prior matters most.
- Priors are regularisers: a Gaussian prior on regression weights gives L2 (ridge) regularisation; a Laplace prior gives L1 (lasso).

- **Generative vs discriminative:** generative models learn $P(x | y)$ and $P(y)$ and classify with Bayes (Naive Bayes, Gaussian discriminant analysis). Discriminative models learn $P(y | x)$ or a boundary directly (logistic regression, SVM, trees).
- **Bayes decision boundary** for two Gaussian classes with equal variance σ^2 and equal priors: the midpoint of the means. Unequal priors shift the boundary away from the more common class, towards the rarer one's mean, by $\frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(1)}{P(2)}$.

Where this shows up in ML and at a company like Naukri

- Precision of a fraud or fake profile detector is a Bayes posterior: with rare fraud, even a good model produces many false alarms. That is why thresholds are tuned on precision and recall, not accuracy.
- Spam and resume category filters are often Naive Bayes baselines.
- Bayesian A/B testing reports "probability that B beats A" directly.
- CTR smoothing with a Beta prior is MAP estimation: a job with 1 click in 2 views should not be ranked above one with 300 in 1000.

Formula sheet

$$P(A | B) = \frac{P(B|A)P(A)}{P(B)} \quad P(B) = \sum P(B | A_j)P(A_j) \text{ (needs MECE)} \quad \text{posterior odds} = \text{prior odds} \times \text{LR}$$

$$\text{odds} = \frac{p}{1-p}, \quad p = \frac{o}{1+o} \quad \text{PPV} = \frac{\text{sens} \cdot \pi}{\text{sens} \cdot \pi + (1 - \text{spec})(1 - \pi)} \quad \text{NB: } P(c | x) \propto P(c) \prod P(x_j | c)$$

$$\text{Laplace } \frac{n_{v,c} + 1}{n_c + K} \quad \text{uniform prior: MAP} = \text{MLE}.$$

15 second recall card: memorise every line

- Bayes needs the hypotheses to be mutually exclusive and collectively exhaustive, and $P(B) > 0$. It does not need independence.
- Naive Bayes assumes features are independent *given the class*. Fix zeros with Laplace smoothing; fix underflow with logs.
- 1% prevalence, 99% sensitivity, 95% specificity: $P(\text{sick} | +) = \frac{1}{6}$. A second positive: about 0.80.
- Precision rises with prevalence; sensitivity and specificity do not change with prevalence.
- Odds form: prior odds $\times$ LR. Double headed coin after HHH: $\frac{8}{9}$. Cab problem: about 0.41.
- $P(A | B) = P(B | A)$ exactly when $P(A) = P(B)$.
- Uniform prior makes MAP equal to MLE. Gaussian prior is L2, Laplace prior is L1.
- Generative: Naive Bayes, LDA/GDA. Discriminative: logistic regression, SVM, trees.

2.8 Question bank: Bayes' theorem

Q2.1 RECALL · about 15 s

★ **PYQ (INT and OA: formula and assumptions of Bayes' theorem)** In $P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_j P(B | A_j)P(A_j)}$,

which assumption is required for the denominator to equal $P(B)$?

- (A) The events A_j are independent of each other
- (B) The events A_j are mutually exclusive and collectively exhaustive
- (C) The events A_j are equally likely
- (D) B is independent of every A_j

Solution

Fast route: The denominator is the law of total probability, which needs a partition (MECE).

Full working:

- Mutually exclusive: no double counting. Exhaustive: nothing missed. Then $\sum P(B \cap A_j) = P(B)$.

Why the other options fail: (A) Mutually exclusive events with positive probability are in fact dependent. (D) If B were independent of every A_j , the posterior would equal the prior and the evidence would be useless. (C) Priors can be anything.

Answer: B.

Q2.2 **RECALL** · about 15 s

★ **PYQ (INT: real life use of Bayes, Naive Bayes)** The Naive Bayes classifier assumes that

- (A) features are independent of each other unconditionally
- (B) features are conditionally independent given the class
- (C) features are independent of the class
- (D) classes are equally likely

Solution

Fast route: “Naive” = conditional independence given the class.

Full working:

- It lets us write $P(x_1, \dots, x_d | c) = \prod_j P(x_j | c)$.

Why the other options fail: (A) is a different, stronger claim. (C) would make features useless. (D) Priors are estimated from class frequencies.

Answer: B.

Q2.3 **CALC** · 1.5 to 2 min

Disease prevalence is 1%. A test has sensitivity 99% and specificity 95%. A person tests positive. The probability they have the disease is closest to

- (A) 0.50
- (B) 0.17
- (C) 0.95
- (D) 0.99

Solution

Fast route: Per 10,000: 99 true positives vs 495 false positives. $\frac{99}{594} = \frac{1}{6}$.

Full working:

- Numerator $0.99 \times 0.01 = 0.0099$. False positives $0.05 \times 0.99 = 0.0495$.
- Posterior $= \frac{0.0099}{0.0594} = 0.1667$.

Why the other options fail: (D) and (C) are the base rate fallacy (repeating the test’s accuracy).

Answer: B.

Q2.4 **CALC** · 1.5 to 2 min

The same person takes a second, independent test with the same accuracy and is positive again. The new probability of disease is closest to

- (A) 0.33
- (B) 0.80
- (C) 0.50
- (D) 0.97

Solution

Fast route: Odds after test 1 = $\frac{1}{5}$; $LR = \frac{0.99}{0.05} = 19.8$; new odds 3.96; $P = \frac{3.96}{4.96}$.

Full working:

- After one positive, $P = \frac{1}{6}$, odds $\frac{1}{5/6} = 0.2$.
- Multiply by LR 19.8: odds 3.96. Probability ≈ 0.798 .

Answer: B.

Q2.5 REASON · about 45 s

Same test (1% prevalence, sensitivity 0.99, specificity 0.95). If the result is **negative**, $P(\text{disease} \mid -)$ is closest to

- (A) 0.01
- (B) 0.0001
- (C) 0.001
- (D) 0.05

Solution

Fast route: Sick and negative: $0.01 \times 0.01 = 0.0001$; healthy and negative: $0.99 \times 0.95 \approx 0.94$. Ratio about 0.0001.

Full working:

- $\frac{0.0001}{0.0001 + 0.9405} = 0.000106$. A negative is very reassuring (NPV about 99.99%).

Answer: B.

Q2.6 REASON · about 45 s

Urn 1 has 3 red and 2 blue balls; urn 2 has 1 red and 4 blue. An urn is picked at random and a red ball is drawn. $P(\text{urn 1} \mid \text{red})$ is

- (A) 0.75
- (B) 0.80
- (C) 0.50
- (D) 0.60

Solution

Fast route: Equal priors, so compare likelihoods $\frac{3}{5}$ vs $\frac{1}{5}$: 3 : 1.

Full working:

- $\frac{0.5 \times 0.6}{0.5 \times 0.6 + 0.5 \times 0.2} = \frac{0.3}{0.4} = 0.75$.

Why the other options fail: (D) is $P(\text{red} \mid \text{urn 1})$, the reversed conditional.

Answer: A.

Q2.7 CALC · 1.5 to 2 min

Factories A, B, C produce 50%, 30%, 20% of bulbs, with defect rates 1%, 2%, 3%. A bulb is defective. The probability it came from C is closest to

- (A) 0.35
- (B) 0.30
- (C) 0.20
- (D) 0.50

Solution

Fast route: Joint rates: A 0.005, B 0.006, C 0.006; C's share $\frac{6}{17}$.

Full working:

- $P(\text{def}) = 0.005 + 0.006 + 0.006 = 0.017$. $P(C \mid \text{def}) = \frac{0.006}{0.017} = 0.353$.

Why the other options fail: (C) is the prior for C. (B) confuses B's share.

Answer: A.

Q2.8 REASON · about 45 s

A bag has one fair coin and one double headed coin. A coin is picked at random and tossed 3 times, giving HHH. The probability it is the double headed coin is

- (A) $7/8$
- (B) $1/2$
- (C) $8/9$
- (D) $2/3$

Solution

Fast route: LR = $\frac{1}{1/8} = 8$; prior odds 1; posterior odds 8; $P = \frac{8}{9}$.

Full working:

- $\frac{0.5 \times 1}{0.5 \times 1 + 0.5 \times \frac{1}{8}} = \frac{1}{1.125} = \frac{8}{9}$.

Answer: C.

Q2.9 CALC · 1.5 to 2 min

40% of email is spam. The word "free" appears in 30% of spam and 2% of non spam. Given that an email contains "free", the probability it is spam is closest to

- (A) 0.30
- (B) 0.98
- (C) 0.91
- (D) 0.60

Solution

Fast route: Spam and free 0.12; ham and free 0.012. Ratio 10 : 1, about 0.91.

Full working:

- $\frac{0.4 \times 0.3}{0.12 + 0.6 \times 0.02} = \frac{0.12}{0.132} = 0.909$.

Answer: C.

Q2.10 CALC · 1.5 to 2 min

In a city 85% of cabs are green and 15% blue. A witness who is right 80% of the time says the cab in an accident was blue. The probability it really was blue is closest to

- (A) 0.59
- (B) 0.15
- (C) 0.41
- (D) 0.80

Solution

Fast route: Says blue and blue: 0.12. Says blue and green: 0.17. $\frac{12}{29}$.

Full working:

- $\frac{0.15 \times 0.8}{0.15 \times 0.8 + 0.85 \times 0.2} = \frac{0.12}{0.29} = 0.414$.

Why the other options fail: (D) is the witness accuracy: base rate fallacy.

Answer: C.

Q2.11 RECALL · about 15 s

Monty Hall: you pick door 1; the host (who never reveals the car, and chooses randomly between two goat doors) opens door 3. $P(\text{car behind door 2} \mid \text{host opens door 3})$ is

- (A) $1/3$
- (B) $1/2$
- (C) 1
- (D) $2/3$

Solution

Fast route: Switch wins $\frac{2}{3}$.

Full working:

- Likelihoods of “opens 3”: car at 1: $\frac{1}{2}$; car at 2: 1; car at 3: 0. Posterior for 2 = $\frac{1}{\frac{1}{2}+1} = \frac{2}{3}$.

Answer: D.

Q2.12 REASON · about 45 s

Prior odds of fraud are 1 : 4. A signal has likelihood ratio 6 in favour of fraud. The posterior probability of fraud is

- (A) 0.6
- (B) 0.86
- (C) 0.75
- (D) 0.2

Solution

Fast route: Odds $\frac{1}{4} \times 6 = 1.5$; $P = \frac{1.5}{2.5}$.

Full working:

- $P = \frac{o}{1+o} = 0.6$.

Why the other options fail: (B) converts 6 to probability ignoring the prior.

Answer: A.

Q2.13 RECALL · about 15 s

Which best describes the base rate fallacy?

- (A) Confusing correlation with causation
- (B) Using a one sided test instead of a two sided test
- (C) Ignoring the prior prevalence and treating $P(+ | D)$ as $P(D | +)$
- (D) Using too small a sample

Solution

Fast route: Base rate = prior prevalence.

Full working:

- The fallacy is to judge $P(D | +)$ by the test accuracy alone.

Answer: C.

Q2.14 **CALC · 1.5 to 2 min**

Using the anemia table (Section 2.6) and Naive Bayes without smoothing, a patient has Appetite Low, Weight Low, BP Low. The prediction and $P(\text{Anemia} | x)$ are

- (A) Anemia, about 0.94
- (B) No anemia, about 0.60
- (C) Anemia, about 0.72
- (D) No anemia, about 0.28

Solution

Fast route: Anemia $\frac{4}{7} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = 0.080$; No $\frac{3}{7} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = 0.032$; share ≈ 0.72 .

Full working:

- Anemia rows 2, 3, 5, 6: Appetite Low only in row 2 ($\frac{1}{4}$); Weight Low in 2, 3, 5 ($\frac{3}{4}$); BP Low in 2, 3, 6 ($\frac{3}{4}$). Score $\frac{4}{7} \times \frac{9}{64} = 0.0804$.
- No anemia rows 1, 4, 7: Appetite Low in 1, 4 ($\frac{2}{3}$); Weight Low in 4 ($\frac{1}{3}$); BP Low in 1 ($\frac{1}{3}$). Score $\frac{3}{7} \times \frac{2}{27} = 0.0317$.
- $P(A | x) = \frac{0.0804}{0.1121} = 0.717$: predict Anemia.

Why the other options fail: (A) is the teaching example (Appetite Normal, Weight Low, BP Low), a different patient.

Answer: C.

Q2.15 **CALC · 1.5 to 2 min**

Same table and method. A patient has Appetite Normal, Weight Normal, BP Normal. The prediction and $P(\text{Anemia} | x)$ are

- (A) No anemia, about 0.10
- (B) Anemia, about 0.55
- (C) Anemia, about 0.70
- (D) No anemia, about 0.30

Solution

Fast route: Anemia $\frac{4}{7} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = 0.027$; No $\frac{3}{7} \cdot \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} = 0.063$.

Full working:

- Anemia: Appetite Normal in 3, 5, 6 ($\frac{3}{4}$); Weight Normal in 6 ($\frac{1}{4}$); BP Normal in 5 ($\frac{1}{4}$). Score $\frac{4}{7} \times \frac{3}{64} = 0.0268$.
- No anemia: Appetite Normal in 7 ($\frac{1}{3}$); Weight Normal in 1, 7 ($\frac{2}{3}$); BP Normal in 4, 7 ($\frac{2}{3}$).

$$\text{Score } \frac{3}{7} \times \frac{4}{27} = 0.0635.$$

- $P(A | x) = \frac{0.0268}{0.0903} = 0.297$: predict No anemia.

Answer: D.

Q2.16 REASON · about 45 s

With Laplace (add one) smoothing, the estimate of $P(\text{Appetite} = \text{Low} \mid \text{Anemia})$ in the same table (Appetite has 2 possible values) is

- (A) $1/6$
- (B) $2/5$
- (C) $1/3$
- (D) $1/4$

Solution

Fast route: $\frac{1+1}{4+2}$.

Full working:

- Raw count 1 of 4 anemia rows. Add 1 to the count and $K = 2$ to the denominator: $\frac{2}{6} = \frac{1}{3}$.

Why the other options fail: (D) is the unsmoothed value. (B) adds 1 to the denominator instead of $K = 2$.

Answer: C.

Q2.17 CALC · 1.5 to 2 min

A coin is either fair or biased with $P(H) = 0.8$, each equally likely a priori. You observe H, H, T. The posterior probability that it is biased is closest to

- (A) 0.80
- (B) 0.64
- (C) 0.25
- (D) 0.51

Solution

Fast route: Likelihoods: fair 0.125, biased $0.8 \cdot 0.8 \cdot 0.2 = 0.128$. Nearly equal, so about one half.

Full working:

- Posterior = $\frac{0.128}{0.128+0.125} = 0.506$. The tail almost cancels the evidence of the two heads.

Answer: D.

Q2.18 REASON · about 45 s

It rains on 10% of days. A forecaster predicts rain on 90% of rainy days and on 20% of dry days. Given a rain forecast, the probability of rain is

- (A) 0.09
- (B) 0.90
- (C) 0.50
- (D) $1/3$

Solution

Fast route: Rain and forecast 0.09; dry and forecast 0.18. $\frac{0.09}{0.27}$.

Full working:

$$\bullet P = \frac{0.1 \times 0.9}{0.1 \times 0.9 + 0.9 \times 0.2} = \frac{1}{3}.$$

Answer: D.

Q2.19 REASON · about 45 s

On a 4 option MCQ a student knows the answer with probability 0.6 and otherwise guesses uniformly. Given the answer is correct, the probability the student knew it is

- (A) 0.86
- (B) 0.75
- (C) 0.90
- (D) 0.60

Solution

Fast route: Correct by knowing 0.6; by guessing $0.4 \times \frac{1}{4} = 0.1$. $\frac{0.6}{0.7}$.

Full working:

$$\bullet \frac{6}{7} \approx 0.857.$$

Answer: A.

Q2.20 RECALL · about 15 s

Which statement is correct?

- (A) With a uniform prior, the MAP estimate equals the MLE
- (B) MAP estimation ignores the likelihood
- (C) As data grows, the prior dominates the posterior more and more
- (D) A Laplace prior on regression weights gives L2 regularisation

Solution

Fast route: Flat prior adds a constant to the log posterior, so the maximiser is unchanged.

Full working:

- MAP maximises log likelihood + log prior; a uniform prior's log is constant.

Why the other options fail: (D) Laplace gives L1; Gaussian gives L2. (B) MAP uses likelihood times prior. (C) The opposite: data eventually dominates.

Answer: A.

Q2.21 REASON · about 45 s

A fraud model has TPR 0.9 and FPR 0.1. Fraud prevalence is 5%. Its precision is closest to

- (A) 0.32
- (B) 0.50
- (C) 0.90
- (D) 0.05

Solution

Fast route: TP $0.9 \times 0.05 = 0.045$; FP $0.1 \times 0.95 = 0.095$. $\frac{45}{140}$.

Full working:

- Precision = $\frac{0.045}{0.045+0.095} = 0.321$.

Why the other options fail: (C) is recall, not precision.

Answer: A.

Q2.22 REASON · about 45 s

The test in Q2.3 (sensitivity 0.99, specificity 0.95) is used where prevalence is 10%. The positive predictive value becomes closest to

- (A) 0.50
- (B) 0.95
- (C) 0.17
- (D) 0.69

Solution

Fast route: TP 0.099; FP 0.045. $\frac{99}{144}$.

Full working:

- $\frac{0.099}{0.099+0.05 \times 0.9} = \frac{0.099}{0.144} = 0.6875$. Same test, much higher PPV: prevalence drives precision.

Answer: D.

Q2.23 RECALL · about 15 s

For events with positive probability, $P(A | B) = P(B | A)$ holds exactly when

- (A) A and B are mutually exclusive
- (B) $P(A) = P(B)$
- (C) always
- (D) A and B are independent

Solution

Fast route: $\frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{P(A)}$ when $P(A \cap B) > 0$ forces $P(A) = P(B)$.

Full working:

- When $P(A \cap B) > 0$, cancel it from both sides: equality holds if and only if $P(A) = P(B)$.

Why the other options fail: (D) Independence gives $P(A | B) = P(A)$ and $P(B | A) = P(B)$, equal only if $P(A) = P(B)$. (C) is the famous mistake.

Answer: B.

Q2.24 CALC · 1.5 to 2 min

Heights in class 1 are $N(160, 10^2)$ and in class 2 $N(175, 10^2)$. With priors $P(1) = 0.8$ and $P(2) = 0.2$, the Bayes decision boundary (classify as class 2 above it) is closest to

- (A) 158.3
- (B) 170.0
- (C) 167.5
- (D) 176.7

Solution

Fast route: Equal priors give the midpoint 167.5. Shift towards class 2 by $\frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{0.8}{0.2} = \frac{100}{15} \times 1.386 = 9.24$.

Full working:

- Choose class 2 when $0.2f_2(x) > 0.8f_1(x)$. Taking logs, the squared terms give $\frac{(x-160)^2 - (x-175)^2}{200} > \ln 4$.
- $(x-160)^2 - (x-175)^2 = 30x - 5025$. So $30x > 5025 + 200 \times 1.386 = 5302.3$, $x > 176.7$.
- Intuition: class 1 is 4 times more common, so you need stronger evidence (a taller person) to say class 2.

Why the other options fail: (C) ignores the priors. (A) shifts in the wrong direction.

Answer: D.

Q2.25 **RECALL** · about 15 s

Which model is generative (models $P(x | y)$ and $P(y)$, then classifies with Bayes' rule)?

- (A) Gaussian Naive Bayes
- (B) Logistic regression
- (C) Random forest
- (D) Linear SVM

Solution

Fast route: Naive Bayes is the textbook generative classifier.

Full working:

- Logistic regression and SVM model the boundary or $P(y | x)$ directly; trees partition feature space.

Answer: A.

References for this chapter

- [3Blue1Brown: Bayes theorem; The medical test paradox.](#)
- [StatQuest: Naive Bayes.](#)
- [Seeing Theory: Bayesian inference.](#)
- [Wikipedia: Sensitivity and specificity.](#)

Chapter 3

Probability Distributions

Companion resource for this section: [Seeing Theory, Probability Distributions](#) (move the sliders and watch every distribution change shape)

3.1 What a distribution is, and why we give them names

Start here: the everyday picture

- Ask 1,000 job posts how many applications they got in their first hour, and draw a bar chart: how many posts got 0, how many got 1, and so on. That bar chart is a **distribution**: the full list of possible values and how likely each one is.
 - The same shapes keep showing up in completely different places. Counts of arrivals always look one way; waiting times another; averages another. So we name those shapes and learn their formulas once.
 - The single most useful skill in this chapter: hear a **story**, name the **distribution**. Once you name it, the mean, variance and probabilities come for free.
- **Discrete** distributions (countable values) are described by a PMF, $P(X = k)$. **Continuous** ones by a density $f(x)$, where probability is area (Chapter 1).
 - **Parameters** are the knobs that set the exact shape: a coin's p , a rate λ , a mean μ and SD σ .
 - InfoEdge flavoured stories: did this user apply (yes or no); how many of 20 recommended jobs were clicked; applications per hour; minutes until the next application; salaries; exam scores.

3.2 Discrete distributions

3.2.1 Bernoulli(p): one yes or no

- Story: a single trial that is a success (value 1) with probability p or a failure (value 0) with probability $1 - p$. Did the user click?
- Mean $E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$. Since $X^2 = X$ for 0/1 values, $E[X^2] = p$, so $\text{Var}(X) = p - p^2 = p(1 - p)$.
- $p(1 - p)$ is largest at $p = 0.5$ (value 0.25): a fair coin is the most unpredictable. That is why "assume $p = 0.5$ " gives the safest sample size in surveys (Chapter 4).

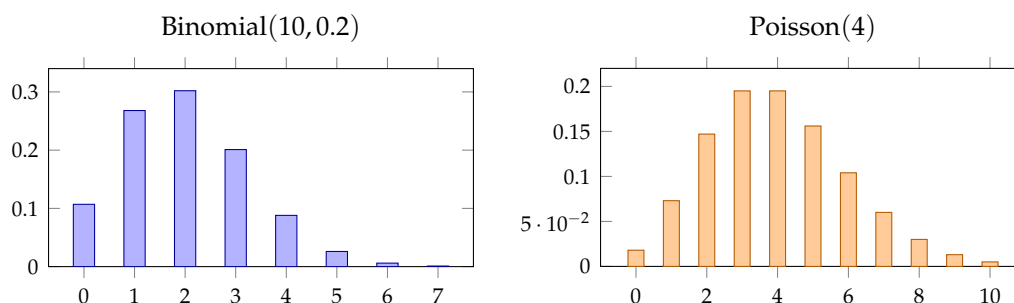
3.2.2 Binomial(n, p): counting successes in n tries

- Story: n **independent** Bernoulli trials, each with the **same** p ; X = number of successes. Clicks out of 20 recommendations shown.
- Building the formula from nothing: take $n = 4, k = 2$. One particular sequence with 2 successes, like SSFF, has probability $p \cdot p \cdot (1 - p)(1 - p) = p^2(1 - p)^2$. How many such sequences? Choose which 2 of the 4 positions are successes: $\binom{4}{2} = 6$. So $P(X = 2) = 6p^2(1 - p)^2$.

In general

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

- Mean and variance by adding n Bernoullis: $E[X] = np$, $\text{Var}(X) = np(1-p)$.
- Example: 10 users, each applies with probability 0.2. $P(\text{exactly } 2) = 45 \times 0.04 \times 0.8^8 = 45 \times 0.04 \times 0.168 = 0.302$. Mean 2, variance 1.6.



3.2.3 Geometric(p): waiting for the first success (a reported InfoEdge derivation)

- Story: repeat independent trials until the first success.
- Two conventions (always check which one a question uses):
 - Y = number of **failures before** the first success: $P(Y = n) = (1-p)^n p$, $n = 0, 1, 2, \dots$; mean $\frac{1-p}{p}$.
 - X = number of **trials including** the success: $P(X = k) = (1-p)^{k-1} p$, $k = 1, 2, \dots$; mean $\frac{1}{p}$.
- The derivation in one sentence: “ n failures in a row have probability $(1-p)^n$ by independence, then one success with probability p ; multiply.”
- Mean by first step thinking (Chapter 1): $E[Y] = (1-p)(1 + E[Y])$ gives $E[Y] = \frac{1-p}{p}$.
- Variance (both conventions): $\frac{1-p}{p^2}$.
- **Memoryless**: having failed 10 times does not change the distribution of how many more failures you will see. The process “forgets”.
- Example: each call to a candidate is accepted with probability 0.25. Probability of exactly 3 declines before the first acceptance: $0.75^3 \times 0.25 = 0.105$. Expected declines: $\frac{0.75}{0.25} = 3$.

3.2.4 Negative Binomial(r, p): waiting for the r -th success

- $P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$: the k -th trial must be the r -th success, and the other $r-1$ successes sit somewhere in the first $k-1$ trials.
- Example: probability the 3rd head comes exactly on toss 5: $\binom{4}{2} \left(\frac{1}{2}\right)^5 = \frac{6}{32} = 0.1875$.
- Mean $\frac{r}{p}$ (sum of r geometric waits). Also used for **over dispersed counts**, where the variance is bigger than the mean.

3.2.5 Hypergeometric: drawing without replacement

- Story: N items of which K are “special”. Draw n **without replacement**. $P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$.
- Example: 10 laptops, 3 defective, inspect 4. $P(\text{exactly 1 defective}) = \frac{\binom{3}{1} \binom{7}{3}}{\binom{10}{4}} = \frac{3 \times 35}{210} = 0.5$.

- It is the “without replacement” version of the Binomial; when N is huge compared with n the two are nearly identical.

3.2.6 Poisson(λ): counts of events at a steady rate (a reported InfoEdge question)

- Story: the **number of events in a fixed interval** of time or space, when events occur independently at a constant average rate λ . Customers per hour, applications per minute, server errors per day, typos per page.
- Where the formula comes from: chop one hour into n tiny slots; in each slot an arrival happens with probability $\frac{\lambda}{n}$. The count is Binomial($n, \frac{\lambda}{n}$). Let n grow: the binomial formula becomes

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- Signature property: **mean = variance** = λ . If your data's variance is much larger than its mean, it is not Poisson (use Negative Binomial).
- Scaling with time: 4 customers per hour means $\lambda = 2$ for 30 minutes. $P(\text{nobody in 30 min}) = e^{-2} = 0.135$.
- Adding: independent Poissons add their rates. Errors at 3 per day on server A and 2 per day on B: total Poisson(5); error free day $e^{-5} = 0.0067$.
- Approximating the Binomial: large n , small p , use $\lambda = np$. 1000 trials with $p = 0.002$: $\lambda = 2$, $P(X \leq 1) = e^{-2}(1 + 2) = 0.406$.
- Example: $\lambda = 4$ per hour, $P(\text{exactly } 2) = \frac{e^{-4} 4^2}{2!} = 8e^{-4} = 0.147$.

3.3 Continuous distributions

3.3.1 Uniform(a, b)

- Every value in $[a, b]$ equally likely: a flat density of height $\frac{1}{b-a}$.
- Probability of a sub interval = its length over the total length. $U(0, 10)$: $P(3 < X < 7) = \frac{4}{10}$.
- Mean $\frac{a+b}{2}$ (the middle), variance $\frac{(b-a)^2}{12}$. For $U(2, 8)$: mean 5, variance $\frac{36}{12} = 3$.
- Discrete uniform on $1, \dots, n$ (a die): variance $\frac{n^2-1}{12}$, which gives $\frac{35}{12}$ for a die.

3.3.2 Exponential(λ): waiting times

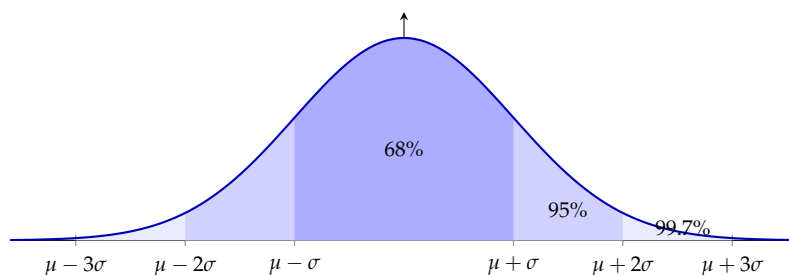
- Story: **time until the next event** in a Poisson process with rate λ . If applications arrive at 6 per hour, the wait for the next one is Exponential with mean $\frac{1}{6}$ hour = 10 minutes.
- The link, derived: $P(\text{wait} > t) = P(\text{no events in time } t) = P(\text{Poisson}(\lambda t) = 0) = e^{-\lambda t}$.
- Density $\lambda e^{-\lambda x}$; mean $\frac{1}{\lambda}$; variance $\frac{1}{\lambda^2}$; median $\frac{\ln 2}{\lambda}$ (the median is below the mean because of the long right tail). Mean 10 minutes $\Rightarrow$ median 6.93 minutes.
- **Memoryless**: $P(X > s + t \mid X > s) = P(X > t)$. A bulb with exponential life that has worked 500 hours is “as good as new”. It is the only continuous memoryless distribution (the geometric is the discrete one).
- **Competing exponentials**: with independent rates λ_1, λ_2 , the first event happens at rate $\lambda_1 + \lambda_2$ (mean $\frac{1}{\lambda_1 + \lambda_2}$), and it is event 2 with probability $\frac{\lambda_2}{\lambda_1 + \lambda_2}$.

3.3.3 Normal (Gaussian) $N(\mu, \sigma^2)$

- Story: the bell curve. It appears whenever a quantity is the **sum or average of many small independent effects** (the Central Limit Theorem, Chapter 5): measurement errors, heights,

test scores, averages of anything.

- Two parameters: the centre μ and the spread σ . Notation: the second number is the **variance**. $N(100, 15^2)$ has SD 15.
- Symmetric about μ : mean = median = mode.
- **The empirical rule**: about 68% of values lie within 1σ of the mean, 95% within 2σ , 99.7% within 3σ .



- **Standardising**: any normal becomes the standard normal $Z \sim N(0, 1)$ via $z = \frac{x - \mu}{\sigma}$: “how many SDs above the mean”. Then read $\Phi(z) = P(Z \leq z)$ from the table on the front pages.
- Example: $\text{IQ} \sim N(100, 15^2)$. $P(\text{IQ} > 130)$: $z = 2$, $P = 1 - 0.9772 = 0.0228$. Top 5% cut off: $z = 1.645$, score $100 + 1.645 \times 15 = 124.7$.
- **Linear change**: if $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$. $X \sim N(10, 4)$ gives $3X - 5 \sim N(25, 36)$.
- **Sums** (a reported InfoEdge OA topic): independent normals add to a normal; means add and **variances add**: $X \sim N(2, 9)$, $Y \sim N(3, 16)$ give $X + Y \sim N(5, 25)$, SD 5; also $X - Y \sim N(-1, 25)$.
- **Continuity correction**: when a normal approximates a count (Binomial), use ± 0.5 . “At least 60 heads in 100 tosses” becomes $P(Y > 59.5)$ with $Y \sim N(50, 25)$: $z = 1.9$, $P \approx 0.029$.

3.3.4 Log normal

- If $\ln Y$ is normal, Y is **log normal**: positive and right skewed. Salaries, prices, time spent on a page.
- If $Y = e^X$ with $X \sim N(\mu, \sigma^2)$: median e^μ , mean $e^{\mu + \sigma^2/2}$ (bigger than the median, pulled up by the right tail).
- This is why we take logs of salaries before modelling: the log is roughly normal, and effects become multiplicative (percent changes).

3.3.5 Beta and Gamma, one line each

- **Beta**(α, β) lives on $[0, 1]$: a distribution for an unknown probability such as a click through rate. Mean $\frac{\alpha}{\alpha + \beta}$. Read Beta(2, 5) as “2 pseudo clicks and 5 pseudo non clicks”: mean $\frac{2}{7}$.
- **Gamma**(k, λ): the waiting time until the k -th Poisson event (a sum of k exponentials). Mean $\frac{k}{\lambda}$.

3.4 The three test distributions: χ^2 , t and F

- They are all built from standard normals, which is why they appear in hypothesis tests (Chapter 7).
- **Chi-square with k degrees of freedom**: $Z_1^2 + \dots + Z_k^2$ with independent standard normals. Mean k , variance $2k$, always positive, right skewed.

- **Student's t with ν df:** $\frac{Z}{\sqrt{\chi^2_\nu/\nu}}$. Looks like a normal but with **heavier tails**, because it accounts for the extra uncertainty of estimating σ from data. As $\nu \rightarrow \infty$ it becomes $N(0,1)$.
- **F with (d_1, d_2) df:** a ratio of two independent chi-squares each divided by its df. Used in ANOVA and to compare two variances.

3.5 Shape: skewness and kurtosis (a reported InfoEdge OA topic)

- **Skewness** measures asymmetry.
 - **Right (positive) skew:** a long tail to the right. Typically mean $>$ median $>$ mode. Incomes: a few very high earners pull the mean up. The median is then the better “typical” value.
 - **Left (negative) skew:** a long tail to the left. Typically mean $<$ median $<$ mode. Scores on an easy exam.
 - Memory hook: the mean is pulled towards the tail.
- **Kurtosis** measures how heavy the tails are (how prone to extreme values), $\frac{E[(X-\mu)^4]}{\sigma^4}$.
 - The normal has kurtosis 3. **Excess kurtosis** = kurtosis -3 , so the normal has 0 (mesokurtic). pandas .kurt() reports excess kurtosis.
 - Excess > 0 : **leptokurtic**, heavier tails, more outliers (the t distribution, returns, revenue per user).
 - Excess < 0 : **platykurtic**, lighter tails (the uniform, excess -1.2).
- **Q-Q plot:** plot the data's quantiles against a normal's. A straight line means normal. Points steeper than the line at both ends mean heavy tails.

3.6 Mixtures and the law of total variance

- If X comes from $N(0,1)$ half the time and from $N(4,1)$ half the time, it is a **mixture**: two bumps.
- Mean: average of the component means, 2.
- Variance = average within component variance + variance of the component means = $1 + \frac{(0-2)^2 + (4-2)^2}{2} = 1 + 4 = 5$. This is the law of total variance: $\text{Var}(X) = E[\text{Var}(X | C)] + \text{Var}(E[X | C])$, the maths behind Gaussian Mixture Models.

3.7 Summary table

Distribution	Story	Mean	Variance
Bernoulli(p)	one yes/no	p	$p(1-p)$
Binomial(n, p)	successes in n trials	np	$np(1-p)$
Geometric(p)	trials to first success	$1/p$	$(1-p)/p^2$
Poisson(λ)	count at a rate	λ	λ
Uniform(a, b)	flat	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	waiting time	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	sums, averages	μ	σ^2
χ^2_k	sum of k squared Z	k	$2k$
Beta(α, β)	unknown probability	$\alpha/(\alpha+\beta)$	

Where this shows up in ML and at a company like Naukri

- Poisson regression (log link) predicts counts such as applications per job; Negative Binomial handles over dispersion.
- Losses are distributions in disguise: squared error assumes normal noise, log loss assumes Bernoulli labels.
- Heavy tails (high kurtosis) in revenue or latency data break mean based metrics; use medians, logs or robust losses.
- Exponential and Poisson model arrivals for capacity planning, notification timing, and queueing.
- Beta priors smooth click through rates for jobs with few impressions.

15 second recall card: memorise every line

- Bernoulli variance $p(1-p)$, maximum 0.25 at $p = 0.5$. Binomial mean np , variance $np(1-p)$.
- Geometric: failures before success mean $\frac{1-p}{p}$; trials mean $\frac{1}{p}$; variance $\frac{1-p}{p^2}$; memoryless.
- Poisson: mean = variance = λ ; rates scale with time and add across independent sources; $P(0) = e^{-\lambda}$.
- Exponential: mean $\frac{1}{\lambda}$, median $\frac{\ln 2}{\lambda}$, $P(X > t) = e^{-\lambda t}$, memoryless. Geometric and exponential are the only memoryless distributions.
- Uniform(a, b): mean $\frac{a+b}{2}$, variance $\frac{(b-a)^2}{12}$.
- Normal: 68, 95, 99.7; $P(Z > 1) = 0.1587$, $P(Z > 2) = 0.0228$; top 5% at $z = 1.645$. $aX + b \sim N(a\mu + b, a^2\sigma^2)$. Independent normals: means add, variances add (also for differences).
- Right skew: mean > median > mode. Normal kurtosis 3, excess 0; leptokurtic = heavy tails.
- Log normal: median e^μ , mean $e^{\mu+\sigma^2/2}$. Beta mean $\frac{\alpha}{\alpha+\beta}$.
- χ_k^2 : mean k , variance $2k$. t : heavier tails, tends to normal. F : ratio of scaled chi-squares, used in ANOVA.
- Poisson(λ): $E[X^2] = \lambda + \lambda^2$.

3.8 Question bank: Probability distributions**Q3.1 REASON · about 45 s**

★ **PYQ (INT and OA: distributions in the data we handle daily)** Which pairing of a story with a first choice distribution is the **least** appropriate?

- (A) Clicks out of 20 recommended jobs shown to a user: Binomial
- (B) Job applications received by a post in one hour: Poisson
- (C) Monthly salaries of employees in India: Normal
- (D) Time between two consecutive logins: Exponential

Solution

Fast route: Salaries are positive and right skewed: log normal, not normal.

Full working:

- (B) counts at a rate: Poisson. (D) waiting time between Poisson events: Exponential. (A) fixed number of yes/no trials: Binomial (if clicks are independent with a common rate).

- (C) A normal would allow negative salaries and would miss the long right tail.

Answer: C.

Q3.2 **CALC** · 1.5 to 2 min

★ **PYQ (OA: sum of two independent normals)** $X \sim N(2, 9)$ and $Y \sim N(3, 16)$ are independent (second parameter is the variance). $P(X + Y > 10)$ is closest to

- (A) 0.0228
- (B) 0.1587
- (C) 0.0062
- (D) 0.2375

Solution

Fast route: $X + Y \sim N(5, 25)$, SD 5; 10 is one SD above: 0.1587.

Full working:

- Means add: 5. Variances add: 25, SD 5.
- $z = \frac{10-5}{5} = 1$, $P(Z > 1) = 1 - 0.8413 = 0.1587$.

Why the other options fail: (D) adds the SDs ($3 + 4 = 7$, so $z \approx 0.71$) instead of the variances. (A) and (C) correspond to $z = 2$ and $z = 2.5$, which come from other wrong spreads. Only variances add.

Answer: B.

Q3.3 **RECALL** · about 15 s

★ **PYQ (OA: kurtosis)** A distribution with excess kurtosis +4 is best described as

- (A) right skewed
- (B) platykurtic, with lighter tails than a normal
- (C) mesokurtic, like a normal
- (D) leptokurtic, with heavier tails than a normal

Solution

Fast route: Positive excess kurtosis = heavy tails = leptokurtic.

Full working:

- Normal: kurtosis 3, excess 0. Positive excess means more extreme values than a normal.

Why the other options fail: (A) Kurtosis is about tails, not asymmetry; skewness measures asymmetry.

Answer: D.

Q3.4 **REASON** · about 45 s

★ **PYQ (INT: derive the geometric distribution)** A recruiter calls candidates one by one; each independently accepts with probability 0.25. The probability that exactly 3 decline before the first acceptance, and the expected number of declines, are

- (A) 0.105 and 3
- (B) 0.42 and 4
- (C) 0.25 and 3
- (D) 0.105 and 4

Solution

Fast route: $0.75^3 \times 0.25$; mean failures $\frac{1-p}{p} = 3$.

Full working:

- Three declines then an acceptance: $0.421875 \times 0.25 = 0.105$.
- Expected declines = $\frac{0.75}{0.25} = 3$. (Expected calls including the acceptance = 4.)

Why the other options fail: (D) mixes the two conventions: 4 is the expected number of calls, not declines.

Answer: A.

Q3.5 **CALC** · 1.5 to 2 min

★ **PYQ (INT: Poisson for customer arrivals)** Customers arrive at 4 per hour (Poisson). The probability of exactly 2 arrivals in an hour, and the probability of nobody in a 30 minute window, are closest to

- (A) 0.073 and 0.368
 (B) 0.147 and 0.018
 (C) 0.238 and 0.135
 (D) 0.147 and 0.135

Solution

Fast route: $\frac{e^{-4}4^2}{2} = 8e^{-4}$; half hour $\lambda = 2$, e^{-2} .

Full working:

- $8 \times 0.0183 = 0.147$.
- 30 minutes: $\lambda = 2$. $P(0) = e^{-2} = 0.135$.

Why the other options fail: (B) uses $\lambda = 4$ for the half hour. (A) uses e^{-1} .

Answer: D.

Q3.6 **RECALL** · about 15 s

★ **PYQ (OA: variance over an interval, uniform version)** X is uniform on $[2, 8]$. $E[X]$ and $\text{Var}(X)$ are

- (A) 5 and 3
 (B) 5 and 12
 (C) 5 and 6
 (D) 4 and 3

Solution

Fast route: Mean at the middle; variance $\frac{6^2}{12} = 3$.

Full working:

- $\frac{2+8}{2} = 5$; $\frac{(8-2)^2}{12} = 3$.

Why the other options fail: (B) forgets to divide by 12 correctly (divides 36 by 3). (C) divides by 6.

Answer: A.

Q3.7 **REASON** · about 45 s

$X \sim \text{Binomial}(10, 0.2)$. $P(X = 2)$ is closest to

- (A) 0.45
- (B) 0.30
- (C) 0.20
- (D) 0.04

Solution

Fast route: $45 \times 0.04 \times 0.8^8$, with $0.8^8 \approx 0.168$.

Full working:

- $\binom{10}{2} = 45$; $45 \times 0.04 = 1.8$; $1.8 \times 0.168 = 0.302$.

Why the other options fail: (D) forgets the $\binom{10}{2}$ and the 0.8^8 factors. (C) confuses with p .

Answer: B.

Q3.8 RECALL · about 15 s

$X \sim \text{Binomial}(100, 0.3)$. Mean, variance and SD are

- (A) 30, 9, 3
- (B) 30, 21, 4.58
- (C) 70, 21, 4.58
- (D) 30, 30, 5.48

Solution

Fast route: $np = 30$; $np(1 - p) = 21$; $\sqrt{21} \approx 4.58$.

Full working:

- Straight from the formulas.

Why the other options fail: (D) uses the Poisson rule variance = mean.

Answer: B.

Q3.9 REASON · about 45 s

A component is defective with probability 0.05. In a batch of 20, the probability of at least one defective is closest to

- (A) 0.05
- (B) 0.36
- (C) 0.64
- (D) 1.00

Solution

Fast route: $1 - 0.95^{20} = 1 - 0.358$.

Full working:

- $0.95^{20} \approx 0.358$ (front table). Answer 0.642.

Why the other options fail: (D) adds 20×0.05 , which double counts. (B) is the complement.

Answer: C.

Q3.10 CALC · 1.5 to 2 min

A rare event has $p = 0.002$ per trial and there are 1000 independent trials. Using the Poisson approximation, $P(\text{at most 1 event})$ is closest to

- (A) 0.135
- (B) 0.406
- (C) 0.271
- (D) 0.677

Solution

Fast route: $\lambda = 2$: $e^{-2}(1 + 2) = 3 \times 0.135$.

Full working:

- $P(0) + P(1) = e^{-2} + 2e^{-2} = 0.406$.

Why the other options fail: (A) is only $P(0)$. (C) is only $P(1)$.

Answer: B.

Q3.11 REASON · about 45 s

Server A logs errors at 3 per day and server B at 2 per day, independently (Poisson). The probability of an error free day across both is closest to

- (A) 0.0067
- (B) 0.050
- (C) 0.0025
- (D) 0.135

Solution

Fast route: Total Poisson(5): e^{-5} .

Full working:

- Equivalently $e^{-3} \times e^{-2} = e^{-5} = 0.0067$.

Answer: A.

Q3.12 RECALL · about 15 s

Daily complaint counts have sample mean 5 and sample variance 18. The best conclusion is

- (A) The data is over dispersed; a Negative Binomial is a better count model
- (B) The data is under dispersed
- (C) The data must be normal
- (D) The data fits a Poisson well

Solution

Fast route: Poisson needs variance $\approx$ mean; $18 \gg 5$.

Full working:

- Over dispersion often comes from heterogeneity (some days or products generate many complaints). Negative Binomial adds a parameter for it.

Answer: A.

Q3.13 CALC · 1.5 to 2 min

A box has 10 items, 3 defective. 4 are drawn without replacement. $P(\text{exactly 1 defective})$ is

- (A) 0.50

- (B) 0.67
(C) 0.30
(D) 0.41

Solution

Fast route: $\frac{\binom{3}{1}\binom{7}{3}}{\binom{10}{4}} = \frac{3 \times 35}{210}$.

Full working:

- $\binom{7}{3} = 35$, $\binom{10}{4} = 210$. $\frac{105}{210} = 0.5$.

Why the other options fail: (D) is the Binomial answer with replacement: $4 \times 0.3 \times 0.7^3 = 0.41$.

Answer: A.

Q3.14 REASON · about 45 s

Bulb lifetimes are exponential with mean 1000 hours. A bulb has already worked 500 hours. The probability it lasts beyond 1500 hours in total is closest to

- (A) 0.607
(B) 0.5
(C) 0.223
(D) 0.368

Solution

Fast route: Memoryless: need 1000 more hours; $e^{-1000/1000} = e^{-1}$.

Full working:

- $P(X > 1500 \mid X > 500) = P(X > 1000) = e^{-1} = 0.368$.

Why the other options fail: (C) is $e^{-1.5}$, which ignores the 500 hours already survived.

Answer: D.

Q3.15 RECALL · about 15 s

An exponential waiting time has mean 10 minutes. Its median is closest to

- (A) 5 minutes
(B) 10 minutes
(C) 14.4 minutes
(D) 6.9 minutes

Solution

Fast route: Median = mean $\times \ln 2 = 6.93$.

Full working:

- Solve $e^{-m/10} = \frac{1}{2}$: $m = 10 \ln 2$.

Answer: D.

Q3.16 REASON · about 45 s

Applications arrive as a Poisson process at 6 per hour. The probability that the wait for the next application exceeds 15 minutes is closest to

- (A) 0.25

- (B) 0.223
- (C) 0.135
- (D) 0.368

Solution

Fast route: $\lambda t = 6 \times \frac{1}{4} = 1.5$; $e^{-1.5}$.

Full working:

- $P(T > t) = e^{-\lambda t} = e^{-1.5} = 0.223$.

Answer: B.

Q3.17 **RECALL** · about 15 s

$IQ \sim N(100, 15^2)$. $P(IQ > 130)$ is closest to

- (A) 0.0228
- (B) 0.0455
- (C) 0.0013
- (D) 0.1587

Solution

Fast route: $z = 2$, one tail: 0.0228.

Full working:

- $1 - \Phi(2) = 0.0228$.

Why the other options fail: (B) is the two tailed value. (C) is $z = 3$.

Answer: A.

Q3.18 **RECALL** · about 15 s

Same IQ distribution. $P(85 < IQ < 115)$ is closest to

- (A) 0.84
- (B) 0.95
- (C) 0.50
- (D) 0.68

Solution

Fast route: Within one SD: 68%.

Full working:

- $\Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = 0.6826$.

Answer: D.

Q3.19 **REASON** · about 45 s

Same IQ distribution. The score needed to be in the top 5% is closest to

- (A) 115.0
- (B) 130.0
- (C) 129.4
- (D) 124.7

Solution

Fast route: One sided 5%: $z = 1.645$; $100 + 1.645 \times 15$.

Full working:

- $100 + 24.7 = 124.7$.

Why the other options fail: (C) uses $z = 1.96$, which is for a two sided 5% (top 2.5%).

Answer: D.

Q3.20 **CALC** · 1.5 to 2 min

A fair coin is tossed 100 times. Using the normal approximation with continuity correction, the probability of at least 60 heads is closest to

- (A) 0.0287
- (B) 0.0228
- (C) 0.0446
- (D) 0.1587

Solution

Fast route: Mean 50, SD 5; $P(Y > 59.5)$: $z = 1.9$.

Full working:

- Binomial mean 50, SD $\sqrt{100 \times 0.25} = 5$.
- “At least 60” for a count becomes “more than 59.5” for the continuous normal.
- $z = \frac{9.5}{5} = 1.9$; $1 - \Phi(1.9) = 1 - 0.9713 = 0.0287$.

Why the other options fail: (B) skips the continuity correction ($z = 2$).

Answer: A.

Q3.21 **RECALL** · about 15 s

$X \sim N(10, 4)$ (variance 4). The distribution of $Y = 3X - 5$ is

- (A) $N(25, 12)$
- (B) $N(30, 36)$
- (C) $N(25, 36)$
- (D) $N(25, 31)$

Solution

Fast route: Mean $3(10) - 5$; variance $3^2 \times 4$.

Full working:

- Shifts move the mean only; scaling multiplies the variance by a^2 .

Why the other options fail: (A) multiplies the variance by 3 instead of 9.

Answer: C.

Q3.22 **REASON** · about 45 s

25 independent observations are drawn from $N(50, 100)$. The probability their mean exceeds 52 is closest to

- (A) 0.0228
- (B) 0.1587

- (C) 0.42
(D) 0.3085

Solution

Fast route: Mean of 25 has SD $\frac{10}{5} = 2$; 52 is one SD up.

Full working:

- $\bar{X} \sim N(50, \frac{100}{25}) = N(50, 4)$. $z = 1$. 0.1587.

Why the other options fail: (C) uses the SD of one observation (10), giving $z = 0.2$.

Answer: B.

Q3.23 RECALL · about 15 s

For a strongly right skewed distribution such as household income, which is typically true?

- (A) mean = median
(B) mean > median
(C) mean < median
(D) the mode exceeds the mean

Solution

Fast route: The mean is pulled towards the long right tail.

Full working:

- Typical ordering: mode < median < mean. Report the median as the typical value.

Answer: B.

Q3.24 RECALL · about 15 s

A left skewed distribution has median 60. The most likely ordering is

- (A) mode < 60 < mean
(B) mean = 60 = mode
(C) mean < 60 < mode
(D) 60 < mean < mode

Solution

Fast route: Left tail pulls the mean down below the median; the peak (mode) is on the right.

Full working:

- Mean < median < mode.

Answer: C.

Q3.25 REASON · about 45 s

$Y = e^X$ with $X \sim N(\mu, \sigma^2)$. Which statement is true?

- (A) Y can be negative
(B) Y is symmetric
(C) The median of Y is e^μ and its mean is $e^{\mu+\sigma^2/2}$
(D) The mean of Y is e^μ

Solution

Fast route: The exponential function keeps order, so the median maps to e^μ ; the right tail lifts the mean.

Full working:

- $P(Y \leq e^\mu) = P(X \leq \mu) = \frac{1}{2}$, so the median is e^μ .
- $E[e^X] = e^{\mu + \sigma^2/2} > e^\mu$ (the MGF of the normal).

Why the other options fail: (A) $e^X > 0$ always. (B) log normal is right skewed.

Answer: C.

Q3.26 RECALL · about 15 s

For a Bernoulli variable, the maximum possible variance and the p at which it occurs are

- (A) 0.25 at $p = 0.5$
- (B) 1 at $p = 0.5$
- (C) 0.5 at $p = 0.5$
- (D) 0.25 at $p = 1$

Solution

Fast route: $p(1-p)$ peaks at $p = \frac{1}{2}$ with value $\frac{1}{4}$.

Full working:

- A parabola with roots 0 and 1 peaks halfway.

Answer: A.

Q3.27 RECALL · about 15 s

The variance of a fair die, from the discrete uniform formula $\frac{n^2-1}{12}$, is

- (A) 35/12
- (B) 3.5
- (C) 25/12
- (D) 36/12

Solution

Fast route: $\frac{36-1}{12}$.

Full working:

- $\frac{35}{12} \approx 2.92$, matching Chapter 1.

Why the other options fail: (D) uses the continuous uniform formula on a length of 6.

Answer: A.

Q3.28 RECALL · about 15 s

$X \sim \text{Uniform}(0, 10)$. $P(3 < X < 7)$ is

- (A) 0.3
- (B) 0.5
- (C) 0.7
- (D) 0.4

Solution**Fast route:** Length ratio $\frac{4}{10}$.**Full working:**

- Flat density $\frac{1}{10}$ times width 4.

Answer: D.**Q3.29** **RECALL · about 15 s**

Z_1, Z_2, Z_3 are independent standard normals and $Q = Z_1^2 + Z_2^2 + Z_3^2$. Then Q follows

- (A) $N(0, 3)$
 (B) t with 3 df
 (C) χ^2 with 3 df, mean 3 and variance 6
 (D) χ^2 with 3 df, mean 3 and variance 3

Solution**Fast route:** Sum of k squared standard normals: χ_k^2 , mean k , variance $2k$.**Full working:**

- $E[Z^2] = 1$ so the mean is 3; $\text{Var}(Z^2) = 2$ so the variance is 6.

Answer: C.**Q3.30** **RECALL · about 15 s**

Compared with the standard normal, Student's t distribution

- (A) is skewed to the right
 (B) has lighter tails and approaches the normal as df decrease
 (C) has heavier tails and approaches the normal as df increase
 (D) has a smaller variance

Solution**Fast route:** Extra uncertainty from estimating σ fattens the tails; it vanishes with many df.**Full working:**

- Variance $\frac{\nu}{\nu-2} > 1$ for $\nu > 2$; symmetric around 0.

Answer: C.**Q3.31** **RECALL · about 15 s**

Which statement about the F distribution is correct?

- (A) It is the square of a chi-square
 (B) It can take negative values
 (C) It is the ratio of two independent chi-squares, each divided by its df, and is used in ANOVA
 (D) It is symmetric around 0

Solution**Fast route:** $F = \frac{\chi_{d_1}^2/d_1}{\chi_{d_2}^2/d_2}$; ANOVA and variance ratio tests.**Full working:**

- Ratios of positive quantities are positive and right skewed. (Also t_v^2 is $F_{1,v}$.)

Answer: C.

Q3.32 REASON · about 45 s

A fair coin is tossed repeatedly. The probability that the 3rd head appears exactly on the 5th toss is

- (A) 0.3125
- (B) 0.1875
- (C) 0.0625
- (D) 0.25

Solution

Fast route: Toss 5 is a head, and exactly 2 heads among the first 4: $\binom{4}{2}/2^5$.

Full working:

- $\frac{6}{32} = 0.1875$ (Negative Binomial).

Why the other options fail: (A) is “exactly 3 heads in 5 tosses”, which does not require the last toss to be the 3rd head.

Answer: B.

Q3.33 RECALL · about 15 s

Each ad impression converts with probability 0.2. The expected number of impressions until the first conversion, and the variance, are

- (A) 5 and 4
- (B) 5 and 20
- (C) 4 and 20
- (D) 5 and 25

Solution

Fast route: $\frac{1}{p} = 5$; $\frac{1-p}{p^2} = \frac{0.8}{0.04} = 20$.

Full working:

- Geometric (trials convention).

Why the other options fail: (C) gives the failures before success mean, $\frac{1-p}{p} = 4$.

Answer: B.

Q3.34 RECALL · about 15 s

Which pair of distributions are the memoryless ones?

- (A) Normal and uniform
- (B) Exponential and normal
- (C) Geometric and exponential
- (D) Poisson and binomial

Solution

Fast route: Geometric (discrete) and exponential (continuous) are the only memoryless distributions.

Full working:

- $P(X > s + t \mid X > s) = P(X > t)$ holds only for these.

Answer: C.**Q3.35 RECALL · about 15 s**

A click through rate is given a Beta(2, 5) prior. Its prior mean is

- (A) $2/7$
 (B) 0.2
 (C) $5/7$
 (D) 0.4

Solution**Fast route:** $\frac{\alpha}{\alpha + \beta}$.**Full working:**

- $\frac{2}{2+5} = 0.286$: “2 pseudo clicks, 5 pseudo non clicks”.

Why the other options fail: (D) is $\frac{2}{5}$, the ratio α/β .**Answer:** A.**Q3.36 REASON · about 45 s**

$X \sim \text{Poisson}(3)$. $E[X^2]$ is

- (A) 6
 (B) 12
 (C) 9
 (D) 3

Solution**Fast route:** $\text{Var} + (\text{mean})^2 = 3 + 9$.**Full working:**

- From $\text{Var} = E[X^2] - (E[X])^2$.

Answer: B.**Q3.37 CALC · 1.5 to 2 min**

Two machines fail independently with exponential lifetimes, rates 1 and 2 per year. The expected time until the first failure, and the probability that machine 2 fails first, are

- (A) 1.5 years and $1/2$
 (B) $1/3$ year and $1/3$
 (C) $1/3$ year and $2/3$
 (D) $1/2$ year and $2/3$

Solution**Fast route:** Minimum of exponentials: rate $1 + 2$; winner probability proportional to its rate.**Full working:**

- $P(\min > t) = e^{-t}e^{-2t} = e^{-3t}$: mean $\frac{1}{3}$.

- $P(\text{machine 2 first}) = \frac{2}{1+2} = \frac{2}{3}$: the faster failing machine usually fails first.

Answer: C.

Q3.38 REASON · about 45 s

Which pairing of data and first choice model is the **poorest**?

- (A) Whether a candidate is shortlisted: Bernoulli
- (B) Time a user spends on a job page: Log normal
- (C) Number of typos on a page: Poisson
- (D) User rating on a 1 to 5 star scale: Normal

Solution

Fast route: Five ordered discrete values bounded on both sides are not normal.

Full working:

- Use a categorical or ordinal model for star ratings.

Answer: D.

Q3.39 REASON · about 45 s

On a normal Q-Q plot of residuals, points lie above the line at the right end and below it at the left end (steeper than the line at both ends). This suggests

- (A) left skew only
- (B) light tails
- (C) perfect normality
- (D) heavy tails (positive excess kurtosis)

Solution

Fast route: Both extremes are more extreme than a normal predicts: heavy tails.

Full working:

- Sample quantiles beyond theoretical quantiles at both ends means extra mass in the tails, i.e. outliers.

Answer: D.

Q3.40 CALC · 1.5 to 2 min

X is drawn from $N(0, 1)$ with probability 0.5 and from $N(4, 1)$ with probability 0.5. $E[X]$ and $\text{Var}(X)$ are

- (A) 2 and 4
- (B) 2 and 1
- (C) 2 and 2
- (D) 2 and 5

Solution

Fast route: Total variance = within (1) + between (4).

Full working:

- Mean $0.5(0) + 0.5(4) = 2$.

- Within component variance: 1. Between: the component means 0 and 4 vary around 2 with variance 4.
- Total 5. Check via $E[X^2]$: $0.5(0 + 1) + 0.5(16 + 1) = 9$; $9 - 4 = 5$.

Why the other options fail: (B) ignores the spread between components. (A) ignores the within component noise.

Answer: D.

References for this chapter

- [Seeing Theory: Probability Distributions](#).
- [3Blue1Brown: Binomial distributions; Why Gaussian plus Gaussian is Gaussian](#).
- [StatQuest: The normal distribution](#).
- [Scribbr: Poisson distribution; Skewness](#).

Chapter 4

Sampling

Companion resource for this section: [Scribbr: Sampling methods \(probability vs non probability sampling, with examples\)](#)

4.1 Why we sample at all

Start here: the everyday picture

- A cook tastes one spoon of soup to decide whether the whole pot needs salt. She does not drink the pot. But she stirs first, because a spoon from the unstirred top could be misleading.
- That is all of sampling in one picture: look at a small part (the spoon) to learn about the whole (the pot), and make sure the part is **representative** (stir first).
- Naukri cannot survey 9 crore job seekers about a new feature, so it surveys 2,000. The questions of this chapter: how to pick those 2,000, and how wrong the answer might be.

4.2 Population, sample, parameter, statistic

- **Population:** every unit you care about (all registered job seekers).
- **Sample:** the units you actually observe (the 2,000 surveyed).
- **Parameter:** a fixed but unknown number about the population, such as the true average session time μ , the true SD σ , or the true share p who like a feature. Greek letters by convention.
- **Statistic:** a number computed from the sample, such as the sample mean $\bar{x}$, the sample SD s , or the sample proportion $\hat{p}$. It changes from sample to sample, so it is itself a random variable.
- Inference means: use a statistic to estimate a parameter, and say how far off we might be.

4.2.1 A sampling distribution you can see completely

- Tiny population: three people with values $\{2, 4, 6\}$. True mean $\mu = 4$; population variance $\sigma^2 = \frac{(2-4)^2 + 0 + (6-4)^2}{3} = \frac{8}{3}$.
- Draw samples of size 2 **with replacement**. There are $3 \times 3 = 9$ equally likely samples:

Sample	(2,2)	(2,4)	(2,6)	(4,2)	(4,4)	(4,6)	(6,2)	(6,4)	(6,6)
$\bar{x}$	2	3	4	3	4	5	4	5	6

- The nine sample means form the **sampling distribution** of $\bar{x}$. Their average is exactly $4 = \mu$. That is what " $\bar{x}$ is an **unbiased** estimator of μ " means: on average over all possible samples, it hits the target.
- Their variance is $\frac{4}{3}$, exactly $\frac{\sigma^2}{n} = \frac{8/3}{2}$. The sample mean is *less* spread out than single values, because averaging cancels some of the randomness.

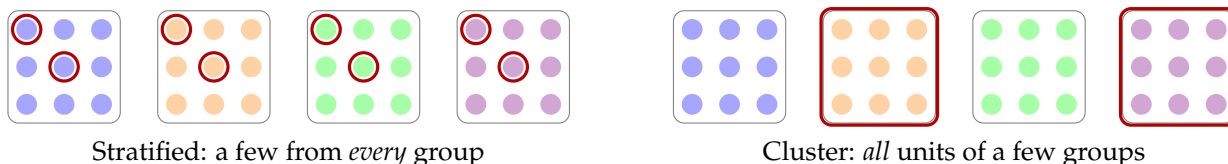
The idea that runs through Chapters 4 to 9

- Every statistic has its own distribution over repeated samples: its **sampling distribution**.
- The SD of that sampling distribution is called the **standard error** (SE). For the mean,

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}.$$

- **SD vs SE:** the SD describes how much *individual data points* vary; the SE describes how much an *estimate* (like the mean) would vary if you repeated the whole study. More data leaves the SD unchanged but shrinks the SE.
- Every confidence interval and test in this book has the form: estimate $\pm$ (critical value) $\times$ SE.

4.3 Probability sampling: every unit has a known chance



4.3.1 Simple random sampling (SRS)

- Every subset of size n is equally likely, like drawing names from a well mixed hat. Each unit's chance of being included is $\frac{n}{N}$.
- Number of possible samples of 3 from 10 (without replacement): $\binom{10}{3} = 120$.

4.3.2 Systematic sampling

- Sort the list, pick a random start, then take every k -th unit with $k = \frac{N}{n}$. 1,000 resumes, need 50: $k = 20$; start at 7: resumes 7, 27, 47, ...
- Easy to run. Danger: **periodicity**. If every 20th row happens to be a Monday, or the same branch office, the sample is badly biased.

4.3.3 Stratified sampling

- Split the population into groups (**strata**) that are similar inside and different from each other (experience band, city tier). Take a random sample **inside every stratum**.
- **Proportional allocation:** $n_h = n \frac{N_h}{N}$. Strata of 5,000, 3,000, 2,000 and $n = 200$: 100, 60, 40.
- Estimate: weight each stratum mean by its population share: $\bar{x}_{st} = \sum_h W_h \bar{x}_h$ with $W_h = \frac{N_h}{N}$. Weights 0.5, 0.3, 0.2 and means 20, 40, 35: $10 + 12 + 7 = 29$. (The unweighted average of the three means, 31.67, would be wrong.)
- Why it helps: differences *between* strata no longer add noise; only variation *within* strata does. Most useful when strata means differ a lot.
- **Neyman allocation:** sample more from strata that are large *and* noisy: $n_h \propto N_h \sigma_h$.

4.3.4 Cluster sampling

- Split into natural groups (**clusters**: colleges, offices, cities), randomly pick some clusters, and survey everyone in them (or a subsample: **multistage**).
- Cheap: one visit per college instead of travelling to thousands of random students.
- Cost: people in the same cluster are alike, so each extra person adds less information. The variance is inflated by the **design effect** $DEFF \approx 1 + (m - 1)\rho$, where m is the cluster size and ρ the intra cluster correlation. Effective sample size = $n/DEFF$.

- Memory hook: **strata**: sample *from every* group; **clusters**: sample *whole* groups.

4.4 Non probability sampling and the biases it brings

- **Convenience**: whoever is easy to reach (your hostel friends). **Voluntary response**: people choose to answer (app reviews), so strong opinions dominate. **Snowball**: participants recruit others. **Quota**: fill counts per group but not randomly inside the group.
- **Selection bias**: the way units enter the sample is related to the outcome.
- **Survivorship bias**: you only see units that survived a filter. WW2 story: returning bombers had bullet holes in the wings, and someone proposed armouring the wings. Abraham Wald said armour the engines, because planes hit there never came back. ML version: a loan default model trained only on approved loans (rejected applicants never get a label), or a churn model trained only on users still active.
- **Non response bias**: those who do not answer differ from those who do. A 10% response rate is a warning sign.
- **Undercoverage**: part of the population is missing from the list you sample from (an online survey misses people without internet).

Bias does not shrink with n

- More data shrinks the **random error** (the SE falls like $1/\sqrt{n}$). It does nothing to **bias**.
- A huge biased sample gives a very precise wrong answer. The 1936 Literary Digest poll had 2.4 million responses and predicted the wrong US president.
- The margin of error covers random error only; it says nothing about bias.

4.5 Estimating the mean and the variance, and why $n - 1$

- Sample mean $\bar{x} = \frac{1}{n} \sum x_i$: unbiased for μ , with variance $\frac{\sigma^2}{n}$.
- Sample variance: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$. Dividing by $n - 1$ (not n) makes it unbiased: $E[s^2] = \sigma^2$.
- Why $n - 1$, in plain words: we measure distances from $\bar{x}$, which sits in the middle of *our* data. The data is always a little closer to its own mean than to the true μ , so $\sum (x_i - \bar{x})^2$ is a bit too small. Precisely, its expected value is $(n - 1)\sigma^2$, so we divide by $n - 1$.
- Another view: the n deviations $x_i - \bar{x}$ always sum to zero, so only $n - 1$ of them are free. We say $n - 1$ **degrees of freedom**.
- Worked example: 2, 4, 4, 4, 5, 5, 7, 9. Mean = $\frac{40}{8} = 5$. Squared deviations 9, 1, 1, 1, 0, 0, 4, 16: total 32. Divide by $n = 8$: 4 (SD 2). Divide by $n - 1 = 7$: 4.571 (SD 2.14).
- Dividing by n is the maximum likelihood estimate: biased low by the factor $\frac{n-1}{n}$, but the bias vanishes as n grows.
- Even though s^2 is unbiased, s is slightly biased low (the square root is curved).
- **Python gotcha**: `np.var(x)` and `np.std(x)` divide by n (ddof=0) by default; `pandas df.var()` and `df.std()` divide by $n - 1$ (ddof=1).

4.6 Standard error, margin of error and sample size

- $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$ (use s if σ is unknown). For a proportion, $SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$.
- **Square root law**: to halve the SE you need 4 times the data; to cut it by 10 you need 100 times. $\sigma = 20, n = 100$: SE 2. For SE 1 you need $n = 400$.

- **Margin of error** at 95%: $\text{MOE} = 1.96 \times \text{SE}$. A poll of 1,000 with $\hat{p} = 0.5$: $\text{SE} = \sqrt{0.25/1000} = 0.0158$; $\text{MOE} = 0.031$, the famous “plus or minus 3 points”.
- **Sample size** for a desired MOE E :
 - mean: $n = \left(\frac{z\sigma}{E}\right)^2$; $\sigma = 15$, $E = 2$: $(14.7)^2 = 216.1$, round **up** to 217.
 - proportion: $n = \frac{z^2 p(1-p)}{E^2}$; with p unknown use 0.5 (the worst case): $\pm 3\%$ needs $\frac{3.8416 \times 0.25}{0.0009} = 1067.1$, so 1,068.
- **Finite population correction** (FPC): when you sample a large fraction of a finite population without replacement, multiply the SE by $\sqrt{\frac{N-n}{N-1}}$. $N = 1000$, $n = 200$: $\sqrt{800/999} = 0.895$. Ignore it when $n/N < 5\%$.
- A surprising fact: the needed sample size hardly depends on the population size. 1,068 people give $\pm 3\%$ for a city or for all of India (as long as they are a random sample).

Try it now (answer right below, cover it first)

$\sigma = 10$ and you want a 95% MOE of 1. How many observations?

Answer: $(1.96 \times 10/1)^2 = 384.2$, round up to 385.

4.7 Resampling: the bootstrap and cross validation

- **Bootstrap**: pretend your sample is the population. Draw n points **with replacement** from it, recompute the statistic, repeat 1,000s of times. The spread of those values estimates the SE, and its 2.5th and 97.5th percentiles give a 95% CI. Works for medians, AUC, ratios, where no formula exists.
- How many original points are missing from one bootstrap sample? A given point is missed by one draw with probability $1 - \frac{1}{n}$, by all n draws with probability $(1 - \frac{1}{n})^n \approx e^{-1} = 0.368$. So about 63.2% of points appear and 36.8% are **out of bag** (OOB). Random Forests use OOB rows as a free validation set.
- The bootstrap cannot fix a biased sample: every resample inherits the bias.
- **k-fold cross validation**: split into k parts, train on $k - 1$, test on the remaining one, rotate. **Stratified** folds keep class ratios (essential for rare fraud labels). **Grouped** folds keep all rows of one user together.
- **Time series**: always train on the past and test on the future (expanding or rolling windows). A random split leaks the future into training.

4.8 Sampling inside algorithms

- **Inverse transform sampling**: if U is uniform on $(0, 1)$ and F is a CDF, then $X = F^{-1}(U)$ has CDF F . Exponential(λ): solve $1 - e^{-\lambda x} = u$ to get $x = -\frac{\ln(1-u)}{\lambda}$. With $\lambda = 2$ and $u = 0.5$: $x = \frac{\ln 2}{2} = 0.347$ (the median).
- **Rejection sampling**: propose from an easy distribution g and accept with probability $\frac{f(x)}{Mg(x)}$, where $f \leq Mg$ everywhere. Acceptance rate $\frac{1}{M}$, so on average M proposals per accepted sample.
- **Importance sampling**: estimate an average under p using samples from q , weighting each by $\frac{p(x)}{q(x)}$.
- **Reservoir sampling**: keep a uniform random sample of k items from a stream of unknown length. Item number i (for $i > k$) enters with probability $\frac{k}{i}$ and replaces a random slot. At the end every item is in the sample with probability $\frac{k}{n}$.

- **Class imbalance:** undersampling the majority class (or oversampling the minority) changes the class ratio the model sees, so its predicted probabilities are too high for the rare class and must be recalibrated. Ranking metrics such as AUC are unaffected.

Where this shows up in ML and at a company like Naukri

- Train, validation and test splits are samples: stratify by label, group by user, split by time.
- Mini batch SGD is sampling: the batch gradient is an unbiased but noisy estimate of the full gradient.
- A/B tests assume random assignment; any logging filter (only logged in users) is a sampling bias.
- Offline evaluation on items the old recommender chose to show carries exposure bias.

Formula sheet

$$\begin{aligned} SE(\bar{x}) &= \frac{\sigma}{\sqrt{n}} & SE(\hat{p}) &= \sqrt{\frac{p(1-p)}{n}} & MOE &= z \cdot SE & n &= \left(\frac{z\sigma}{E}\right)^2 & n &= \frac{z^2 p(1-p)}{E^2} & FPC &= \sqrt{\frac{N-n}{N-1}} \\ s^2 &= \frac{1}{n-1} \sum (x_i - \bar{x})^2 & E[\hat{\sigma}_{MLE}^2] &= \frac{n-1}{n} \sigma^2 & \bar{x}_{st} &= \sum W_h \bar{x}_h & n_h &= n \frac{N_h}{N} & \text{Neyman } n_h &\propto N_h \sigma_h \\ DEFF &= 1 + (m-1)\rho & OOB &\approx 0.368 & \text{inclusion in SRS} && \frac{n}{N}. \end{aligned}$$

15 second recall card: memorise every line

- Parameter (population, fixed, Greek) vs statistic (sample, random, Latin).
- SD: spread of data. SE: spread of an estimate; $\frac{\sigma}{\sqrt{n}}$. Halving the SE needs 4 times the data.
- s^2 with $n-1$ is unbiased; dividing by n is the MLE, biased low. `np.var` uses n ; `pandas .var()` uses $n-1$.
- $\pm 3\%$ at 95% needs 1,068; MOE of a 1,000 person poll at 50% is about 3.1%.
- Stratified: sample every stratum, weight by stratum share. Cluster: whole groups, cheap, $DEFF = 1 + (m-1)\rho$.
- Systematic sampling's risk: periodicity. Bias is not fixed by larger n .
- Survivorship bias: WW2 bombers, loans only from approved applicants.
- Bootstrap: sample with replacement; about 36.8% OOB; cannot fix bias.
- Reservoir: item i enters with probability $\frac{k}{i}$. Rejection: M proposals per accepted sample.
- Time series: never random split.

4.9 Question bank: Sampling

No InfoEdge PYQ found for this topic

The syllabus lists Sampling, but no public candidate report names a specific sampling question. The bank follows the reported OA style: short recall items and one step calculations.

Q4.1 REASON · about 45 s

For the data 2, 4, 4, 4, 5, 5, 7, 9, the population variance (divide by n) and the sample variance (divide by $n-1$) are

- (A) 2 and 2.14
- (B) 5 and 5.71
- (C) 4.57 and 4
- (D) 4 and 4.57

Solution

Fast route: Mean 5; squared deviations sum to 32; $\frac{32}{8}$ and $\frac{32}{7}$.

Full working:

- Deviations $-3, -1, -1, -1, 0, 0, 2, 4$; squares 9, 1, 1, 1, 0, 0, 4, 16; total 32.
- $32/8 = 4$; $32/7 = 4.571$.

Why the other options fail: (A) gives the SDs. (C) swaps them: dividing by the smaller number gives the larger value.

Answer: D.

Q4.2 RECALL · about 15 s

For i.i.d. data, which statement about $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ is correct?

- (A) Dividing by n instead gives an unbiased estimator
- (B) The bias of the divide by n estimator grows with n
- (C) $E[s^2] = \sigma^2$
- (D) $E[s] = \sigma$ exactly

Solution

Fast route: The whole point of $n - 1$ is unbiasedness of s^2 .

Full working:

- $E[\sum (x_i - \bar{x})^2] = (n - 1)\sigma^2$.

Why the other options fail: (D) s is slightly biased low (square root is concave). (A) that estimator is biased by $\frac{n-1}{n}$. (B) the bias shrinks to zero.

Answer: C.

Q4.3 REASON · about 45 s

Session times have $\sigma = 20$ s. The SE of the mean with $n = 100$ users, and the number of users needed to make the SE 1 s, are

- (A) 20 s and 2000
- (B) 2 s and 200
- (C) 2 s and 400
- (D) 0.2 s and 400

Solution

Fast route: $\frac{20}{10} = 2$; halving needs 4 times the data.

Full working:

- $20/\sqrt{n} = 1 \Rightarrow \sqrt{n} = 20 \Rightarrow n = 400$.

Why the other options fail: (B) assumes the SE falls in proportion to n , not $\sqrt{n}$.

Answer: C.

Q4.4 REASON · about 45 s

A poll of 1,000 people finds 50% support. The 95% margin of error is closest to

- (A) 5%
- (B) 0.5%

- (C) 1.6%
(D) 3.1%

Solution

Fast route: $SE = \sqrt{0.25/1000} = 0.0158$; times 1.96.

Full working:

- $1.96 \times 0.0158 = 0.031$.

Why the other options fail: (C) is the SE, not the MOE.

Answer: D.

Q4.5 **CALC** · 1.5 to 2 min

How many respondents are needed to estimate a proportion within $\pm 3\%$ at 95% confidence when nothing is known about p ?

- (A) 2,401
(B) 385
(C) 1,067
(D) 1,068

Solution

Fast route: $p = 0.5$: $\frac{1.96^2 \times 0.25}{0.03^2} = 1067.1$, round up.

Full working:

- $\frac{3.8416 \times 0.25}{0.0009} = \frac{0.9604}{0.0009} = 1067.1$. Always round up: 1,068.

Why the other options fail: (C) rounds down (then the MOE slightly exceeds 3%). (B) is for $\pm 5\%$. (A) is for $\pm 2\%$.

Answer: D.

Q4.6 **CALC** · 1.5 to 2 min

Salaries have $\sigma = 15$ (thousand rupees). The sample size needed for a 95% CI for the mean with half width 2 is

- (A) 57
(B) 217
(C) 216
(D) 15

Solution

Fast route: $\left(\frac{1.96 \times 15}{2}\right)^2 = 14.7^2 = 216.1$, round up.

Full working:

- $14.7^2 = 216.09$; the next integer is 217.

Why the other options fail: (C) rounds down. (A) forgets to square.

Answer: B.

Q4.7 **REASON** · about 45 s

You sample $n = 200$ without replacement from $N = 1000$. The finite population correction factor is closest to

- (A) 0.895
- (B) 0.80
- (C) 0.95
- (D) 1.00

Solution

Fast route: $\sqrt{800/999} \approx \sqrt{0.8} = 0.894$.

Full working:

- $FPC = \sqrt{\frac{N-n}{N-1}} = 0.895$. Having seen 20% of the population leaves less to be uncertain about.

Why the other options fail: (B) forgets the square root.

Answer: A.

Q4.8 RECALL · about 15 s

A company has 5,000 freshers, 3,000 mid level and 2,000 senior employees. A stratified sample of 200 with proportional allocation takes

- (A) 67, 67, 66
- (B) 120, 50, 30
- (C) 100, 60, 40
- (D) 50, 75, 75

Solution

Fast route: Shares 50%, 30%, 20% of 200.

Full working:

- $n_h = 200 \times \frac{N_h}{10000}$.

Why the other options fail: (A) is equal allocation.

Answer: C.

Q4.9 REASON · about 45 s

Continuing, the stratum sample means of weekly hours on the portal are 20, 40 and 35. The stratified estimate of the population mean is

- (A) 29
- (B) 32.5
- (C) 35
- (D) 31.67

Solution

Fast route: $0.5(20) + 0.3(40) + 0.2(35)$.

Full working:

- $10 + 12 + 7 = 29$.

Why the other options fail: (D) is the unweighted average of the three means, which over weights the small senior stratum.

Answer: A.

Q4.10 RECALL · about 15 s

Which statement correctly contrasts stratified and cluster sampling?

- (A) Stratified samples from every group; cluster sampling selects whole groups and samples only some of them
- (B) Cluster sampling samples from every group; stratified selects whole groups
- (C) Cluster sampling is usually more statistically efficient than SRS of the same size
- (D) Stratification helps most when strata means are all equal

Solution

Fast route: Strata: from every group. Clusters: whole groups, some of them.

Full working:

- Cluster sampling is chosen for cost; with positive intra cluster correlation it is less efficient than SRS.

Why the other options fail: (C) the design effect exceeds 1. (D) stratification helps most when strata means *differ*.

Answer: A.

Q4.11 REASON · about 45 s

Cluster sampling with 20 students per college and intra cluster correlation $\rho = 0.1$. The design effect and the effective sample size of 400 students are

- (A) 2.9 and about 138
- (B) 2.0 and 200
- (C) 3.0 and about 133
- (D) 1.9 and about 210

Solution

Fast route: $1 + 19(0.1) = 2.9$; $400/2.9$.

Full working:

- $DEFF = 1 + (m - 1)\rho = 2.9$. Effective $n = 138$.

Why the other options fail: (C) uses m instead of $m - 1$.

Answer: A.

Q4.12 RECALL · about 15 s

A bank trains a loan default model using only loans that were approved in the past. The main problem is

- (A) Rejected applicants never received a label, so the training data is selection biased
- (B) The model will be too simple
- (C) Collecting 10 times more approved loans will remove the bias
- (D) The data violates the Central Limit Theorem

Solution

Fast route: Labels exist only for units that passed a filter related to risk: selection (survivorship) bias.

Full working:

- The model has never seen outcomes for profiles that used to be rejected (“reject inference”

problem).

Why the other options fail: (C) more biased data shrinks variance, not bias.

Answer: A.

Q4.13 **RECALL** · about 15 s

Returning WW2 bombers had bullet holes mostly on the wings and fuselage, few on the engines. The correct conclusion is to

- (A) collect more returning planes
- (B) armour the engines
- (C) armour the wings
- (D) armour nothing

Solution

Fast route: Planes hit in the engines did not return: survivorship bias.

Full working:

- The absence of engine holes among survivors means engine hits were fatal.

Answer: B.

Q4.14 **RECALL** · about 15 s

A simple random sample of 3 users is drawn from 10. The number of possible samples, and the probability that a particular user is included, are

- (A) 1000 and 0.3
- (B) 120 and 0.3
- (C) 120 and 0.1
- (D) 720 and 0.3

Solution

Fast route: $\binom{10}{3} = 120$; inclusion $\frac{n}{N} = 0.3$.

Full working:

- Samples containing a given user: $\binom{9}{2} = 36$; $\frac{36}{120} = 0.3$.

Why the other options fail: (D) counts ordered samples.

Answer: B.

Q4.15 **CALC** · 1.5 to 2 min

A batch of 20 laptops has 4 defective. An inspector picks 5 without replacement. The probability of no defective laptop is closest to

- (A) 0.20
- (B) 0.28
- (C) 0.41
- (D) 0.33

Solution

Fast route: $\frac{16}{20} \cdot \frac{15}{19} \cdot \frac{14}{18} \cdot \frac{13}{17} \cdot \frac{12}{16}$.

Full working:

- Product = 0.2817. Equivalently $\binom{16}{5} / \binom{20}{5} = \frac{4368}{15504}$.

Why the other options fail: (D) is with replacement: $0.8^5 = 0.328$.

Answer: B.

Q4.16 **RECALL** · about 15 s

In a bootstrap sample of size n drawn from n points, the fraction of original points that do **not** appear is, for large n , about

- (A) 0.368
- (B) 0.1
- (C) 0.5
- (D) 0.632

Solution

Fast route: $(1 - \frac{1}{n})^n \rightarrow e^{-1}$.

Full working:

- These are the out of bag points (36.8%); 63.2% appear at least once.

Why the other options fail: (D) is the fraction that *do* appear.

Answer: A.

Q4.17 **RECALL** · about 15 s

Which is **not** a valid use of the bootstrap?

- (A) A confidence interval for a model's test AUC
- (B) Correcting a biased sampling frame
- (C) A confidence interval for the median session time
- (D) Creating the training sets for bagging

Solution

Fast route: Resampling a biased sample reproduces the bias.

Full working:

- (C) and (A) are statistics without easy SE formulas, ideal for the bootstrap. (D) bagging means bootstrap aggregating.

Answer: B.

Q4.18 **REASON** · about 45 s

Using inverse transform sampling, a uniform draw $U = 0.5$ is converted into an Exponential($\lambda = 2$) sample. The value produced is closest to

- (A) 0.69
- (B) 0.50
- (C) 0.35
- (D) 0.25

Solution

Fast route: $x = -\frac{\ln(1-u)}{\lambda} = \frac{\ln 2}{2}$.

Full working:

- $\frac{0.693}{2} = 0.347$, which is the median of Exponential(2).

Why the other options fail: (B) is the mean. (A) forgets to divide by λ .

Answer: C.

Q4.19 **RECALL** · about 15 s

Reservoir sampling keeps $k = 10$ items from a stream. When the 50th item arrives, it enters the reservoir with probability

- (A) $1/10$
- (B) $10/49$
- (C) $10/50$
- (D) $1/50$

Solution

Fast route: Item i enters with probability $\frac{k}{i}$.

Full working:

- $\frac{10}{50} = 0.2$. This keeps every item's final inclusion probability at $\frac{k}{n}$.

Answer: C.

Q4.20 **RECALL** · about 15 s

For a NumPy array x and a pandas Series s holding the same numbers, which is true by default?

- (A) Both divide by $n - 1$
- (B) $\text{np.var}(x)$ divides by $n - 1$ and $s.\text{var}()$ divides by n
- (C) $\text{np.var}(x)$ divides by n and $s.\text{var}()$ divides by $n - 1$
- (D) Both divide by n

Solution

Fast route: NumPy: $\text{ddof}=0$. pandas: $\text{ddof}=1$.

Full working:

- So the two give different answers on small data. Pass ddof explicitly when it matters.

Answer: C.

Q4.21 **RECALL** · about 15 s

You build a model predicting next month's job applications from 3 years of daily data. The appropriate validation scheme is

- (A) an expanding window: train on the past, test on the following period, roll forward
- (B) a random 80/20 split of days
- (C) stratified k -fold on the target
- (D) leave one out cross validation on shuffled days

Solution

Fast route: Time series: never let the future into training.

Full working:

- Random or shuffled splits leak future information and neighbouring correlated days.

Answer: A.

Q4.22 REASON · about 45 s

A fraud model is trained after undersampling the negatives from 99:1 to 50:50. On the real population, its raw predicted fraud probabilities will be

- (A) always 0.5
- (B) too high
- (C) exactly calibrated
- (D) too low

Solution

Fast route: The model saw fraud 50% of the time instead of 1%.

Full working:

- Recalibrate using the sampling rate or on a natural held out sample. Ranking metrics like AUC are unaffected in expectation.

Answer: B.

Q4.23 RECALL · about 15 s

A list of 1,000 records is sorted so that every 10th record comes from the same branch office. Systematic sampling with $k = 10$ is used. The main risk is

- (A) a too small sample
- (B) it cannot be computed
- (C) it equals SRS
- (D) periodicity: the sample may contain only one branch

Solution

Fast route: The list's period equals the step.

Full working:

- One random start then picks the same position in every cycle.

Answer: D.

Q4.24 RECALL · about 15 s

A survey emailed to 10,000 users gets 800 replies. Which statement is **wrong**?

- (A) Non response bias is a serious concern
- (B) Comparing respondents and non respondents on known attributes helps assess bias
- (C) Weighting respondents to match the population can reduce bias
- (D) The margin of error computed from $n = 800$ fully describes the uncertainty

Solution

Fast route: MOE covers random error only, not bias.

Full working:

- With an 8% response rate, the responders are likely the more engaged users.

Answer: D.

Q4.25 **RECALL** · about 15 s

The standard deviation and the standard error of the mean differ because

- (A) the SD shrinks as $1/\sqrt{n}$
- (B) they are the same thing
- (C) the SE grows with sample size
- (D) the SD describes spread of individual values, the SE describes spread of the sample mean across repeated samples

Solution

Fast route: $SE = \frac{SD}{\sqrt{n}}$: estimate uncertainty, not data spread.

Full working:

- More data does not change how variable individuals are; it makes the average more precise.

Answer: D.

Q4.26 **CALC** · 1.5 to 2 min

Neyman allocation with two strata: $N_1 = 6000$, $\sigma_1 = 10$ and $N_2 = 4000$, $\sigma_2 = 30$. With $n = 300$ the allocation is

- (A) 180 and 120
- (B) 100 and 200
- (C) 150 and 150
- (D) 75 and 225

Solution

Fast route: $N_h\sigma_h$: 60000 vs 120000, a 1 : 2 split.

Full working:

- $n_1 = 300 \times \frac{1}{3} = 100$, $n_2 = 200$.

Why the other options fail: (A) is proportional allocation (ignores σ).

Answer: B.

References for this chapter

- [Scribbr: Sampling methods; Sampling bias.](#)
- [StatQuest: Standard deviation vs standard error; Bootstrapping.](#)
- [Wikipedia: Reservoir sampling; Survivorship bias; Bessel's correction.](#)

Chapter 5

Central Limit Theorem

Companion resource for this section: [3Blue1Brown: But what is the Central Limit Theorem?](#) (the best visual intuition available)

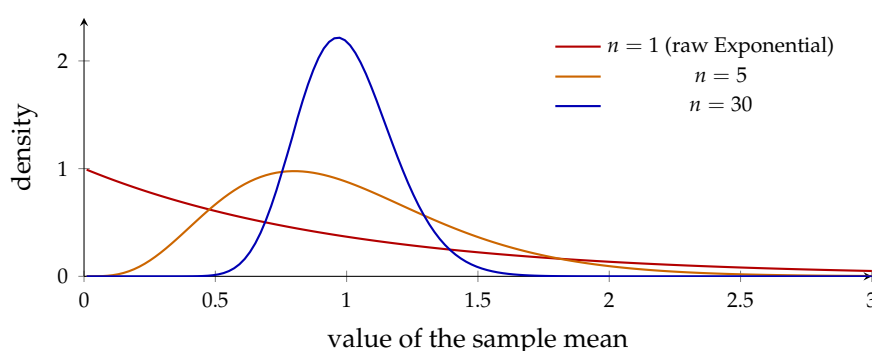
5.1 The surprise: averages forget where they came from

Start here: the everyday picture

- Roll one die: every face 1 to 6 is equally likely. The bar chart is flat, nothing like a bell.
- Now roll two dice and take their *average*. Averages near 3.5 are common (many combinations give them) and extremes like 1 or 6 are rare (only one combination each). The chart becomes a triangle.
- Average three dice, then ten. The chart turns into a smooth bell, centred at 3.5 and getting narrower.
- The original die was flat, but its averages became bell shaped anyway. That is the Central Limit Theorem (CLT): **averages of many independent things look normal, whatever the things themselves look like.**

5.1.1 Seeing it with two dice, by hand

- Count the ways to get each sum of two dice: sum 2 in 1 way, sum 3 in 2 ways, ..., sum 7 in 6 ways, ..., sum 12 in 1 way. The counts 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 form a triangle.
- Three dice: the counts for sums 3 to 10 are 1, 3, 6, 10, 15, 21, 25, 27: already curved like a bell.
- The spread of the average shrinks: one die has SD $\sqrt{35/12} = 1.71$; the average of n dice has SD $\frac{1.71}{\sqrt{n}}$: 1.21 for 2 dice, 0.99 for 3, 0.54 for 10.



The mean of n Exponential(1) values. The raw data is very skewed, yet by $n = 30$ the mean is nearly a symmetric bell centred at 1 with SD $1/\sqrt{30} = 0.18$.

5.2 The precise statement

Central Limit Theorem

Take n independent draws from **any** population with mean μ and finite variance σ^2 . For large n :

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{and} \quad S_n = X_1 + \cdots + X_n \approx N(n\mu, n\sigma^2).$$

Equivalently $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is approximately standard normal.

- Three facts packed inside:
 - **Centre**: the average is centred on the true mean μ (always true, even for small n).
 - **Spread**: its SD is $\frac{\sigma}{\sqrt{n}}$, the standard error (always true for independent draws).
 - **Shape**: it becomes normal (this is the part that needs large n).
- For **sums**, the mean grows like n and the SD like $\sqrt{n}$. Variances add, SDs do not.
- What the CLT does **not** say: it does not make the raw data normal. Salaries stay skewed forever; only their averages become bell shaped.
- If the population is itself normal, $\bar{X}$ is exactly normal for every n ; the CLT matters when the population is not normal.

5.3 When it works, and when it fails

- **Independence** (or only weak dependence) is needed. Daily active users on consecutive days are correlated; then $\frac{s}{\sqrt{n}}$ understates the true SE.
- **Finite variance** is needed. The **Cauchy** distribution has no mean and no variance: the average of n Cauchy values is again standard Cauchy, no matter how large n is. Averaging buys nothing.
- **How large is large?** The textbook “ $n \geq 30$ ” is a rule of thumb for mildly skewed data. Very skewed data (revenue per user, where a few “whales” dominate) may need thousands. The error of the normal approximation shrinks like $\frac{1}{\sqrt{n}}$ and is larger when the data is more skewed (Berry Esseen theorem).
- For proportions, check $np \geq 10$ and $n(1-p) \geq 10$.
- Not everything is covered: the *maximum* of a sample does not become normal (it follows extreme value distributions).

5.4 Law of Large Numbers vs Central Limit Theorem

- **LLN**: $\bar{X} \rightarrow \mu$ as n grows. It says *where* the average ends up.
- **CLT**: the error $\bar{X} - \mu$ is about $\frac{\sigma}{\sqrt{n}}$ in size and normally shaped. It says *how fast and in what shape* the average gets there.
- Casino picture: LLN says the house wins in the long run; CLT tells the house how much its nightly profit wobbles.
- **Gambler’s fallacy** is not the LLN: after 10 heads, tails is not “due”. Past excess heads are not cancelled; they are diluted by the many future tosses.

5.5 Using the CLT: a four step recipe

1. Decide what the question is about: a **mean**, a **sum**, or a **proportion**.
2. Write its centre and SD:
 - mean: centre μ , SD $\frac{\sigma}{\sqrt{n}}$;
 - sum: centre $n\mu$, SD $\sigma\sqrt{n}$;
 - proportion: centre p , SD $\sqrt{\frac{p(1-p)}{n}}$.
3. Standardise: $z = \frac{\text{value} - \text{centre}}{\text{SD}}$ and read $\Phi(z)$.
4. For whole number counts (heads, claims, dice sums) apply a **continuity correction** of ± 0.5 .
 - Example (sum): parcels weigh on average 5 kg with SD 1 kg. A truck loads 100. Total has centre 500 and SD $1 \times \sqrt{100} = 10$. $P(\text{total} > 515)$: $z = 1.5$, $P = 1 - 0.9332 = 0.0668$.
 - Example (mean, same question): mean weight must exceed 5.15; SD of the mean $\frac{1}{10} = 0.1$; $z = 1.5$. Same answer, as it must be.
 - Example (proportion): true click rate 20%, 100 impressions. SD of $\hat{p} = \sqrt{\frac{0.2 \times 0.8}{100}} = 0.04$. $P(\hat{p} > 0.28)$: $z = 2$, 0.0228.
 - Example (counts): a fair coin tossed 400 times; $P(190 \leq \text{heads} \leq 210)$. Mean 200, SD 10. With the correction use 189.5 to 210.5: $z = \pm 1.05$, $P = 2(0.8531) - 1 = 0.706$.

Try it now (answer right below, cover it first)

Scores have mean 60 and SD 12. For the mean of 36 students, find $P(\bar{X} < 58)$.

Answer: SD of the mean $= \frac{12}{6} = 2$; $z = -1$; $P = 0.1587$.

5.6 What the CLT powers

- **Confidence intervals:** $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ is justified by the CLT. $\bar{x} = 100$, $\sigma = 20$, $n = 100$: 100 ± 3.92 .
- **z and t tests** and A/B tests on conversion rates (Chapter 7).
- **Normal approximations:** $\text{Binomial}(n, p) \approx N(np, np(1-p))$ and $\text{Poisson}(\lambda) \approx N(\lambda, \lambda)$ for large n or λ .
- **Difference of two independent means:** $\text{SE} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ (variances add even for a difference).
- **Old trick:** $U_1 + \dots + U_{12} - 6$ with uniforms has mean 0 and variance $12 \times \frac{1}{12} = 1$: roughly standard normal.
- **Delta method:** a smooth function of an average is also roughly normal, with

$$\text{SE}(g(\bar{X})) \approx |g'(\mu)| \text{SE}(\bar{X}).$$

For $g = \ln$: $\text{SE of } \ln \bar{X} \approx \frac{\text{SE}(\bar{X})}{\mu}$, a relative error. Used for CIs on “percentage lift”.

Where this shows up in ML and at a company like Naukri

- **Mini batch SGD:** a batch gradient is an average of B per example gradients, so its noise has $\text{SD} \propto \frac{1}{\sqrt{B}}$. Going from batch 32 to 512 (16 times) only cuts noise by 4.
- **Reporting accuracy:** on 10,000 test examples at 90% accuracy, $\text{SE} = \sqrt{0.9 \times 0.1 / 10000} = 0.3$ percentage points. A 0.2 point gain is within noise.
- **Ensembles:** averaging many roughly independent models reduces variance.
- **Weight initialisation:** a neuron’s input is a sum of many weight times input terms, roughly normal with variance proportional to the number of inputs; Xavier and He ini-

tialisation scale weights by $\frac{1}{n_{\text{in}}}$ to keep that variance stable.

Formula sheet

$\bar{X} \approx N(\mu, \frac{\sigma^2}{n})$ $S_n \approx N(n\mu, n\sigma^2)$ $\hat{p} \approx N(p, \frac{p(1-p)}{n})$ $\text{SE}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ Binomial
 $\approx N(np, np(1-p))$ with ± 0.5 Poisson(λ) $\approx N(\lambda, \lambda)$ delta: $\text{SE}(g(\bar{X})) \approx |g'(\mu)|\text{SE}(\bar{X})$ CI:
 $\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$ $n = (\frac{z\sigma}{E})^2$.

15 second recall card: memorise every line

- CLT: the mean of many independent finite variance values is approximately normal with SD $\frac{\sigma}{\sqrt{n}}$, whatever the population shape. The data itself does not become normal.
- Needs independence and finite variance. Fails for Cauchy (the average stays Cauchy). Sample maximum is not normal.
- LLN: where the mean goes. CLT: the shape and size of its error.
- Sum of n : mean $n\mu$, SD $\sigma\sqrt{n}$. Variances add, SDs do not.
- n from 25 to 100: SE halves. Batch 32 to 512: gradient noise divides by 4.
- Normal population: $\bar{X}$ exactly normal for any n . Unknown σ with small n : t distribution.
- Positive autocorrelation: $\frac{s}{\sqrt{n}}$ is too small (effective n is lower).
- Sum of 12 uniforms minus 6: mean 0, variance 1.

5.7 Question bank: Central Limit Theorem

No InfoEdge PYQ found for this topic

No public InfoEdge report names a CLT question, but the reported “sum of two independent normals” and “p-value” items rest on it. Questions follow the reported OA style.

Q5.1 RECALL · about 15 s

Which statement about the Central Limit Theorem is correct?

- (A) The CLT requires the population to be normal
- (B) The SD of the sample mean is σ/n
- (C) For large n the data itself becomes normally distributed
- (D) For large n the sample mean is approximately normal whatever the population shape, provided the variance is finite

Solution

Fast route: It is about the *mean*, needs finite variance, and the SE is $\sigma/\sqrt{n}$.

Full working:

- The shape of the sampling distribution of $\bar{X}$ becomes normal.

Why the other options fail: (C) the raw data keeps its shape. (A) if the population were normal we would not need the CLT. (B) the SE uses $\sqrt{n}$.

Answer: D.

Q5.2 REASON · about 45 s

Scores have mean 70 and SD 8. A random sample of 16 students is drawn. $P(\bar{X} > 72)$ is closest to

- (A) 0.3085
- (B) 0.40
- (C) 0.1587
- (D) 0.0228

Solution

Fast route: $SE = \frac{8}{4} = 2$; 72 is one SE above: 0.1587.

Full working:

- $z = \frac{72-70}{2} = 1$; $1 - 0.8413$.

Why the other options fail: (B) uses the SD of one student, $z = 0.25$.

Answer: C.

Q5.3 REASON · about 45 s

Parcel weights have mean 5 kg and SD 1 kg. The probability that 100 parcels weigh more than 520 kg in total is closest to

- (A) 0.0228
- (B) 0.0013
- (C) 0.4207
- (D) 0.1587

Solution

Fast route: Total: mean 500, $SD \sqrt{100} = 10$; $z = 2$.

Full working:

- $P(Z > 2) = 0.0228$.

Why the other options fail: (C) uses $SD 100 \times 1 = 100$ (adding SDs instead of variances): $z = 0.2$.

Answer: A.

Q5.4 CALC · 1.5 to 2 min

An elevator's limit is 1,250 kg. Adult weights have mean 75 kg and SD 10 kg. If 16 adults board, the probability the limit is exceeded is closest to

- (A) 0.0228
- (B) 0.2266
- (C) 0.1056
- (D) 0.0062

Solution

Fast route: Total mean 1200, $SD 10 \times 4 = 40$; $z = \frac{50}{40} = 1.25$.

Full working:

- $\Phi(1.25) = 0.8944$; $1 - 0.8944 = 0.1056$.
- Same via the mean: $\frac{1250}{16} = 78.125$; SD of mean 2.5; $z = 1.25$.

Answer: C.

Q5.5 REASON · about 45 s

A job post's true click through rate is 20%. It is shown 100 times. The probability that the observed

CTR exceeds 28% is closest to

- (A) 0.28
- (B) 0.1587
- (C) 0.0228
- (D) 0.0062

Solution

Fast route: $SE = \sqrt{0.16/100} = 0.04$; $\frac{0.08}{0.04} = 2$.

Full working:

- $np = 20$ and $n(1 - p) = 80$ are both at least 10, so the normal approximation is fine.

Answer: C.

Q5.6 CALC · 1.5 to 2 min

A fair die is rolled 100 times. Using a continuity correction, $P(\text{sum} \geq 380)$ is closest to

- (A) 0.042
- (B) 0.004
- (C) 0.36
- (D) 0.16

Solution

Fast route: Mean 350, SD $\sqrt{100 \times 2.917} = 17.08$; $z = \frac{29.5}{17.08} \approx 1.73$.

Full working:

- One roll: $\mu = 3.5$, $\sigma^2 = \frac{35}{12}$. Sum: variance 291.7.
- “ ≥ 380 ” becomes “ > 379.5 ”. $1 - \Phi(1.73) \approx 0.042$.

Answer: A.

Q5.7 CALC · 1.5 to 2 min

A fair coin is tossed 400 times. The probability that the number of heads is between 190 and 210 inclusive is closest to

- (A) 0.68
- (B) 0.50
- (C) 0.71
- (D) 0.95

Solution

Fast route: Mean 200, SD 10; with correction ± 10.5 : $z = \pm 1.05$.

Full working:

- $2\Phi(1.05) - 1 = 2(0.8531) - 1 = 0.706$.

Why the other options fail: (A) skips the continuity correction ($z = \pm 1$).

Answer: C.

Q5.8 CALC · 1.5 to 2 min

Waiting times are Exponential with mean 1 minute. For the mean of 50 waiting times, $P(\bar{X} > 1.2)$ is closest to

- (A) 0.08
- (B) 0.30
- (C) 0.42
- (D) 0.01

Solution

Fast route: Exponential SD = mean = 1; $SE = \frac{1}{\sqrt{50}} = 0.141$; $z = 1.41$.

Full working:

- $1 - \Phi(1.41) \approx 0.079$. (The exact Gamma answer is about 0.084: the CLT is an approximation.)

Why the other options fail: (C) uses the SD of a single wait instead of the SE ($z = 0.2$ gives 0.42).

Answer: A.

Q5.9 REASON · about 45 s

$\bar{x} = 100$, $\sigma = 20$, $n = 100$. The 95% confidence interval for μ is

- (A) (98, 102)
- (B) (96.08, 103.92)
- (C) (94.85, 105.15)
- (D) (60.8, 139.2)

Solution

Fast route: $SE = 2$; $\pm 1.96 \times 2 = \pm 3.92$.

Full working:

- 100 ± 3.92 .

Why the other options fail: (A) is ± 1 SE (68%). (D) uses σ instead of the SE. (C) uses 2.576 (99%).

Answer: B.

Q5.10 REASON · about 45 s

What sample size makes a 95% CI half width at most 1 if $\sigma = 5$?

- (A) 97
- (B) 25
- (C) 100
- (D) 96

Solution

Fast route: $(1.96 \times 5)^2 = 96.04$; round up.

Full working:

- $\frac{9.8}{\sqrt{n}} \leq 1 \Rightarrow n \geq 96.04$, so 97.

Answer: A.

Q5.11 RECALL · about 15 s

You average n i.i.d. standard Cauchy variables. As n grows, the distribution of the average

- (A) collapses to 0

- (B) approaches $N(0, 1/n)$
- (C) approaches $N(0, 1)$
- (D) stays standard Cauchy for every n

Solution

Fast route: No finite mean or variance: LLN and CLT both fail.

Full working:

- The characteristic function argument shows the mean of n standard Cauchys is exactly standard Cauchy.

Answer: D.

Q5.12 **RECALL** · about 15 s

Which pairing is correct?

- (A) LLN: the sample becomes normal
- (B) CLT: the sample mean converges to the median
- (C) LLN: after many heads, tails becomes more likely
- (D) LLN: the sample mean converges to μ ; CLT: the scaled error is approximately normal

Solution

Fast route: LLN = where; CLT = shape of the error.

Full working:

- $\sqrt{n}(\bar{X} - \mu)/\sigma \rightarrow N(0, 1)$ in distribution.

Why the other options fail: (C) is the gambler's fallacy.

Answer: D.

Q5.13 **REASON** · about 45 s

Two independent groups: $\sigma_1 = 10$, $n_1 = 50$ and $\sigma_2 = 12$, $n_2 = 60$. The SE of $\bar{X}_1 - \bar{X}_2$ is closest to

- (A) 2.10
- (B) 0.35
- (C) 4.40
- (D) 1.34

Solution

Fast route: $\sqrt{\frac{100}{50} + \frac{144}{60}} = \sqrt{2 + 2.4}$.

Full working:

- $\sqrt{4.4} = 2.10$. Variances add even for a difference.

Why the other options fail: (C) forgets the square root. (D) subtracts the variances.

Answer: A.

Q5.14 **CALC** · 1.5 to 2 min

An insurer has 10,000 independent policies, each with claim probability 0.01. The reserve covers 120 claims. The probability that claims exceed the reserve is closest to

- (A) 0.10

- (B) 0.002
- (C) 0.20
- (D) 0.02

Solution

Fast route: Mean 100, SD $\sqrt{99} \approx 9.95$; $z = \frac{20.5}{9.95} \approx 2.06$.

Full working:

- “More than 120” = “at least 121” $\rightarrow > 120.5$ with the correction.
- $1 - \Phi(2.06) = 0.0197$.

Answer: D.

Q5.15 REASON · about 45 s

Applications arrive at a Poisson rate of 4 per hour. Ignoring the continuity correction, the probability of more than 440 applications in 100 hours is closest to

- (A) 0.0013
- (B) 0.0228
- (C) 0.1587
- (D) 0.3085

Solution

Fast route: Poisson(400): mean 400, SD 20; $z = 2$.

Full working:

- For a Poisson, variance = mean, so SD = $\sqrt{400} = 20$.

Answer: B.

Q5.16 RECALL · about 15 s

$Y = U_1 + \dots + U_{12} - 6$ with U_i i.i.d. Uniform(0, 1). The mean and variance of Y are

- (A) 0 and 1
- (B) 0 and 12
- (C) 6 and 1
- (D) 0 and 1/12

Solution

Fast route: $12 \times \frac{1}{2} - 6 = 0$; $12 \times \frac{1}{12} = 1$.

Full working:

- By the CLT Y is roughly standard normal: an old way to simulate $N(0, 1)$.

Answer: A.

Q5.17 RECALL · about 15 s

The noise (SD) of a mini batch gradient with batch size 32 is s . With batch size 512 it is about

- (A) $s/2$
- (B) $s/4$
- (C) $s/16$

(D) s **Solution****Fast route:** Noise $\propto 1/\sqrt{B}$; $\sqrt{16} = 4$.**Full working:**

- 16 times the batch, 4 times less noise.

Answer: B.**Q5.18 RECALL · about 15 s**

Revenue per user is extremely right skewed. With 5,000 users per arm, which claim is **wrong**?

- (A) A few very large spenders can flip the sign of the observed difference
- (B) Winsorising or a log transform can stabilise the comparison
- (C) The CLT guarantees the t test is exact whenever $n > 30$
- (D) The normal approximation of the mean may still be poor

Solution**Fast route:** “30” is a rule of thumb for mild skew, never a guarantee.**Full working:**

- Approximation error grows with skewness (Berry Esseen). Bootstrap CIs and robust metrics help.

Answer: C.**Q5.19 RECALL · about 15 s**

You compute $s/\sqrt{60}$ as the SE of the mean of 60 consecutive days of daily active users, which are strongly positively autocorrelated. This SE is most likely

- (A) zero
- (B) correct
- (C) too large
- (D) too small

Solution**Fast route:** Correlated days carry overlapping information: effective $n < 60$.**Full working:**

- For an AR(1) with coefficient ρ , $n_{\text{eff}} \approx n \frac{1-\rho}{1+\rho}$.

Answer: D.**Q5.20 REASON · about 45 s**

Delta method: $\bar{X}$ is the mean of $n = 100$ values with $\mu = 10$ and $\sigma = 5$. The approximate SE of $\ln \bar{X}$ is

- (A) 0.005
- (B) 0.05
- (C) 0.23
- (D) 0.5

Solution

Fast route: $SE(\bar{X}) = 0.5$; multiply by $g'(\mu) = \frac{1}{10}$.

Full working:

- 0.05: a 5% relative uncertainty.

Answer: B.

Q5.21 RECALL · about 15 s

If the population is normal with known σ , the mean of $n = 4$ observations is

- (A) uniform
- (B) t distributed
- (C) exactly normal
- (D) approximately normal only if $n \geq 30$

Solution

Fast route: Linear combinations of independent normals are exactly normal.

Full working:

- The t appears only when σ is replaced by s .

Answer: C.

Q5.22 REASON · about 45 s

A classifier has 90% accuracy on 10,000 test examples. The standard error of this accuracy is closest to

- (A) 3 percentage points
- (B) 0.3 percentage points
- (C) 0.9 percentage points
- (D) 0.03 percentage points

Solution

Fast route: $\sqrt{0.9 \times 0.1/10000} = \frac{0.3}{100}$.

Full working:

- 0.003, i.e. 0.3 points. A 0.2 point gain is inside the noise.

Answer: B.

Q5.23 RECALL · about 15 s

As the sample size increases from 25 to 100, the sampling distribution of $\bar{X}$

- (A) keeps the same centre and its SD quarters
- (B) keeps the same centre and its SD halves
- (C) keeps the same SD
- (D) moves towards the population median

Solution

Fast route: $\frac{\sigma}{5} \rightarrow \frac{\sigma}{10}$.

Full working:

- The centre is μ for every n .

Answer: B.

Q5.24 **RECALL** · about 15 s

Which quantity is **not** approximately normal in large samples by a Central Limit argument?

- (A) The sample maximum of n uniform values
- (B) The sample mean of exponential waiting times
- (C) The OLS regression slope estimate
- (D) The sample proportion $\hat{p}$

Solution

Fast route: Extremes follow extreme value laws, not the CLT.

Full working:

- $\hat{p}$ is a mean of Bernoullis; OLS slopes are weighted sums of errors; both are asymptotically normal.

Answer: A.

Q5.25 **CALC** · 1.5 to 2 min

$X \sim \text{Binomial}(50, 0.4)$. Using the normal approximation with continuity correction, $P(X \leq 15)$ is closest to

- (A) 0.15
- (B) 0.20
- (C) 0.07
- (D) 0.10

Solution

Fast route: Mean 20, SD $\sqrt{12} = 3.46$; $z = \frac{15.5-20}{3.46} = -1.30$.

Full working:

- $\Phi(-1.30) = 0.0968$.

Why the other options fail: (C) skips the correction ($z = -1.44$ gives 0.075).

Answer: D.

References for this chapter

- [3Blue1Brown: But what is the Central Limit Theorem?](#)
- [StatQuest: The Central Limit Theorem.](#)
- [Scribbr: Central limit theorem.](#)
- [Wikipedia: Berry Esseen theorem.](#)

Chapter 6

Correlations

Companion resource for this section: [StatQuest: Pearson's Correlation, Clearly Explained](#) (then Covariance and R-squared from the same channel)

6.1 Do two things move together?

Start here: the everyday picture

- Plot each candidate as a dot: years of experience across, salary up. The cloud slopes upwards: more experience tends to come with more salary. Plot hours of sleep against typing errors: the cloud slopes down. Plot shoe size against exam marks: no pattern.
- Correlation turns “the cloud slopes up, down, or not at all, and how tightly” into one number between -1 and $+1$.

6.2 Covariance: built from scratch

- Step 1: for every point, measure how far x is from its mean and how far y is from its mean.
- Step 2: multiply the two distances. If both are above average, or both below, the product is **positive**. If one is above and the other below, it is **negative**.
- Step 3: average the products. Mostly positive products mean the variables rise together.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$

The sample version divides the sum of products by $n - 1$.

- Worked example: $x = 1, 2, 3, 4, 5$ and $y = 2, 4, 5, 4, 5$. Means 3 and 4.

x	y	$x - \bar{x}$	$y - \bar{y}$	product	$(x - \bar{x})^2, (y - \bar{y})^2$
1	2	-2	-2	4	4, 4
2	4	-1	0	0	1, 0
3	5	0	1	0	0, 1
4	4	1	0	0	1, 0
5	5	2	1	2	4, 1
				$S_{xy} = 6$	$S_{xx} = 10, S_{yy} = 6$

Sample covariance = $\frac{6}{4} = 1.5$.

- The problem with covariance: it has units (years $\times$ rupees) and its size depends on the scale. Measure salary in thousands instead of rupees and the covariance shrinks 1,000 times though nothing real changed. $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$.
- Useful identities: $\text{Cov}(X, X) = \text{Var}(X)$; $\text{Var}(X \pm Y) = \text{Var}X + \text{Var}Y \pm 2\text{Cov}(X, Y)$.

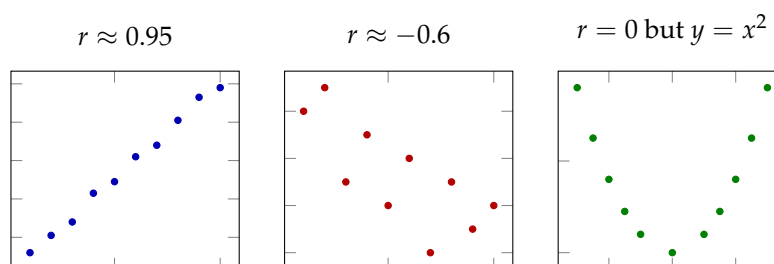
6.3 Pearson correlation r : covariance without units

- Divide the covariance by both SDs:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

Stuck on this page? See: [StatQuest: Pearson's correlation, clearly explained](#)

- Example: $r = \frac{6}{\sqrt{10 \times 6}} = \frac{6}{7.746} = 0.775$.
- Always $-1 \leq r \leq 1$. $r = \pm 1$ only when the points lie exactly on a line. A value like 1.02 is impossible: an instant elimination in the exam.
- Unit free: shifting or rescaling by a positive number does not change r ; multiplying one variable by a negative number flips its sign. $\text{corr}(3X + 5, -2Y + 1) = -\text{corr}(X, Y)$.
- Geometry: after centring, r is the cosine of the angle between the two data vectors.
- Reading values: about 0.1 weak, 0.3 moderate, 0.5 and above strong (rough guide, depends on the field).



6.3.1 What r cannot see

- r measures only **straight line** association. With $x = -2, -1, 0, 1, 2$ and $y = x^2$, the covariance is 0, so $r = 0$, although y is completely determined by x . **Zero correlation does not mean independence.**
- One outlier can create or destroy a correlation. **Anscombe's quartet**: four datasets with the same means, variances, $r = 0.816$ and regression line, but completely different pictures. Always plot.

6.4 Correlation and the regression line

- The least squares line $\hat{y} = a + bx$ has slope $b = r \frac{s_y}{s_x} = \frac{S_{xy}}{S_{xx}}$ and passes through $(\bar{x}, \bar{y})$, so $a = \bar{y} - b\bar{x}$. Example: $b = \frac{6}{10} = 0.6, a = 4 - 1.8 = 2.2$.
- Standardised reading: if x is 1 SD above its mean, the prediction is r SDs above the mean of y . With $r = 0.5, \bar{x} = 10, s_x = 2, \bar{y} = 30, s_y = 8$: at $x = 12, \hat{y} = 30 + 0.5 \times 8 = 34$.
- $R^2 = r^2$ in simple regression: the fraction of the variance of y explained by the line. $r = 0.7$ explains only 49%.
- Regressing y on x and x on y give different lines; their slopes multiply to r^2 .
- **Regression to the mean**: because $|r| < 1$, extreme values are followed by less extreme ones on average (a student 2 SD above on test 1 is predicted $2r$ SD above on test 2).
- **Is r significant?** Under $H_0 : \rho = 0, t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ with $n - 2$ df. With $r = 0.5$ and $n = 27$: $t = \frac{0.5 \times 5}{0.866} = 2.89 > 2.06$: significant. With a million users even $r = 0.01$ is significant but useless: significance is not strength.

6.5 Rank correlations and other relatives

- **Spearman ρ_s** : replace each value by its rank, then compute Pearson on the ranks. It captures any **monotonic** relationship ($y = e^x$ gives $\rho_s = 1$) and is robust to outliers. Without ties: $\rho_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$, where d_i is the rank difference.

- Example: two interviewers rank 5 candidates as 1, 2, 3, 4, 5 and 3, 1, 4, 5, 2. $d = -2, 1, -1, -1, 3$, $\sum d^2 = 16$, $\rho_s = 1 - \frac{96}{120} = 0.2$.
- **Kendall τ** : look at every pair of items. A pair is **concordant** if both rankings order it the same way. $\tau = \frac{C-D}{\binom{n}{2}}$. Orderings (1, 2, 3, 4) and (1, 3, 2, 4): 6 pairs, one discordant: $\tau = \frac{5-1}{6} = \frac{2}{3}$.
- **Phi (ϕ)** for two yes/no variables with table $\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}$: $\phi = \frac{ad-bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}$. **Point biserial**: Pearson between a yes/no and a number.
- **Partial correlation**: what is left after removing a third variable z from both:

$$r_{xy \cdot z} = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}.$$

A large drop suggests z is a confounder.

- **Mutual information**: zero if and only if the variables are independent; it catches $y = x^2$.

6.6 Correlation is not causation, and other traps

- **Confounding**: ice cream sales and drownings rise together because summer drives both.
- **Reverse causality**: active users are satisfied, or satisfied users become active?
- **Simpson's paradox**: a trend inside every group reverses when the groups are pooled, because group sizes differ. A new model can be better on mobile and on desktop, yet worse overall, if it received more of the low converting mobile traffic.
- **Range restriction**: among hired employees (all high scorers), test score and job rating correlate weakly, although among all applicants they correlate strongly. Restricting the range of x shrinks $|r|$.
- **Spurious trends**: two series that both grow over time (a company's revenue and smartphone sales) correlate at 0.95. Correlate the month to month changes instead.

6.7 Heteroscedasticity (a reported InfoEdge OA and interview topic)

- Words: "homo" = same, "hetero" = different, "scedastic" = scatter.
- **Homoscedasticity**: the spread of the residuals (errors) is the same at every value of x . **Heteroscedasticity**: the spread changes with x , typically a **funnel** in the residual vs fitted plot.
- Everyday example: predicting monthly spending from income. Low income people all spend similar amounts; high income people vary enormously. Errors grow with income.
- Consequences for ordinary least squares (OLS): the coefficient estimates stay **unbiased**, but the usual **standard errors are wrong**, so t tests, p -values and confidence intervals mislead. OLS is also no longer the most efficient estimator.
- Detection: residual vs fitted plot; Breusch Pagan test; White test.
- Fixes: transform y (log, square root, Box Cox); **weighted least squares** (weights $\propto \frac{1}{\text{variance}}$); **robust (White, HC) standard errors**; a model whose variance depends on the mean (Poisson, Gamma GLMs). More data from the same process does not fix it.

6.8 Multicollinearity and VIF (a reported InfoEdge interview topic)

- **Multicollinearity**: predictors are strongly correlated with each other (age and years of experience; total pay and basic pay).
- Effect: the model cannot tell which of two near copies deserves the credit, so individual coefficients become unstable (large SEs, strange signs), though predictions can remain fine.

- **VIF** (variance inflation factor) for predictor j : regress x_j on all the other predictors, get R_j^2 , then $VIF_j = \frac{1}{1-R_j^2}$. Its square root is the factor by which the coefficient's SE is inflated.
- $R_j^2 = 0.8 \Rightarrow VIF = 5$ ($SE \times 2.24$); $R_j^2 = 0.9 \Rightarrow 10$. Common thresholds: 5 or 10.
- Fixes: drop or combine features, PCA, **ridge regression**, more varied data. Tree models predict fine but split feature importance between correlated features.

6.9 Two more diagnostics

- **Durbin Watson** checks whether consecutive residuals are correlated (time series): $DW \approx 2(1 - \hat{\rho})$. 2 means none, near 0 strong positive, near 4 strong negative. $DW = 0.8$ gives $\hat{\rho} \approx 0.6$.
- A **valid correlation matrix** must be symmetric, have 1s on the diagonal, entries in $[-1, 1]$, and be **positive semidefinite** (all eigenvalues ≥ 0). You cannot have x_1, x_2 at $+0.9$, x_2, x_3 at $+0.9$ and x_1, x_3 at -0.9 .

Where this shows up in ML and at a company like Naukri

- Feature selection drops one of two features with $|r| > 0.95$; a low r with the target does not prove a feature useless (non linear effects, interactions).
- Ensembles: two models with error variance σ^2 and error correlation ρ average to $\sigma^2 \frac{1+\rho}{2}$. Random Forests lower ρ by random feature subsets.
- Cosine similarity between centred embeddings is Pearson correlation.
- A suspiciously high correlation between a feature and the label often signals target leakage.

Formula sheet

$$\begin{aligned} \text{Cov} &= \frac{S_{xy}}{n-1} & r &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} & b &= r \frac{s_y}{s_x} & R^2 &= r^2 & b_{yx}b_{xy} &= r^2 & t &= \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} & \rho_s &= \\ 1 - \frac{6\sum d^2}{n(n^2-1)} & \tau &= \frac{C-D}{\binom{n}{2}} & \text{Var}(X \pm Y) &= \sigma_X^2 + \sigma_Y^2 \pm 2r\sigma_X\sigma_Y & \text{VIF} &= \frac{1}{1-R_j^2} & DW &\approx 2(1-\rho) \\ \phi &= \frac{ad-bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}. \end{aligned}$$

15 second recall card: memorise every line

- $|r| \leq 1$; r is unit free; a negative scale flips its sign.
- $r = 0$ does not mean independent ($y = x^2$). Correlation is not causation.
- $R^2 = r^2$: $r = 0.7$ explains 49%.
- Spearman: Pearson on ranks, monotonic, robust; $y = e^x$ gives 1. Kendall: concordant minus discordant pairs.
- Heteroscedasticity: non constant residual variance, funnel plot. OLS estimates stay unbiased; SEs are wrong. Fix: log/Box Cox, WLS, robust SEs.
- Multicollinearity: unstable coefficients, predictions fine. $VIF = \frac{1}{1-R^2}$: 0.8 gives 5, 0.9 gives 10. Fix: drop, combine, ridge, PCA.
- Simpson's paradox: subgroup trend reverses when pooled. Range restriction shrinks r .
- Durbin Watson 2 means no autocorrelation.
- Correlation matrices are positive semidefinite.

6.10 Question bank: Correlations

Q6.1 RECALL · about 15 s

★ **PYQ (OA: homoscedasticity vs heteroscedasticity)** In a linear regression, which statement about heteroscedasticity is correct?

- (A) It makes the usual OLS standard errors unreliable, while the coefficient estimates remain unbiased
- (B) It cannot be detected from data
- (C) It makes the OLS coefficient estimates biased
- (D) It means the residuals have constant variance at all fitted values

Solution

Fast route: Unequal scatter hurts the SEs, not the unbiasedness of the estimates.

Full working:

- Unbiasedness needs $E[\varepsilon | X] = 0$, not equal variance. The textbook SE formula assumes one common σ^2 , so it is wrong under heteroscedasticity.

Why the other options fail: (C) is the classic misconception. (D) defines homoscedasticity. (B) residual plots and the Breusch Pagan test detect it.

Answer: A.

Q6.2 RECALL · about 15 s

★ **PYQ (INT: how do you fix heteroscedasticity?)** The residual spread of a salary model grows with the predicted salary. Which is **not** a remedy?

- (A) Heteroscedasticity robust (White) standard errors
- (B) Adding more rows from the same distribution
- (C) Weighted least squares with weights inversely proportional to the error variance
- (D) Model $\log(\text{salary})$ instead of salary

Solution

Fast route: More of the same data keeps the same variance pattern.

Full working:

- Log turns multiplicative errors into constant ones; WLS gives precise points more weight; robust SEs fix the inference.

Answer: B.

Q6.3 REASON · about 45 s

★ **PYQ (INT: multicollinearity and VIF)** Regressing feature x_3 on all the other features gives $R^2 = 0.8$. The VIF of x_3 , and the factor by which its coefficient's SE is inflated, are

- (A) 0.2 and 0.45
- (B) 5 and 5
- (C) 1.25 and 1.12
- (D) 5 and 2.24

Solution

Fast route: $\frac{1}{1-0.8} = 5$; SE inflation $\sqrt{5}$.

Full working:

- VIF inflates the *variance*; the SE is its square root: 2.24.

Why the other options fail: (C) uses $\frac{1}{0.8}$. (B) forgets the square root for the SE.

Answer: D.

Q6.4 **CALC** · 1.5 to 2 min

For $x = 1, 2, 3, 4, 5$ and $y = 2, 4, 5, 4, 5$, the sample covariance and Pearson r are

- (A) 1.2 and 0.775
- (B) 1.5 and 0.775
- (C) 6 and 0.775
- (D) 1.5 and 0.6

Solution

Fast route: $S_{xy} = 6$, $S_{xx} = 10$, $S_{yy} = 6$ (table in Section 6.2).

Full working:

- Covariance $\frac{6}{n-1} = 1.5$. $r = \frac{6}{\sqrt{60}} = 0.775$.

Why the other options fail: (A) divides by n . (D) gives r^2 . (C) is S_{xy} , not the covariance.

Answer: B.

Q6.5 **REASON** · about 45 s

Using the same data, the least squares line and its R^2 are

- (A) $\hat{y} = 2 + 0.6x$, $R^2 = 0.775$
- (B) $\hat{y} = 1 + x$, $R^2 = 0.5$
- (C) $\hat{y} = 2.2 + 0.6x$, $R^2 = 0.6$
- (D) $\hat{y} = 2.2 + 0.775x$, $R^2 = 0.6$

Solution

Fast route: $b = \frac{6}{10}$; line through $(3, 4)$; $R^2 = r^2$.

Full working:

- $a = 4 - 0.6 \times 3 = 2.2$. $R^2 = 0.775^2 = 0.6$.

Why the other options fail: (D) uses r as the slope, which is only right when $s_x = s_y$.

Answer: C.

Q6.6 **RECALL** · about 15 s

$\text{corr}(X, Y) = 0.6$. Then $\text{corr}(3X + 5, -2Y + 1)$ equals

- (A) -0.36
- (B) 0.6
- (C) -3.6
- (D) -0.6

Solution

Fast route: Shifts do nothing; a negative scale flips the sign; size unchanged.

Full working:

- $\text{corr}(aX + b, cY + d) = \text{sign}(ac) \text{corr}(X, Y).$

Answer: D.

Q6.7 REASON · about 45 s

$r = 0.5$, $\bar{x} = 10$, $s_x = 2$, $\bar{y} = 30$, $s_y = 8$. The least squares prediction of y at $x = 12$ is

- (A) 31
- (B) 34
- (C) 32
- (D) 38

Solution

Fast route: x is 1 SD up, so $\hat{y}$ is $r = 0.5$ SD up: $30 + 4$.

Full working:

- Slope $b = 0.5 \times \frac{8}{2} = 2$; $\hat{y} = 30 + 2 \times 2 = 34$.

Why the other options fail: (D) ignores regression to the mean (assumes $r = 1$).

Answer: B.

Q6.8 RECALL · about 15 s

The correlation between hours studied and exam score is 0.7. Which statement is correct?

- (A) About 49% of the variation in scores is explained by a linear function of hours
- (B) Studying more causes higher scores
- (C) 70% of students improved
- (D) 70% of the variation in scores is explained

Solution

Fast route: Variance explained is r^2 .

Full working:

- $0.7^2 = 0.49$.

Why the other options fail: (B) correlation cannot establish causation. (D) confuses r with r^2 .

Answer: A.

Q6.9 RECALL · about 15 s

X takes values $-2, -1, 0, 1, 2$ with equal probability and $Y = X^2$. Which is true?

- (A) $\text{Cov}(X, Y) = 0$ but X, Y are dependent
- (B) $\text{Cov}(X, Y) = 0$ and X, Y are independent
- (C) $\text{Cov}(X, Y) > 0$
- (D) $r = 1$ because Y is a function of X

Solution

Fast route: Symmetric X with $Y = X^2$: zero correlation, perfect dependence.

Full working:

- $E[X] = 0$, $E[XY] = E[X^3] = 0$, so $\text{Cov} = 0$. But knowing X tells you Y exactly.

Answer: A.

Q6.10 CALC · 1.5 to 2 min

Two interviewers rank 5 candidates. Interviewer A: 1, 2, 3, 4, 5. Interviewer B: 3, 1, 4, 5, 2. Spearman's ρ_s is

- (A) 0.8
- (B) 0.2
- (C) 0.5
- (D) -0.2

Solution

Fast route: $\sum d^2 = 16$; $1 - \frac{6 \times 16}{5 \times 24}$.

Full working:

- $d = -2, 1, -1, -1, 3$; squares 4, 1, 1, 1, 9; $1 - \frac{96}{120} = 0.2$.

Why the other options fail: (A) is $\frac{96}{120}$, forgetting to subtract from 1.

Answer: B.

Q6.11 REASON · about 45 s

Kendall's τ between the orderings (1, 2, 3, 4) and (1, 3, 2, 4) is

- (A) $2/3$
- (B) $5/6$
- (C) $1/2$
- (D) 1

Solution

Fast route: 6 pairs, 1 discordant: $\frac{5-1}{6}$.

Full working:

- Only items 2 and 3 are swapped.

Why the other options fail: (B) is the share of concordant pairs, not τ .

Answer: A.

Q6.12 RECALL · about 15 s

For $y = e^x$ on any set of distinct x values, the Spearman correlation is

- (A) exactly 1
- (B) 0
- (C) less than Pearson
- (D) undefined

Solution

Fast route: Monotone increasing: the ranks are identical.

Full working:

- Pearson would be below 1 because the relationship is curved.

Answer: A.

Q6.13 CALC · 1.5 to 2 min

A correlation $r = 0.5$ is computed from $n = 27$ pairs. The t statistic for $H_0 : \rho = 0$, and the decision at 5% ($t_{0.025,25} = 2.06$), are

- (A) 1.44, do not reject
- (B) 2.50, reject
- (C) 1.73, do not reject
- (D) 2.89, reject

Solution

Fast route: $\frac{0.5 \times 5}{\sqrt{0.75}} = \frac{2.5}{0.866}$.

Full working:

- $t = 2.887$ with 25 df; exceeds 2.06; $p \approx 0.008$.

Why the other options fail: (B) forgets the $\sqrt{1 - r^2}$ denominator.

Answer: D.

Q6.14 REASON · about 45 s

$\text{Var}(X) = 16$, $\text{Var}(Y) = 9$, $\text{corr}(X, Y) = 0.5$. $\text{Var}(X + Y)$ and $\text{Var}(X - Y)$ are

- (A) 31 and 19
- (B) 37 and 25
- (C) 37 and 13
- (D) 25 and 25

Solution

Fast route: $\text{Cov} = 0.5 \times 4 \times 3 = 6$; 25 ± 12 .

Full working:

- Positive correlation makes sums noisier and differences quieter (the idea behind paired tests, Chapter 9).

Why the other options fail: (A) adds only one covariance instead of two.

Answer: C.

Q6.15 REASON · about 45 s

Two models each have prediction error variance σ^2 , and the correlation between their errors is 0.5. The error variance of their simple average is

- (A) $0.25\sigma^2$
- (B) σ^2
- (C) $0.5\sigma^2$
- (D) $0.75\sigma^2$

Solution**Fast route:** $\sigma^2 \frac{1+\rho}{2}$.**Full working:**

- $\frac{1}{4}(\sigma^2 + \sigma^2 + 2 \times 0.5\sigma^2) = 0.75\sigma^2$. Independent errors ($\rho = 0$) would give $0.5\sigma^2$.

Answer: D.**Q6.16** **CALC · 1.5 to 2 min**

Which is a valid correlation matrix for three variables?

(A)
$$\begin{bmatrix} 1 & 1.2 & 0 \\ 1.2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(B)
$$\begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{bmatrix}$$

(C)
$$\begin{bmatrix} 1 & 0.5 & 0.3 \\ 0.4 & 1 & 0.2 \\ 0.3 & 0.2 & 1 \end{bmatrix}$$

(D)
$$\begin{bmatrix} 1 & 0.9 & -0.9 \\ 0.9 & 1 & 0.9 \\ -0.9 & 0.9 & 1 \end{bmatrix}$$

Solution**Fast route:** (A) has an entry above 1, (C) is not symmetric, (D) is internally impossible.**Full working:**

- (D): if $x_1 \approx x_2$ and $x_2 \approx x_3$, then x_1 cannot be strongly opposite to x_3 . Its determinant is $-2.888 < 0$, so it is not positive semidefinite.
- (B): eigenvalues 2, 0.5, 0.5, all positive.

Answer: B.**Q6.17** **REASON · about 45 s** $Y = X + N$, where X and the noise N are independent with equal variance. $\text{corr}(X, Y)$ is

(A) 0.707

(B) 1

(C) 0.25

(D) 0.5

Solution**Fast route:** $\frac{\sigma^2}{\sigma \cdot \sqrt{2}\sigma} = \frac{1}{\sqrt{2}}$.**Full working:**

- $\text{Cov}(X, X + N) = \sigma^2$; $\text{SD}(Y) = \sqrt{2}\sigma$.

Why the other options fail: (D) is r^2 .**Answer:** A.

Q6.18 REASON · about 45 s

The slope of y on x is 0.8 and the slope of x on y is 0.45. The correlation r is

- (A) 0.36
- (B) 0.6
- (C) 0.625
- (D) 1.25

Solution

Fast route: $r^2 = 0.8 \times 0.45 = 0.36$.

Full working:

- Both slopes are positive, so $r = +0.6$.

Answer: B.

Q6.19 CALC · 1.5 to 2 min

$r_{xy} = 0.5$, $r_{xz} = 0.6$, $r_{yz} = 0.8$. The partial correlation $r_{xy \cdot z}$ is closest to

- (A) 0.50
- (B) -0.30
- (C) 0.04
- (D) 0.26

Solution

Fast route: Numerator $0.5 - 0.48 = 0.02$; denominator $0.8 \times 0.6 = 0.48$.

Full working:

- $\frac{0.02}{0.48} = 0.042$. Almost all of the x to y link runs through z : a likely confounder.

Answer: C.

Q6.20 CALC · 1.5 to 2 min

A 2×2 table of “saw new UI” vs “applied” has $a = 40$ (yes, yes), $b = 10$ (yes, no), $c = 20$ (no, yes), $d = 30$ (no, no). The phi coefficient is closest to

- (A) 0.60
- (B) 0.80
- (C) 0.41
- (D) 0.17

Solution

Fast route: $\frac{1200 - 200}{\sqrt{50 \cdot 50 \cdot 60 \cdot 40}} = \frac{1000}{2449}$.

Full working:

- $\phi = 0.408$. Also $\chi^2 = n\phi^2 = 100 \times 0.167 = 16.7$ (links to Chapter 7).

Why the other options fail: (D) is ϕ^2 .

Answer: C.

Q6.21 RECALL · about 15 s

A new recommender has a higher click rate on mobile and on desktop separately, but a lower click rate overall. This is an example of

- (A) multicollinearity
- (B) heteroscedasticity
- (C) Simpson's paradox
- (D) regression to the mean

Solution

Fast route: Trend reverses on pooling because of a different traffic mix.

Full working:

- Old model: desktop 90/1000, mobile 10/200: overall 8.3%. New: desktop 20/200 (10%), mobile 60/1000 (6%): overall 6.7%. Better in both, worse overall.

Answer: C.

Q6.22 **RECALL** · about 15 s

Among employees hired through a test, the correlation between test score and job rating is 0.15, though among all applicants it would be about 0.5. The most likely reason is

- (A) range restriction
- (B) heteroscedasticity
- (C) multicollinearity
- (D) Simpson's paradox

Solution

Fast route: Only high scorers were hired, so x barely varies.

Full working:

- Less spread in x means a weaker observed correlation.

Answer: A.

Q6.23 **REASON** · about 45 s

The Durbin Watson statistic of a regression's residuals is 0.8. The lag 1 residual autocorrelation is about

- (A) -0.6
- (B) 0.8
- (C) 0.6
- (D) 0

Solution

Fast route: $0.8 = 2(1 - \rho) \Rightarrow \rho = 0.6$.

Full working:

- Positive autocorrelation: SEs are too small and significance is overstated.

Answer: C.

Q6.24 **RECALL** · about 15 s

Two features in a linear regression have correlation 0.98. Which consequence is **least** likely?

- (A) Ridge regression stabilises the coefficients
- (B) Individual coefficients have large standard errors

- (C) Coefficients may have counter intuitive signs
- (D) Predictive accuracy on similar new data is badly damaged

Solution

Fast route: Multicollinearity hurts interpretation, not prediction on similar data.

Full working:

- The combined effect of the two features is well estimated even though the split between them is not.

Answer: D.

Q6.25 **RECALL** · about 15 s

Monthly revenue of a company and monthly global smartphone sales over 10 years show $r = 0.95$. The best next step is to

- (A) conclude smartphone sales cause the revenue
- (B) correlate the month to month changes to remove the shared trend
- (C) square it to get the causal effect
- (D) report the correlation as strong causal evidence because $n = 120$

Solution

Fast route: Two trending series correlate spuriously; difference them.

Full working:

- This is the same idea as testing for stationarity (Dickey Fuller) before time series modelling.

Answer: B.

Q6.26 **RECALL** · about 15 s

Which value is impossible for a Pearson correlation?

- (A) 0
- (B) 0.999
- (C) -1
- (D) 1.02

Solution

Fast route: $|r| \leq 1$ always.

Full working:

- Guaranteed by the Cauchy Schwarz inequality.

Answer: D.

References for this chapter

- StatQuest: Pearson's correlation; Covariance; R-squared.
- Wikipedia: Homoscedasticity and heteroscedasticity; Variance inflation factor.
- Wikipedia: Simpson's paradox; Anscombe's quartet.
- Seeing Theory: Regression analysis.

Chapter 7

Hypothesis Testing

Companion resource for this section: [StatQuest: Hypothesis Testing and The Null Hypothesis, Clearly Explained](#)

7.1 The logic, as a courtroom

Start here: the everyday picture

- In court, the accused is presumed innocent. The prosecution brings evidence. If the evidence would be very unlikely for an innocent person, the court convicts. Otherwise it says “not guilty”, which is not the same as “proven innocent”.
- A hypothesis test works the same way. We presume “nothing is going on” and ask: if that were true, how surprising is the data we got?

- **Null hypothesis H_0 :** the boring default. The new apply button changes nothing; the coin is fair; job category and applying are unrelated.
- **Alternative H_1 :** what you suspect instead. The button changes (or increases) applications.
- **Test statistic:** a single number measuring how far the data is from what H_0 predicts, in units of standard error. Almost always

$$\text{test statistic} = \frac{\text{estimate} - \text{value under } H_0}{\text{standard error}}.$$

- **p-value:** the probability, *assuming H_0 is true*, of a test statistic at least as extreme as the one observed. Small p = the data would be rare if nothing were going on.
- **Significance level α** (usually 0.05): the false alarm rate you accept, chosen before seeing data. Rule: reject H_0 if $p \leq \alpha$; otherwise **fail to reject**. We never “accept” or “prove” H_0 .
- Tiny example: a coin gives 9 heads in 10 tosses. Under fairness, $P(\geq 9 \text{ heads}) = \frac{10+1}{1024} = 0.0107$. Two sided (also count ≤ 1 head): 0.021. Below 0.05, so we reject fairness.

7.2 The standard steps (a reported InfoEdge interview question)

1. State H_0 and H_1 , one sided or two sided, **before** looking at the data.
2. Fix the significance level α .
3. Choose the test that fits the data type and design; check its assumptions.
4. Compute the test statistic.
5. Find the p-value (or compare the statistic with a critical value).
6. Decide: reject H_0 if $p \leq \alpha$. Report the effect size and a confidence interval, not just the verdict.

Critical value and p-value give the same decision

- Two sided test at 5%: reject if $|z| > 1.96$. One sided: reject if $z > 1.645$.
- $z = 2$: two sided $p = 2(1 - 0.9772) = 0.0455$; one sided $p = 0.0228$. Both reject at 5%.
- **One or two sided:** use two sided when a change in either direction matters. One sided only when the other direction is impossible or irrelevant *and* decided in advance. For the same

data, the one sided p-value is half the two sided one (if the effect is in the predicted direction). Switching after seeing data is cheating (p-hacking).

7.3 The menu of tests: which test when

Question	Test	Statistic
Mean vs a value, σ known or n large	one sample z	$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$
Mean vs a value, σ unknown	one sample t	$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}, \text{ df } n - 1$
Proportion vs a value	one proportion z	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$
Two proportions (A/B test)	two proportion z	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$
Two independent means, equal variances	pooled t	$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}, \text{ df } n_1 + n_2 - 2$
Two independent means, unequal variances	Welch t	$SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$
Same units measured twice	paired t (Ch. 9)	$t = \bar{d} / (s_d / \sqrt{n})$
Three or more group means	one way ANOVA	$F = \text{MSB} / \text{MSW}$
Counts vs expected proportions	χ^2 goodness of fit	$\sum \frac{(O-E)^2}{E}, \text{ df } k - 1$
Two categorical variables	χ^2 independence	$\text{df } (r - 1)(c - 1)$
Small counts in a 2×2 table	Fisher's exact	exact probabilities
One variance	χ^2	$\frac{(n-1)s^2}{\sigma_0^2}, \text{ df } n - 1$
Two variances	F	$\frac{s_1^2}{s_2^2}$
Same distribution?	Kolmogorov Smirnov	largest CDF gap
Two groups, skewed data	Mann Whitney U	ranks
Stationary time series?	(Augmented) Dickey Fuller	H_0 : unit root

- **z vs t :** use t when σ is estimated by s . The t has heavier tails to allow for that extra uncertainty, and approaches z as n grows (critical value 2.776 at 4 df, 2.131 at 15 df, 1.96 at infinite df).
- **Pooled vs Welch:** Welch does not assume equal variances and loses almost nothing when they are equal: the safe default.
- **Why ANOVA and not many t tests:** three pairwise tests at 5% give a $1 - 0.95^3 = 14.3\%$ chance of at least one false alarm.

7.3.1 Worked: one sample t

- Claimed mean screening time 50 s. Sample of 16: $\bar{x} = 52, s = 4$. $SE = \frac{4}{\sqrt{16}} = 1; t = \frac{52-50}{1} = 2$, df 15.
- Two sided critical 2.131: fail to reject. One sided ($H_1 : \mu > 50$) critical 1.753: reject. Same data, different conclusions: that is why the direction must be decided in advance.

7.3.2 Worked: A/B test with two proportions

- Control: 200 applications from 2,000 visitors (10%). Treatment: 260 from 2,000 (13%).
- Pooled rate under H_0 : $\bar{p} = \frac{460}{4000} = 0.115$. $SE = \sqrt{0.115 \times 0.885 \times (\frac{1}{2000} + \frac{1}{2000})} = 0.0101$.
- $z = \frac{0.03}{0.0101} = 2.97$, two sided $p \approx 0.003$: a real lift of 3 points (30% relative).

7.4 Chi-square tests (a reported InfoEdge interview topic)

- Idea: compare **observed** counts O with the counts E you would **expect** if H_0 were true. Big gaps, relative to E , mean evidence against H_0 : $\chi^2 = \sum \frac{(O-E)^2}{E}$.
- Goodness of fit**: is a die fair? 60 rolls, expected 10 per face. Observed 5, 8, 9, 8, 10, 20. $\chi^2 = \frac{25+4+1+4+0+100}{10} = 13.4$, $df = 6 - 1 = 5$, critical 11.07: reject.
- Independence** in a table: expected count in a cell = $\frac{\text{row total} \times \text{column total}}{n}$ (because under independence $P(\text{row and column}) = P(\text{row})P(\text{column})$). $df = (r - 1)(c - 1)$.
- Worked 2×2 : version A: 30 applied, 20 did not; version B: 20 applied, 30 did not. All margins 50, $n = 100$, every expected count 25. $\chi^2 = 4 \times \frac{25}{25} = 4 > 3.841$: reject independence ($p = 0.046$). For a 2×2 table, χ^2 equals the square of the two proportion z ($z = 2$).
- Assumptions: independent observations (one row per user), raw counts (not percentages), expected counts at least 5; otherwise use **Fisher's exact test**. Yates' correction (subtract 0.5 from each $|O - E|$) is a conservative option for 2×2 .
- ML uses: χ^2 feature selection for categorical features, drift checks, sample ratio mismatch checks in A/B tests.

7.5 ANOVA in one picture

- Split the total variation into **between groups** (how far each group mean is from the grand mean) and **within groups** (noise inside groups).
- $F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(N-k)}$. If all means are equal, $F \approx 1$; large F means the groups differ more than noise explains.
- Example: three layouts, 5 users each, means 10, 12, 14, within variances 1.5, 2, 1.5. $SSB = 5(4 + 0 + 4) = 40$, $MSB = 20$. $SSW = 4(1.5 + 2 + 1.5) = 20$, $MSW = \frac{20}{12} = 1.67$. $F = 12$, far above the critical 3.89.
- Assumptions: independent observations, roughly normal residuals, equal variances. Equal group sizes are not required. A significant F says "some mean differs", not which; follow up with Tukey's test.

7.6 Tests about variances; confidence intervals as tests

- One variance: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$. Two variances: $F = \frac{s_1^2}{s_2^2}$ (very sensitive to non normality; Levene's test is the robust option).
- Duality**: a 95% CI contains exactly the values that a two sided 5% test would not reject. If the CI for a difference is $(2, 8)$, then "difference = 0" is rejected and "difference = 5" is not.
- Correct reading of a 95% CI: if we repeated the study many times, 95% of the intervals built this way would contain the true value.

7.7 Non parametric and resampling tests

- Mann Whitney U**: compares two independent groups through ranks; robust to skew and outliers.
- Kolmogorov Smirnov**: largest vertical gap between two CDFs; detects distribution drift between training and production.
- Permutation test**: shuffle the group labels many times to build the null distribution directly;

the p-value is the fraction of shuffles at least as extreme as the real data. Needs only exchangeability under H_0 .

- **Likelihood ratio test:** for nested models, $-2 \log \Lambda$ is approximately χ^2 with df equal to the difference in the number of parameters.

7.8 Stationarity and the Dickey Fuller test (a reported InfoEdge OA topic)

- A time series is **stationary** if its mean, variance and correlation pattern do not change over time. Daily applications that keep growing are not stationary.
- Model $y_t = \rho y_{t-1} + \varepsilon_t$. If $\rho = 1$ (a **unit root**), every shock stays forever: a random walk.
- **Dickey Fuller:** H_0 : unit root (non stationary). H_1 : stationary. The statistic is a t like ratio but compared with special Dickey Fuller critical values (about -2.86 at 5% with a constant), not the ordinary t table. The **Augmented** version adds lagged differences to handle autocorrelation.
- Reading output: statistic -3.5 , $p = 0.008$: reject the unit root, the series looks stationary. Statistic -1.2 , $p = 0.67$: cannot reject, treat as non stationary.
- **KPSS** flips the hypotheses (H_0 : stationary), so analysts often run both.
- If non stationary: difference the series ($y_t - y_{t-1}$), log it, or remove the trend. The “I” in ARIMA is this differencing.

Where this shows up in ML and at a company like Naukri

- A/B tests on ranking or UI changes are two proportion z tests or t tests on per user metrics.
- Every coefficient's p-value in a regression output is a t test of $H_0 : \beta_j = 0$: $t = \frac{\hat{\beta}}{SE}$.
- scikit-learn's `chi2` and `f_classif` are hypothesis tests used for feature selection.
- Drift monitoring uses KS for numeric and χ^2 for categorical features.

Formula sheet

statistic = $\frac{\text{estimate} - \text{null}}{SE}$ two sided z : 1.96 (5%), 2.576 (1%); one sided 1.645 χ^2 GOF df $k - 1$;
independence df $(r - 1)(c - 1)$; $E = \frac{RC}{n}$ $F = \frac{MSB}{MSW}$ variance test $\frac{(n-1)s^2}{\sigma_0^2}$ 2×2 : $\chi^2 = z^2$
family wise error $1 - (1 - \alpha)^m$ Dickey Fuller H_0 : unit root; KPSS H_0 : stationary LRT: χ^2
with df = parameter difference.

15 second recall card: memorise every line

- Steps: hypotheses, α , test and assumptions, statistic, p-value, decision plus effect size. Never “accept” H_0 .
- $z = 2$: one sided 0.0228, two sided 0.0455. 9 heads in 10: two sided about 0.021.
- σ unknown: t with $n - 1$ df. Unequal variances: Welch. Same units twice: paired. 3 or more groups: ANOVA (equal sizes not needed).
- χ^2 : expected = $\frac{RC}{n}$; df $(r - 1)(c - 1)$; expected counts below 5: Fisher's exact.
- Many pairwise tests inflate false alarms: 3 tests give 14.3%.
- CI and test duality: value outside the 95% CI is rejected at 5%.
- Dickey Fuller H_0 : unit root (non stationary); fix by differencing. KPSS H_0 : stationary.
- Regression coefficient $t = \hat{\beta}/SE$. LRT df = difference in parameters.
- Permutation test: shuffle labels; assumption free except exchangeability.

7.9 Question bank: Hypothesis testing

Q7.1 RECALL · about 15 s

★ PYQ (INT and OA: how do you conduct a hypothesis test?) Which ordering of steps is correct?

- (A) State H_0 and H_1 ; fix α ; choose the test and check assumptions; compute the statistic; get the p-value; decide and report the effect size
- (B) Collect data, look for an interesting pattern, then choose H_0 and α to match it
- (C) Choose α after computing the p-value so that the result is significant
- (D) Compute the p-value; if it is below 0.05, accept H_1 as proven

Solution

Fast route: Hypotheses and α come first, before any data is examined.

Full working:

- Otherwise the real false positive rate is no longer α .

Why the other options fail: (B) and (C) are p-hacking. (D) a small p-value is evidence against H_0 , not proof of H_1 .

Answer: A.

Q7.2 CALC · 1.5 to 2 min

★ PYQ (INT: chi-squared test) 100 users saw version A or B of a job page. A: 30 applied, 20 did not. B: 20 applied, 30 did not. The χ^2 statistic (no continuity correction) and the decision at 5% are

- (A) 4, reject independence
- (B) 2, do not reject
- (C) 16, reject
- (D) 4, do not reject

Solution

Fast route: All margins 50: every expected count 25; each cell contributes $\frac{25}{25} = 1$.

Full working:

- $\chi^2 = 4$, $df = (2 - 1)(2 - 1) = 1$, critical 3.841. $4 > 3.841$: reject; $p = 0.046$.
- Cross check: $\hat{p}_A = 0.6$, $\hat{p}_B = 0.4$, $SE = 0.1$, $z = 2$, $z^2 = 4$.

Why the other options fail: (D) uses the wrong critical value. (C) forgets to divide by E .

Answer: A.

Q7.3 RECALL · about 15 s

★ PYQ (OA: Dickey Fuller test) The null hypothesis of the (augmented) Dickey Fuller test is that the series

- (A) is normally distributed
- (B) has no autocorrelation
- (C) has a unit root (is non stationary)
- (D) is stationary

Solution

Fast route: DF: H_0 unit root. A small p-value suggests stationarity.

Full working:

- It uses its own critical values, not the ordinary t table. KPSS is the test with H_0 : stationary.

Answer: C.

Q7.4 RECALL · about 15 s

A test statistic $z = 2.0$ is observed. The one sided and two sided p-values are

- (A) 0.0455 and 0.0228
- (B) 0.0228 and 0.0455
- (C) 0.0228 and 0.0228
- (D) 0.05 and 0.10

Solution

Fast route: $1 - 0.9772$, then double.

Full working:

- Two sided counts both tails.

Answer: B.

Q7.5 CALC · 1.5 to 2 min

A claimed mean is 50. A sample of 16 gives $\bar{x} = 52$, $s = 4$. At 5%, the two sided test and the one sided test ($H_1 : \mu > 50$) conclude

- (A) two sided fails to reject, one sided rejects
- (B) both fail to reject
- (C) two sided rejects, one sided fails to reject
- (D) both reject

Solution

Fast route: $t = \frac{2}{4/4} = 2$ with 15 df; critical 2.131 (two sided) and 1.753 (one sided).

Full working:

- $2 < 2.131$: fail to reject two sided. $2 > 1.753$: reject one sided.

Answer: A.

Q7.6 REASON · about 45 s

A coin shows 60 heads in 100 tosses. Testing fairness at 5% (two sided):

- (A) $z = 2$, fail to reject
- (B) $z = 2$, reject
- (C) $z = 10$, reject
- (D) $z = 1$, fail to reject

Solution

Fast route: SE under H_0 : $\sqrt{0.25/100} = 0.05$; $\frac{0.1}{0.05} = 2$.

Full working:

- $p = 0.0455 < 0.05$: reject.

Answer: B.

Q7.7 **CALC** · 1.5 to 2 min

A/B test: control 200 applications from 2,000 visitors; treatment 260 from 2,000. The two proportion z statistic is closest to

- (A) 1.5
- (B) 2.0
- (C) 3.0
- (D) 6.0

Solution

Fast route: Pooled 0.115; $SE \approx \sqrt{0.1018 \times 0.001} \approx 0.0101$; $\frac{0.03}{0.0101}$.

Full working:

- $z = 2.97$, $p \approx 0.003$.

Answer: C.

Q7.8 **CALC** · 1.5 to 2 min

Two independent groups of 10: means 52 and 48, SDs 4 and 5. A pooled two sample t test at 5% two sided (critical $t_{18} = 2.101$) gives

- (A) $t \approx 0.9$, fail to reject
- (B) $t \approx 4.0$, reject
- (C) $t \approx 1.98$, reject
- (D) $t \approx 1.98$, fail to reject

Solution

Fast route: $s_p^2 = \frac{9(16)+9(25)}{18} = 20.5$; $SE = 4.53\sqrt{0.2} = 2.02$; $t = \frac{4}{2.02}$.

Full working:

- $t = 1.98 < 2.101$: fail to reject ($p \approx 0.064$).

Answer: D.

Q7.9 **RECALL** · about 15 s

Welch's t test is preferred over the pooled t test mainly when

- (A) the data are not normal
- (B) the group variances are clearly unequal
- (C) there are more than two groups
- (D) the samples are paired

Solution

Fast route: Welch drops the equal variance assumption.

Full working:

- It is a safe default, losing little power when variances happen to be equal.

Answer: B.

Q7.10 CALC · 1.5 to 2 min

A die is rolled 60 times with counts 5, 8, 9, 8, 10, 20 for faces 1 to 6. The χ^2 goodness of fit statistic and the decision at 5% (critical 11.07) are

- (A) 1.34, do not reject
- (B) 134, reject
- (C) 13.4, reject fairness
- (D) 13.4, do not reject

Solution

Fast route: $(O - E)^2$: 25, 4, 1, 4, 0, 100; total 134; divide by $E = 10$.

Full working:

- df 5; $13.4 > 11.07$; $p \approx 0.02$. Face 6 contributes 10 of the 13.4.

Why the other options fail: (B) forgets to divide by E .

Answer: C.

Q7.11 RECALL · about 15 s

The degrees of freedom of a χ^2 independence test on a 3×4 table are

- (A) 6
- (B) 7
- (C) 11
- (D) 12

Solution

Fast route: $(3 - 1)(4 - 1)$.

Full working:

- With row and column totals fixed, only 6 cells are free.

Answer: A.

Q7.12 RECALL · about 15 s

In a contingency table with $n = 100$, a row total is 30 and a column total is 50. The expected count in that cell under independence is

- (A) 30
- (B) 80
- (C) 50
- (D) 15

Solution

Fast route: $\frac{30 \times 50}{100}$.

Full working:

- $0.3 \times 0.5 \times 100 = 15$.

Answer: D.

Q7.13 CALC · 1.5 to 2 min

Three page layouts, 5 users each: means 10, 12, 14; within group variances 1.5, 2.0, 1.5. The one way

ANOVA F is

- (A) 6
- (B) 20
- (C) 3
- (D) 12

Solution

Fast route: $MSB = \frac{5 \times 8}{2} = 20$; $MSW = \frac{4 \times 5}{12} = 1.67$.

Full working:

- Grand mean 12; $SSB = 5(4 + 0 + 4) = 40$; $SSW = 4(1.5) + 4(2) + 4(1.5) = 20$.
- $F = \frac{20}{1.667} = 12$, far above $F_{0.05,2,12} = 3.89$.

Why the other options fail: (B) is MSB alone.

Answer: D.

Q7.14 **RECALL** · about 15 s

Instead of ANOVA, an analyst runs 3 independent pairwise tests, each at $\alpha = 0.05$. If all nulls are true, the probability of at least one false positive is closest to

- (A) 0.14
- (B) 0.15
- (C) 0.10
- (D) 0.05

Solution

Fast route: $1 - 0.95^3 = 0.1426$.

Full working:

- Nearly three times the intended 5%.

Why the other options fail: (B) adds 3×0.05 , a slight over count.

Answer: A.

Q7.15 **RECALL** · about 15 s

A 95% CI for a mean difference is (2, 8) seconds. Which is correct at the 5% level (two sided)?

- (A) There is a 95% probability that the true difference lies in (2, 8)
- (B) H_0 : difference = 5 is rejected
- (C) H_0 : difference = 10 is not rejected
- (D) H_0 : difference = 0 is rejected

Solution

Fast route: Values outside the CI are rejected; inside are not.

Full working:

- 0 lies outside, so it is rejected.

Why the other options fail: (A) is the classic misreading: the 95% describes the procedure over repeated studies.

Answer: D.

Q7.16 REASON · about 45 s

A process should have variance 10. A sample of 21 gives $s^2 = 15$. Testing $H_1 : \sigma^2 > 10$ at 5% ($\chi^2_{0.05,20} = 31.41$):

- (A) $\chi^2 = 1.5$, fail to reject
- (B) $\chi^2 = 30$, reject
- (C) $\chi^2 = 30$, fail to reject
- (D) $\chi^2 = 31.5$, reject

Solution

Fast route: $\frac{20 \times 15}{10} = 30 < 31.41$.

Full working:

- $p \approx 0.07$. A 50% bigger sample variance is not yet conclusive with 21 points.

Answer: C.

Q7.17 REASON · about 45 s

Error variances of two models from 16 batches each are 25 and 10. With $F_{0.05,15,15} = 2.40$ and $F_{0.025,15,15} = 2.86$, the one sided and two sided 5% tests conclude

- (A) neither rejects
- (B) two sided rejects, one sided does not
- (C) one sided rejects, two sided does not
- (D) both reject

Solution

Fast route: $F = 2.5$: above 2.40, below 2.86.

Full working:

- A two sided test at 5% uses the 2.5% upper point.

Answer: C.

Q7.18 RECALL · about 15 s

You have $n = 15$ measurements from a roughly normal population with unknown σ . To test a claim about the mean, use

- (A) an F test
- (B) a χ^2 test
- (C) a one sample t test with 14 df
- (D) a z test

Solution

Fast route: Estimated σ gives t with $n - 1$ df.

Full working:

- $t = \frac{\bar{x} - \mu_0}{s/\sqrt{15}}$.

Answer: C.

Q7.19 RECALL · about 15 s

A 2×2 table of “rare skill present” vs “hired” has an expected count of 2 in one cell. The best test is

- (A) Fisher's exact test
- (B) χ^2 without correction
- (C) ANOVA
- (D) two sample t test

Solution

Fast route: Expected counts below 5: Fisher's exact.

Full working:

- It computes exact hypergeometric probabilities instead of relying on the χ^2 approximation.

Answer: A.

Q7.20 **RECALL** · about 15 s

Which pairing of problem and test is **wrong**?

- (A) Did the same 30 users spend more time after a redesign? Two independent sample t test
- (B) Do mean salaries differ across 4 cities? One way ANOVA
- (C) Is city tier associated with churn (yes or no)? χ^2 test of independence
- (D) Is production user age distributed like training data? Kolmogorov Smirnov

Solution

Fast route: Same users measured twice: paired test.

Full working:

- Ignoring the pairing throws away the fact that each user is compared with themselves (Chapter 9).

Answer: A.

Q7.21 **RECALL** · about 15 s

Revenue per user is heavily skewed with outliers, 40 users per group. A robust test to compare the two groups is

- (A) a χ^2 goodness of fit test
- (B) a pooled t test on raw revenue
- (C) an F test for variances
- (D) the Mann Whitney U test

Solution

Fast route: Rank based tests resist skew and outliers.

Full working:

- A permutation test or a bootstrap CI are also good choices.

Answer: D.

Q7.22 **RECALL** · about 15 s

A test fails to reject H_0 . Which statement is correct?

- (A) The result may be due to low power, for example a small sample
- (B) It proves H_0 is true
- (C) The effect size must be exactly zero

(D) The p-value must be exactly 1

Solution

Fast route: Absence of evidence is not evidence of absence.

Full working:

- To claim “no meaningful difference”, use an equivalence test (TOST) with a pre specified margin.

Answer: A.

Q7.23 RECALL · about 15 s

The KPSS test differs from the Dickey Fuller test because its null hypothesis is that the series

- (A) has a unit root
- (B) is white noise
- (C) is stationary
- (D) is normally distributed

Solution

Fast route: KPSS reverses DF’s hypotheses.

Full working:

- Running both gives a more reliable picture.

Answer: C.

Q7.24 REASON · about 45 s

An ADF test on daily job postings gives statistic -1.4 with $p = 0.58$, and KPSS gives $p = 0.01$. The sensible conclusion is

- (A) the series is white noise
- (B) the series is non stationary; difference it and re test
- (C) the tests contradict each other; ignore both
- (D) the series is stationary; fit ARMA directly

Solution

Fast route: ADF cannot reject a unit root; KPSS rejects stationarity: both say non stationary.

Full working:

- Also use a time respecting validation split for any forecasting model.

Answer: B.

Q7.25 RECALL · about 15 s

In a regression output, “years of experience” has coefficient 2.1 with SE 0.7. With a large sample, the t statistic and conclusion at 5% are

- (A) 2.8, not significant
- (B) 3.0, significant
- (C) 0.33, not significant
- (D) 1.47, not significant

Solution**Fast route:** $\frac{2.1}{0.7} = 3 > 1.96$.**Full working:**

- This tests $H_0 : \beta = 0$.

Answer: B.**Q7.26 RECALL · about 15 s**

A permutation test for a difference in means between two groups

- (A) can only be used for proportions
- (B) gives a p-value that is always below 0.05
- (C) requires normally distributed data
- (D) builds the null distribution by repeatedly shuffling the group labels

Solution**Fast route:** Under H_0 the labels are arbitrary, so shuffle them.**Full working:**

- The p-value is the fraction of shuffles with a difference at least as extreme as observed.

Answer: D.**Q7.27 RECALL · about 15 s**

A likelihood ratio test compares a model with 5 parameters to a nested model with 3. Under H_0 , $-2 \log \Lambda$ is approximately

- (A) $F_{3,5}$
- (B) χ^2 with 2 df
- (C) χ^2 with 5 df
- (D) $N(0, 1)$

Solution**Fast route:** Wilks: $\text{df} = 5 - 3$.**Full working:**

- Used to compare nested logistic regressions.

Answer: B.**Q7.28 RECALL · about 15 s**

Which is **not** an assumption of the one way ANOVA F test?

- (A) Roughly normal residuals within groups
- (B) Equal group sizes
- (C) Independent observations
- (D) Equal group variances

Solution**Fast route:** Unequal group sizes are allowed.**Full working:**

- Equal sizes only make the test more robust to unequal variances.

Answer: B.

Q7.29 REASON · about 45 s

9 heads are observed in 10 tosses of a supposedly fair coin. The two sided exact p-value is closest to

- (A) 0.10
- (B) 0.001
- (C) 0.011
- (D) 0.021

Solution

Fast route: $P(\geq 9) = \frac{11}{1024} \approx 0.0107$; double it.

Full working:

- By symmetry, $P(\leq 1)$ is also 0.0107. Total 0.0215.

Why the other options fail: (C) is the one sided value.

Answer: D.

References for this chapter

- StatQuest: Hypothesis testing; How to calculate p-values.
- Scribbr: Choosing the right statistical test; Chi-square test of independence.
- Wikipedia: Augmented Dickey Fuller test.
- Seeing Theory: Frequentist inference.

Chapter 8

Statistical Significance, Errors and Power

Companion resource for this section: [StatQuest: p-values, clearly explained](#) (then [Statistical Power and Power Analysis](#) from the same channel)

8.1 What “statistically significant” really means

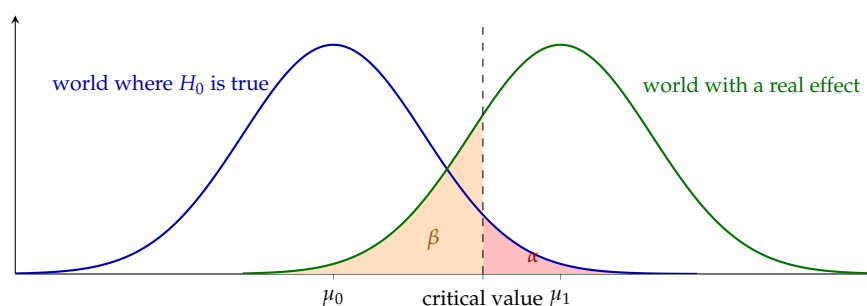
Start here: the everyday picture

- A friend claims a coin is fair. You toss it 10 times and get 9 heads. You think: “if the coin really were fair, 9 or more heads would happen only about 1% of the time. That is too surprising; I doubt the claim.”
- That “1%” is the p-value, and “too surprising” is your significance level. Statistical significance is nothing more than: **the data would be surprising if nothing were going on.**
- **p-value:** the probability, computed **assuming H_0 is true**, of a result at least as extreme as the one observed.
- **Significant at level α :** $p \leq \alpha$. With $\alpha = 0.05$ we accept being fooled by chance in 5% of the experiments where nothing is really going on.
- What a p-value is **not**:
 - not the probability that H_0 is true (that needs Bayes and a prior);
 - not the probability that the result is “due to chance”;
 - not a measure of how big or important the effect is.
- A beautiful fact: when H_0 is true (and the statistic is continuous), the p-value is **uniformly distributed** on $[0, 1]$. So in 1,000 experiments with no real effect, about 50 give $p < 0.05$ and about 100 give $p < 0.10$. Experimentation teams run such “A/A tests” to check their pipelines: the p-value histogram should be flat.

8.2 The two ways to be wrong (a reported InfoEdge OA topic)

	H_0 true (no effect)	H_0 false (real effect)
Reject H_0	Type I error (false positive), prob. α	correct: power = $1 - \beta$
Fail to reject	correct, prob. $1 - \alpha$	Type II error (false negative), prob. β

- $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$. $\beta = P(\text{fail to reject} \mid H_1 \text{ true})$. They are computed under *different* worlds, so $\alpha + \beta$ is not 1 in general (typical: 0.05 and 0.20).
- Courtroom: Type I = convicting an innocent person; Type II = letting a guilty person go.
- Medical test: Type I = telling a healthy person they are sick; Type II = telling a sick person they are healthy (a **false negative**), usually the costlier error for serious diseases.
- Classifier language: with “positive” meaning “reject H_0 ”: false positive rate = α , false negative rate = β , recall = power.
- **The trade off:** for a fixed sample, making α smaller makes β larger. The only way to reduce both is more data, less noise, or a bigger true effect.



Red: α (false positives). Orange: β (false negatives). Moving the dashed line right lowers α and raises β . Pushing the bells apart (bigger effect) or narrowing them (more data) shrinks both.

8.3 Power and sample size

- **Power** = $1 - \beta$: the chance of detecting an effect that is really there. The usual target is 80%.
- Power goes up with: a larger true effect, a larger n , less noise, a larger α , a one sided test in the right direction, and paired or variance reduced designs.
- **Worked power calculation**: $H_0 : \mu = 100$, $\sigma = 10$, $n = 25$ ($SE = 2$), one sided $\alpha = 0.05$. We reject when $\bar{x} > 100 + 1.645 \times 2 = 103.29$. If the truth is $\mu = 104$: power = $P(Z > \frac{103.29 - 104}{2}) = P(Z > -0.355) = 0.64$, so $\beta = 0.36$.
- **Sample size formula** for comparing two means (per group):

$$n = \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2 \sigma^2}{\delta^2} \approx \frac{16\sigma^2}{\delta^2} \quad (\alpha = 0.05, \text{ power } 80\%),$$

because $(1.96 + 0.84)^2 = 7.84$ and $2 \times 7.84 \approx 16$.

- For two proportions: $n = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 [p_1(1-p_1) + p_2(1-p_2)]}{(p_1 - p_2)^2}$. From 10% to 12%: $\frac{7.84 \times 0.1956}{0.0004} \approx 3,834$ per group.
- **Minimum detectable effect (MDE)**: the smallest effect you have 80% power to see with the traffic you have. Halving the MDE needs 4 times the users.

8.4 Effect size: significant is not the same as important

- **Cohen's d** = $\frac{\bar{x}_1 - \bar{x}_2}{s}$: the difference in SD units. Rough labels: 0.2 small, 0.5 medium, 0.8 large.
- With huge n , tiny effects become significant. The test statistic grows like $\sqrt{n}$: four times the data doubles z . $r = 0.05$ with $n = 10,000$ gives $t \approx 5$ (hugely significant) but explains only 0.25% of variance.
- With small n , big effects can be non significant. Always report the effect size with a confidence interval.
- How much wider is a 99% CI than a 95% CI? $\frac{2.576}{1.96} = 1.31$: about 31% wider.

8.5 Multiple testing: the more you look, the more you find

- Test 20 true null hypotheses at 5%: $P(\text{at least one false positive}) = 1 - 0.95^{20} = 0.64$. This is the "green jelly beans cause acne" problem (xkcd 882).
- **Family wise error rate (FWER)** control, the chance of even one false positive:
 - **Bonferroni**: test each hypothesis at $\frac{\alpha}{m}$. Simple and conservative.
 - **Holm**: sort p-values ascending; compare the k -th smallest with $\frac{\alpha}{m-k+1}$; stop at the first failure. Never less powerful than Bonferroni.

- **False discovery rate (FDR)** control, the expected share of false discoveries among the discoveries: **Benjamini Hochberg**: sort p-values, find the largest k with $p_{(k)} \leq \frac{k}{m}q$, reject hypotheses 1.. k . Much more powerful when screening thousands of features or metrics.
- Worked: eight p-values 0.001, 0.008, 0.039, 0.041, 0.042, 0.06, 0.074, 0.205, at level 0.05. Bonferroni threshold 0.00625: 1 rejection. Holm: $0.001 \leq 0.00625$, then $0.008 > 0.00714$: 1 rejection. BH thresholds $\frac{k}{8} \times 0.05$: 0.00625, 0.0125, 0.01875, ...; 0.001 and 0.008 pass, 0.039 fails its threshold 0.01875 and so do all later ones: 2 rejections.
- Hidden multiplicity: many metrics, many segments ("it works for Android users in Pune"), and **peeking** at an A/B test every day and stopping at the first $p < 0.05$, which can push the true false positive rate to 20% or more. Fixes: pre registration, one primary metric, a fixed sample size, or sequential methods.

8.6 How often is a significant result real?

- Suppose 10% of ideas truly work, power is 80%, $\alpha = 0.05$. Out of 1,000 ideas: 100 real, 80 detected; 900 useless, 45 false positives.
- $P(\text{real} \mid \text{significant}) = \frac{80}{80+45} = 0.64$. A third of "wins" are false: Bayes' theorem in disguise.
- **Winner's curse**: in underpowered studies, only the luckily large estimates cross the threshold, so significant results overestimate the true effect (and can even have the wrong sign).

Where this shows up in ML and at a company like Naukri

- Large experimentation platforms (Naukri, LinkedIn, Microsoft) use power analysis, guardrail metrics, FDR control, sample ratio mismatch checks and CUPED. CUPED adjusts each user's metric using their pre experiment value and cuts variance by a factor $1 - \rho^2$ ($\rho = 0.7$ roughly halves the needed sample).
- Classifier thresholds are the same trade off as α vs β : lowering the threshold catches more positives (recall up, false negatives down) at the cost of more false positives.
- Trying 50 hyperparameter settings and reporting the best test score is multiple testing; that is why a final untouched test set exists.

Formula sheet

$$\alpha = P(\text{reject} \mid H_0) \quad \beta = P(\text{not reject} \mid H_1) \quad \text{power} = 1 - \beta \quad n = \frac{2(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2} \approx \frac{16\sigma^2}{\delta^2}$$

$$z_{0.80} = 0.84, z_{0.90} = 1.28 \quad d = \frac{\Delta}{s} \quad \text{Bonferroni } \frac{\alpha}{m} \quad \text{Holm } \frac{\alpha}{m-k+1} \quad \text{BH } p_{(k)} \leq \frac{k}{m}q \quad \text{FWER}$$

$$= 1 - (1 - \alpha)^m \quad \text{under } H_0: p \sim U(0, 1) \quad \text{PPV} = \frac{(1-\beta)\pi}{(1-\beta)\pi + \alpha(1-\pi)} \quad \text{CUPED variance} \times (1 - \rho^2).$$

15 second recall card: memorise every line

- α : Type I, false positive, reject a true H_0 . β : Type II, false negative, miss a real effect. Power $= 1 - \beta$. $\alpha + \beta \neq 1$.
- Smaller α with fixed n means larger β . More data reduces both.
- p-value is $P(\text{data this extreme} \mid H_0)$, not $P(H_0 \mid \text{data})$. Under H_0 it is uniform.
- One sided p is half the two sided p (effect in the predicted direction).
- 20 tests at 5%: 64% chance of a false alarm. Bonferroni $\frac{\alpha}{m}$. BH controls FDR.
- $n \approx \frac{16\sigma^2}{\delta^2}$ per group; halving the effect needs 4 times n ; z grows like $\sqrt{n}$.
- 99% CI is 31% wider than 95%. Cohen's d : 0.2, 0.5, 0.8.
- Peeking inflates false positives. Significant does not mean large.

8.7 Question bank: Statistical significance

Q8.1 RECALL · about 15 s

★ PYQ (OA: alpha and beta for Type I and Type II errors) Which statement is correct?

- (A) α is the probability of rejecting H_0 when H_0 is true
- (B) β is the probability of rejecting H_0 when H_1 is true
- (C) For a fixed sample size, decreasing α also decreases β
- (D) $\alpha + \beta = 1$ always

Solution

Fast route: α is the false positive rate under H_0 .

Full working:

- β is the probability of *failing* to reject when H_1 is true.

Why the other options fail: (D) the two are computed in different worlds. (B) describes power ($1 - \beta$). (C) the opposite: a stricter threshold misses more real effects.

Answer: A.

Q8.2 RECALL · about 15 s

★ PYQ (OA: what is a false negative) A resume screening model flags “good fit” candidates as the positive class. A false negative is

- (A) a poor fit candidate correctly rejected
- (B) a good fit candidate correctly flagged
- (C) a poor fit candidate flagged as good fit
- (D) a good fit candidate rejected by the model

Solution

Fast route: False negative: truly positive, predicted negative.

Full working:

- In testing language it is a Type II error; it lowers recall $= \frac{TP}{TP+FN}$.

Why the other options fail: (C) is a false positive. (A) and (B) are correct decisions.

Answer: D.

Q8.3 RECALL · about 15 s

★ PYQ (INT: p-value and significance level) An A/B test returns $p = 0.03$ with $\alpha = 0.05$. The correct interpretation is

- (A) there is a 97% chance the treatment is better
- (B) there is a 3% chance the null hypothesis is true
- (C) if there were truly no difference, a difference at least this large would be seen about 3% of the time
- (D) the effect is large enough to matter for the business

Solution

Fast route: The p-value is computed assuming H_0 .

Full working:

- Since $0.03 \leq 0.05$, the result is significant at 5%.

Why the other options fail: (B) and (A) confuse $P(\text{data} \mid H_0)$ with $P(H_0 \mid \text{data})$. (D) significance says nothing about size.

Answer: C.

Q8.4 **CALC** · 1.5 to 2 min

$H_0 : \mu = 100$ vs $H_1 : \mu > 100$, $\sigma = 10$, $n = 25$, $\alpha = 0.05$. If the true mean is 104, the power is closest to

- (A) 0.50
- (B) 0.64
- (C) 0.80
- (D) 0.36

Solution

Fast route: Critical $\bar{x} = 100 + 1.645 \times 2 = 103.29$; under $\mu = 104$: $z = -0.355$.

Full working:

- Power = $P(\bar{x} > 103.29 \mid \mu = 104) = \Phi(0.355) = 0.64$.

Why the other options fail: (D) is β .

Answer: B.

Q8.5 **CALC** · 1.5 to 2 min

Continuing, the smallest n giving 80% power (one sided, $\alpha = 0.05$, $\sigma = 10$, true difference 4) is

- (A) 25
- (B) 39
- (C) 62
- (D) 50

Solution

Fast route: $\left(\frac{(1.645 + 0.842) \times 10}{4} \right)^2 = 6.22^2 = 38.7$; round up.

Full working:

- One sample, one sided formula $n = \left(\frac{(z_{1-\alpha} + z_{1-\beta})\sigma}{\delta} \right)^2$.

Answer: B.

Q8.6 **RECALL** · about 15 s

Which change increases the power of a test, all else fixed?

- (A) A smaller α
- (B) A smaller true effect
- (C) A larger sample size
- (D) A noisier metric

Solution

Fast route: More data narrows the bells.

Full working:

- The other three all make a real effect harder to detect.

Answer: C.

Q8.7 **RECALL** · about 15 s

Two groups have means 105 and 100 and a common SD of 10. Cohen's d is

- (A) 1.05
- (B) 5
- (C) 0.05
- (D) 0.5

Solution

Fast route: $\frac{5}{10}$.

Full working:

- A "medium" effect.

Answer: D.

Q8.8 **REASON** · about 45 s

20 independent hypotheses are all truly null and each is tested at $\alpha = 0.05$. The probability of at least one false positive, and the Bonferroni per test threshold, are

- (A) 0.05 and 0.0025
- (B) 0.64 and 0.01
- (C) 1.00 and 0.05
- (D) 0.64 and 0.0025

Solution

Fast route: $1 - 0.95^{20} = 1 - 0.358; \frac{0.05}{20}$.

Full working:

- 0.64 and 0.0025.

Why the other options fail: (C) adds 20×0.05 , which over counts.

Answer: D.

Q8.9 **CALC** · 1.5 to 2 min

Eight metrics give p-values 0.001, 0.008, 0.039, 0.041, 0.042, 0.06, 0.074, 0.205. At level 0.05, the numbers of rejections by Bonferroni, Holm and Benjamini Hochberg are

- (A) 1, 2, 5
- (B) 1, 1, 2
- (C) 2, 2, 2
- (D) 5, 5, 5

Solution

Fast route: Bonferroni cut 0.00625; Holm stops at $0.008 > 0.00714$; BH passes the first two only.

Full working:

- BH thresholds $\frac{k}{8} \times 0.05$: 0.00625, 0.0125, 0.01875, 0.025, 0.03125, 0.0375, 0.04375, 0.05. Only

$p_{(1)}$ and $p_{(2)}$ are below their thresholds, and no later one is.

Why the other options fail: (D) is the uncorrected count ($p < 0.05$).

Answer: B.

Q8.10 **CALC** · 1.5 to 2 min

Baseline apply rate is 10%. With $\alpha = 0.05$ (two sided) and 80% power, the sample size per group to detect an increase to 12% is closest to

- (A) 400
- (B) 1,900
- (C) 15,000
- (D) 3,800

Solution

Fast route: $\frac{7.84 \times (0.09 + 0.1056)}{0.02^2}$.

Full working:

- $\frac{7.84 \times 0.1956}{0.0004} = 3,834$.

Answer: D.

Q8.11 **RECALL** · about 15 s

About 3,800 users per group detect a 2 point lift from 10%. To detect a 1 point lift from the same baseline you need roughly

- (A) 7,500
- (B) 14,700
- (C) 3,800
- (D) 1,900

Solution

Fast route: $n \propto \frac{1}{\delta^2}$: half the effect, about 4 times the users.

Full working:

- Exact: $\frac{7.84 \times (0.09 + 0.0979)}{0.0001} \approx 14,700$.

Answer: B.

Q8.12 **RECALL** · about 15 s

You run 1,000 A/A tests (no real difference) at $\alpha = 0.05$. Which is expected?

- (A) No test shows $p < 0.05$
- (B) Most p-values are near 1
- (C) About 50 tests show $p < 0.05$
- (D) About 500 tests show $p < 0.05$

Solution

Fast route: Under H_0 , p is uniform: $P(p < 0.05) = 0.05$.

Full working:

- The p-value histogram should be flat; this is how platforms validate their statistics.

Answer: C.

Q8.13 **RECALL** · about 15 s

For the same data (normal based), a 99% CI compared with a 95% CI is

- (A) about 4% wider
- (B) about 31% wider
- (C) narrower
- (D) the same width

Solution

Fast route: $\frac{2.576}{1.96} = 1.31$.

Full working:

- More confidence costs width.

Answer: B.

Q8.14 **RECALL** · about 15 s

A two sided test gives $p = 0.08$ and the effect is in the direction predicted in advance. The one sided p-value is

- (A) 0.04
- (B) 0.08
- (C) 0.92
- (D) 0.16

Solution

Fast route: Half of the two sided value.

Full working:

- Legitimate only if the one sided hypothesis was fixed before seeing the data.

Answer: A.

Q8.15 **RECALL** · about 15 s

Keeping the effect fixed, the sample size is multiplied by 4. The z statistic becomes

- (A) 2 times larger
- (B) half
- (C) 4 times larger
- (D) unchanged

Solution

Fast route: $z \propto \sqrt{n}$.

Full working:

- The SE halves, so z doubles.

Answer: A.

Q8.16 CALC · 1.5 to 2 min

A correlation $r = 0.05$ is found across $n = 10,000$ users. The t statistic is closest to, and the conclusion is,

- (A) 5, significant but practically tiny
- (B) 5, large effect
- (C) 0.5, not significant
- (D) 50, huge effect

Solution

Fast route: $t \approx 0.05 \times \sqrt{10000} = 5$.

Full working:

- $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \approx 5.0$; $p < 10^{-6}$; yet $r^2 = 0.25\%$ of variance.

Answer: A.

Q8.17 CALC · 1.5 to 2 min

An A/B test designed as 50/50 on 10,000 users ends with 5,212 in control and 4,788 in treatment. The sample ratio mismatch χ^2 statistic (df 1) is closest to

- (A) 212
- (B) 4
- (C) 0.4
- (D) 18

Solution

Fast route: $2 \times \frac{212^2}{5000} = \frac{89888}{5000}$.

Full working:

- 17.98, far above 3.841 ($p \approx 2 \times 10^{-5}$): randomisation or logging is broken; do not trust the results until fixed.

Answer: D.

Q8.18 REASON · about 45 s

A classifier's confusion matrix: TP = 80, FN = 20, FP = 30, TN = 870. Recall and precision are

- (A) 0.80 and 0.73
- (B) 0.73 and 0.80
- (C) 0.20 and 0.03
- (D) 0.80 and 0.97

Solution

Fast route: Recall $\frac{80}{100}$; precision $\frac{80}{110}$.

Full working:

- False negative rate 0.2 (like β); false positive rate $\frac{30}{900} = 0.033$ (like α).

Why the other options fail: (C) gives the error rates, not recall and precision.

Answer: A.

Q8.19 CALC · 1.5 to 2 min

10% of tested product ideas truly work; tests have 80% power and $\alpha = 0.05$. The fraction of significant results that are real is closest to

- (A) 0.64
- (B) 0.95
- (C) 0.10
- (D) 0.80

Solution

Fast route: Per 1,000: 80 true positives vs 45 false positives.

Full working:

- $\frac{80}{125} = 0.64$.

Why the other options fail: (B) is $1 - \alpha$; (D) is the power. Neither is the share of wins that are real.

Answer: A.

Q8.20 RECALL · about 15 s

A PM checks an A/B dashboard daily and stops as soon as $p < 0.05$. The actual false positive rate is

- (A) exactly 5%
- (B) well above 5%
- (C) zero
- (D) below 5%

Solution

Fast route: Each peek is another chance for noise to cross the line.

Full working:

- Fixes: fixed sample size from a power analysis, or sequential testing methods.

Answer: B.

Q8.21 RECALL · about 15 s

Significant results from underpowered studies tend to

- (A) replicate with high probability
- (B) have exactly the true effect size
- (C) underestimate the true effect
- (D) overestimate the true effect (winner's curse)

Solution

Fast route: Only the lucky large estimates cross the threshold.

Full working:

- They also replicate rarely (replication probability is about the power).

Answer: D.

Q8.22 RECALL · about 15 s

Which statement about statistical significance is **false**?

- (A) It measures how surprising the data is under H_0
- (B) It depends on sample size
- (C) A significant result means the effect is large
- (D) The threshold α should be set before seeing the data

Solution

Fast route: Tiny effects become significant with large n .

Full working:

- Report effect sizes and CIs alongside p-values.

Answer: C.

Q8.23 REASON · about 45 s

CUPED adjusts each user's metric using their pre experiment value, with correlation $\rho = 0.7$ between the two. The variance of the adjusted metric, relative to the original, is about

- (A) 0.49
- (B) 0.70
- (C) 0.51
- (D) 0.30

Solution

Fast route: $1 - \rho^2 = 1 - 0.49$.

Full working:

- The needed sample size roughly halves.

Why the other options fail: (A) is ρ^2 , the part removed, not the part left.

Answer: C.

Q8.24 RECALL · about 15 s

Screening 5,000 candidate features, you want the expected share of false discoveries among selected features to be at most 10%. Use

- (A) Holm at 0.1
- (B) Bonferroni at 0.1
- (C) Benjamini Hochberg with $q = 0.1$
- (D) no correction

Solution

Fast route: "Expected share of false discoveries" is the FDR.

Full working:

- BH controls FDR and keeps far more power than FWER methods at this scale.

Answer: C.

Q8.25 RECALL · about 15 s

A disease screening model's threshold is lowered. Compared with before, it will have

- (A) higher recall
- (B) more false negatives

- (C) higher precision in every case
(D) fewer false positives

Solution

Fast route: Flagging more cases catches more true positives.

Full working:

- False negatives fall and recall rises; false positives rise and precision usually falls.

Answer: A.

References for this chapter

- StatQuest: p-values; Statistical power; Power analysis; False discovery rates.
- Scribbr: Type I and Type II errors; Statistical significance.
- xkcd 882: Significant.

Chapter 9

Paired Means Tests

Companion resource for this section: [Stat Trek: Hypothesis Test for the Difference Between Paired Means](#) (worked example, step by step)

9.1 What “paired” means

Start here: the everyday picture

- You want to know whether a diet works. Option 1: weigh 5 people who dieted and 5 different people who did not, and compare. Option 2: weigh the same 5 people before and after the diet.
- In option 1, the differences between people (some weigh 50 kg, some 90 kg) swamp the small effect of the diet. In option 2, each person is compared with themselves, so those differences disappear. Option 2 is a **paired design**, and it is far more sensitive.
- Data is **paired** when each value in one group has a natural partner in the other: the **same unit measured twice**, or two units deliberately matched.
- Paired examples: a user’s weekly applications before and after a UI change; two models scored on the **same** test examples; left arm vs right arm of the same patient; the same resume scored by two parsers.
- Not paired: 500 Android users vs 500 different iOS users. That is two independent samples (Chapter 7).
- The key move: reduce each pair to one number, the **difference** $d_i = x_i - y_i$. The question “did anything change?” becomes “is the mean difference zero?”, which is a one sample question about the d_i .

9.2 Why pairing is so powerful

- From Chapter 6: $\text{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$.
- With $\sigma_X = \sigma_Y = 10$ and a within pair correlation $\rho = 0.8$ (people are consistent with themselves): $\text{Var}(D) = 100 + 100 - 160 = 40$. If the two groups were independent people ($\rho = 0$), it would be 200. Pairing cut the noise variance by a factor of 5, which means you need about 5 times fewer units for the same power.
- Worked example (reused in the bank): five users’ weekly minutes on the site.

User	1	2	3	4	5
Before	70	72	75	68	80
After	68	68	72	63	79
$d = \text{Before minus After}$	2	4	3	5	1

- Every single user went down, yet the two columns overlap heavily (values range from 63 to 80). An independent two sample t test sees mostly overlap: $t \approx 0.88$, not significant. A paired test sees five consistent drops: $t \approx 4.24$, significant. Same numbers, opposite conclusions.

- When pairing does **not** help: if the within pair correlation is zero or negative, the variance of D is not smaller, and you lose degrees of freedom ($n - 1$ instead of $2n - 2$).

9.3 The paired t test, step by step

1. Compute the differences d_i , their mean $\bar{d}$ and SD s_d (dividing by $n - 1$).
 2. $H_0 : \mu_d = 0$ (no change); $H_1 : \mu_d \neq 0$ (or one sided).
 3. $t = \frac{\bar{d}}{s_d / \sqrt{n}}$ with $n - 1$ degrees of freedom, where n is the number of **pairs**.
 4. CI: $\bar{d} \pm t_{\alpha/2, n-1} \frac{s_d}{\sqrt{n}}$.
 5. Effect size: $d_z = \frac{\bar{d}}{s_d}$ (note $t = d_z \sqrt{n}$).
- On the table: $\bar{d} = \frac{15}{5} = 3$. Deviations from 3: $-1, 1, 0, 2, -2$; $s_d^2 = \frac{1+1+0+4+4}{4} = 2.5$, $s_d = 1.581$. $SE = \frac{1.581}{\sqrt{5}} = 0.707$. $t = \frac{3}{0.707} = 4.24$ with 4 df; critical 2.776: reject ($p \approx 0.013$).
 - CI: $3 \pm 2.776 \times 0.707 = 3 \pm 1.96 = (1.04, 4.96)$ minutes. Zero is outside, consistent with the test.
 - Shortcut from totals: $s_d^2 = \frac{\sum d^2 - (\sum d)^2 / n}{n-1}$. With $\sum d = 30$, $\sum d^2 = 120$, $n = 10$: $s_d^2 = \frac{120-90}{9} = 3.33$, $s_d = 1.83$, $t = \frac{3}{1.83/\sqrt{10}} = 5.2$.
 - Assumptions: different pairs are independent of each other; the **differences** are roughly normal (the raw values need not be); with many pairs the CLT covers it. Equal variances are not needed.
 - Large samples: with 100 pairs, $\bar{d} = 1.2$ and $s_d = 5$: SE 0.5, $z = 2.4$, $p = 0.016$.
 - Sample size for a paired study: $n = \left(\frac{(z_{\alpha/2} + z_{\beta}) \sigma_d}{\delta} \right)^2$. $\sigma_d = 10$, $\delta = 4$: $(2.8 \times 2.5)^2 = 49$ pairs.

9.4 Paired tests without normality, and for yes/no outcomes

- **Sign test**: count how many differences are positive. Under H_0 each sign is a fair coin flip, so the count is Binomial($n, 0.5$). Very robust, but it ignores sizes. Example: a new ranker wins on 8 of 10 query sets. $P(\geq 8) = \frac{45+10+1}{1024} = 0.055$; two sided 0.109: not significant.
- **Wilcoxon signed rank test**: rank the absolute differences (drop zeros), then add up the ranks of positive differences (T^+) and of negative ones (T^-). $T = \min(T^+, T^-)$; small T is evidence against H_0 . It uses sizes through ranks: more powerful than the sign test, robust to outliers.
 - Example: differences 4, 6, -1, 8, -2, 5. Sorted absolute values 1, 2, 4, 5, 6, 8 get ranks 1 to 6. The negatives (-1, -2) have ranks 1 and 2: $T^- = 3$, $T^+ = 18$, $T = 3$. For $n = 6$ the 5% two sided critical value is 0, so we cannot reject ($p = 0.156$).
- **McNemar's test**: paired yes/no outcomes, such as two classifiers judged on the same test examples. Only **discordant** pairs carry information: $b = A$ right and B wrong; $c = A$ wrong and B right. $\chi^2 = \frac{(b-c)^2}{b+c}$ with 1 df (continuity corrected: $\frac{(|b-c|-1)^2}{b+c}$). Examples both models get right (or both wrong) say nothing about which is better.
 - Example: $b = 30$, $c = 10$: $\chi^2 = \frac{400}{40} = 10 > 3.841$: model A is significantly better.
- Three or more measurements on the same units: **repeated measures ANOVA**; the rank based version is the **Friedman test**.

9.5 Design traps in before and after studies

- **No control group:** a drop after a change could be seasonality (festival week), a marketing campaign or a bug. Best design: paired measurements *plus* a randomised control group (difference in differences).
- **Regression to the mean:** if units are selected because they were extreme “before” (the 200 least active users), they drift back towards average “after” even with no treatment.
- **Order and carryover effects:** when every unit gets both treatments (a crossover design), half should get A first and half B first, to balance learning and fatigue.
- **Missing partners:** a user with only a “before” value is dropped; if dropouts relate to the outcome, the result is biased.

Where this shows up in ML and at a company like Naukri

- Comparing two models on the same test set: use McNemar (for accuracy) or a paired t or paired bootstrap on per example losses. Comparing two overall accuracies as if independent wastes power.
- Comparing models across k cross validation folds with a paired t test is over confident, because training sets overlap. The Nadeau and Bengio correction replaces $\frac{s^2}{k}$ by $\left(\frac{1}{k} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)s^2$. With 5 folds and differences 0, 4, 2, 2, 2 ($\bar{d} = 2$, $s^2 = 2$): naive $t = 3.16$, corrected $t = 2.11$.
- LLM evaluation: two models answering the same prompts give paired win or loss data; drop ties and use a sign test.

Formula sheet

$$d_i = x_i - y_i \quad t = \frac{\bar{d}}{s_d/\sqrt{n}}, \text{ df } n - 1 \quad \text{CI } \bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}} \quad d_z = \frac{\bar{d}}{s_d} \quad s_d^2 = \frac{\sum d^2 - (\sum d)^2/n}{n-1} \quad \text{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y \quad \text{sign test: Bin}(n, 0.5) \quad \text{Wilcoxon } T = \min(T^+, T^-) \quad \text{McNemar } \frac{(b-c)^2}{b+c}$$

$$n = \left(\frac{(z_{\alpha/2} + z_{\beta})\sigma_d}{\delta} \right)^2.$$

15 second recall card: memorise every line

- Paired = same units twice or matched units. Reduce to differences; paired t is a one sample t on them.
- df = number of pairs minus 1. Assumes the *differences* are roughly normal and pairs are independent.
- Pairing wins when the within pair correlation is positive: $\text{Var}(D) = 2\sigma^2(1 - \rho)$ for equal SDs.
- Sign test uses signs only; Wilcoxon signed rank uses ranks of $|d|$; McNemar uses only discordant pairs.
- Three or more time points: repeated measures ANOVA or Friedman.
- Before and after without a control: seasonality and regression to the mean can fake an effect.
- Cross validation paired t is over confident (overlapping training sets).

9.6 Question bank: Paired means tests

No InfoEdge PYQ found for this topic

The syllabus names “Paired Means Tests”, but no public report recalls a specific question. The bank follows the reported OA style and adds the model comparison angle InfoEdge interviewers favour.

Q9.1 CALC · 1.5 to 2 min

Five users’ weekly minutes: before 70, 72, 75, 68, 80; after 68, 68, 72, 63, 79. The paired t statistic and the decision at 5% two sided ($t_{0.025,4} = 2.776$) are

- (A) 1.90, do not reject
- (B) 0.88, do not reject
- (C) 4.24, reject
- (D) 9.49, reject

Solution

Fast route: $d = 2, 4, 3, 5, 1$; $\bar{d} = 3$, $s_d = 1.58$, SE 0.707.

Full working:

$$\bullet s_d^2 = \frac{1+1+0+4+4}{4} = 2.5. \quad t = \frac{3}{0.707} = 4.24 > 2.776.$$

Why the other options fail: (B) is the (wrong) independent two sample t . (A) is d_z , not t .

Answer: C.

Q9.2 REASON · about 45 s

For the same data, the 95% CI for the mean drop is closest to

- (A) (2.29, 3.71)
- (B) (0.61, 5.39)
- (C) (1.04, 4.96)
- (D) (1.61, 4.39)

Solution

Fast route: $3 \pm 2.776 \times 0.707 = 3 \pm 1.96$.

Full working:

- Zero is outside the interval, consistent with rejecting H_0 .

Why the other options fail: (D) uses 1.96 instead of the t critical value with 4 df.

Answer: C.

Q9.3 RECALL · about 15 s

On the same data, an independent two sample t test gives only $t \approx 0.88$. The main reason for the difference is that

- (A) the independent test is always more conservative
- (B) the paired test has more degrees of freedom
- (C) there is an arithmetic error
- (D) the independent test treats the large user to user differences as noise, while pairing cancels them

Solution

Fast route: Pairing removes between user variation.

Full working:

- Group SDs are about 4.7 and 6.0, but the differences have SD only 1.58.

Why the other options fail: (B) paired has fewer df (4 vs 8).

Answer: D.

Q9.4 REASON · about 45 s

Scores before and after each have SD 10, and their correlation is 0.8. The variance of the difference is

- (A) 20
- (B) 40
- (C) 160
- (D) 200

Solution

Fast route: $100 + 100 - 2(0.8)(10)(10)$.

Full working:

- 40, five times smaller than 200 for independent groups.

Answer: B.

Q9.5 RECALL · about 15 s

Which design does **not** call for a paired test?

- (A) Session length of 500 Android users vs 500 iOS users
- (B) Blood pressure of the left arm vs right arm of the same patients
- (C) Recruiter response time for 40 recruiters before and after a new dashboard
- (D) Two classifiers evaluated on the same 1,000 test images

Solution

Fast route: Different people in the two groups: independent samples.

Full working:

- The others each have a natural partner for every value.

Answer: A.

Q9.6 RECALL · about 15 s

Which is an assumption of the paired t test?

- (A) The before values and after values are independent of each other
- (B) Both groups have the same sample size by coincidence
- (C) The differences are approximately normal (or there are many pairs)
- (D) The before and after variances are equal

Solution

Fast route: The test only ever looks at the differences.

Full working:

- Pairs must be independent of each other, but within a pair the values are expected to be correlated.

Why the other options fail: (A) is the opposite of the paired idea.

Answer: C.

Q9.7 **CALC** · 1.5 to 2 min

Ten pairs give $\sum d = 30$ and $\sum d^2 = 120$. The paired t statistic is closest to

- (A) 5.20
- (B) 1.64
- (C) 9.49
- (D) 3.00

Solution

Fast route: $s_d^2 = \frac{120 - 90}{9} = 3.33$; $SE = \frac{1.826}{3.162} = 0.577$; $t = \frac{3}{0.577}$.

Full working:

- $t = 5.196$ with 9 df, $p \approx 0.0006$.

Why the other options fail: (C) uses $\sqrt{n} \times \bar{d}$ ignoring s_d .

Answer: A.

Q9.8 **RECALL** · about 15 s

For the five user example ($\bar{d} = 3$, $s_d = 1.581$), the paired effect size d_z is closest to

- (A) 0.53
- (B) 1.90
- (C) 3.00
- (D) 4.24

Solution

Fast route: $\frac{3}{1.581}$.

Full working:

- 1.90: very large. Note $t = d_z \sqrt{n} = 1.90 \times 2.236 = 4.24$.

Answer: B.

Q9.9 **REASON** · about 45 s

A paired t statistic of 2.0 is obtained from 10 pairs. At 5%, the two sided and one sided tests conclude ($t_{0.025,9} = 2.262$, $t_{0.05,9} = 1.833$)

- (A) two sided rejects, one sided does not
- (B) two sided does not reject, one sided rejects
- (C) both reject
- (D) neither rejects

Solution

Fast route: $2 < 2.262$ but $2 > 1.833$.

Full working:

- $df = 10 - 1 = 9$.

Answer: B.

Q9.10 **CALC · 1.5 to 2 min**

A new ranking model beats the old one on 8 of 10 query sets and loses on 2. The two sided sign test p-value is closest to

- (A) 0.011
- (B) 0.109
- (C) 0.344
- (D) 0.055

Solution

Fast route: $P(\geq 8) = \frac{45+10+1}{1024} = 0.055$; double it.

Full working:

- Not significant at 5%, though 8 of 10 feels convincing.

Why the other options fail: (D) is the one sided value.

Answer: B.

Q9.11 **CALC · 1.5 to 2 min**

Wilcoxon signed rank test on differences 4, 6, -1, 8, -2, 5. The statistic $T = \min(T^+, T^-)$ is

- (A) 1
- (B) 21
- (C) 18
- (D) 3

Solution

Fast route: The negatives -1, -2 have the two smallest ranks: $1 + 2 = 3$.

Full working:

- Ranks of $|d|$: 1 (1), 2 (2), 3 (4), 4 (5), 5 (6), 6 (8). $T^- = 3$, $T^+ = 18$. With $n = 6$ the 5% critical value is 0, so not significant.

Why the other options fail: (C) is T^+ ; (B) is the total of all ranks.

Answer: D.

Q9.12 **CALC · 1.5 to 2 min**

Two classifiers on the same 1,000 examples: both correct 850, both wrong 110, A correct and B wrong 30, A wrong and B correct 10. McNemar's χ^2 (no correction) is

- (A) 40
- (B) 20
- (C) 2.5
- (D) 10

Solution

Fast route: $\frac{(30-10)^2}{30+10} = \frac{400}{40}$.

Full working:

- $10 > 3.841$: A is significantly better ($p \approx 0.002$). Corrected version: $\frac{19^2}{40} = 9.0$.

Answer: D.

Q9.13 RECALL · about 15 s

In McNemar's test, the 850 examples that both models classified correctly

- (A) strongly increase significance
- (B) are compared with the 110 both got wrong
- (C) must be at least 5
- (D) do not enter the statistic

Solution

Fast route: Concordant pairs carry no information about which model is better.

Full working:

- Only b and c enter.

Answer: D.

Q9.14 REASON · about 45 s

100 users are measured before and after; the mean difference is 1.2 minutes with SD of differences 5. At 5% (large sample)

- (A) $z = 1.2$, not significant
- (B) $z = 2.4$, significant
- (C) $z = 0.24$, not significant
- (D) $z = 24$, significant

Solution

Fast route: $SE = \frac{5}{10} = 0.5$; $\frac{1.2}{0.5}$.

Full working:

- Two sided $p = 0.016$.

Why the other options fail: (C) uses the SD instead of the SE.

Answer: B.

Q9.15 CALC · 1.5 to 2 min

How many pairs are needed to detect a mean difference of 4 with $\sigma_d = 10$, $\alpha = 0.05$ two sided and power 80%?

- (A) 49
- (B) 98
- (C) 25
- (D) 196

Solution**Fast route:** $\left(\frac{2.8 \times 10}{4}\right)^2 = 7^2$.**Full working:**

- 49 pairs. An independent design with SD 10 per group would need about 98 per group (196 people).

Answer: A.**Q9.16** **CALC · 1.5 to 2 min**

Two models are compared with 5 fold CV; accuracy differences (points) are 0, 4, 2, 2, 2. The naive paired t and the Nadeau Bengio corrected t (test to train ratio $\frac{1}{4}$) are

- (A) 1.41 and 0.94
 (B) 3.16 and 3.16
 (C) 3.16 and 2.11
 (D) 2.11 and 3.16

Solution**Fast route:** $\bar{d} = 2$, $s^2 = 2$. Naive SE $\sqrt{2/5}$; corrected SE $\sqrt{(0.2 + 0.25) \times 2}$.**Full working:**

- Naive: $\frac{2}{0.632} = 3.16$. Corrected: $\frac{2}{0.949} = 2.11$, below 2.776: not significant.
- Folds share 75% of their training data, so their results are correlated and the naive SE is too small.

Answer: C.**Q9.17** **RECALL · about 15 s**

A team shows a new onboarding flow to the 200 least active users, and their activity rises the next month. The first concern is

- (A) too many degrees of freedom
 (B) heteroscedasticity
 (C) multicollinearity
 (D) regression to the mean

Solution**Fast route:** Units picked for extreme values drift back towards average anyway.**Full working:**

- A randomised holdout of equally inactive users would separate the treatment effect from this drift (and from seasonality).

Answer: D.**Q9.18** **RECALL · about 15 s**

The paired t test is mathematically identical to

- (A) a one sample t test on the differences
 (B) a χ^2 test
 (C) a pooled two sample t test

(D) a two group ANOVA

Solution

Fast route: Compute d_i and test $\mu_d = 0$.

Full working:

- A two group ANOVA equals the pooled t test, not the paired one.

Answer: A.

Q9.19 **RECALL** · about 15 s

Paired differences are heavily skewed with one extreme outlier and $n = 12$. The most appropriate test is

- (A) Wilcoxon signed rank
- (B) independent t
- (C) F test
- (D) paired t

Solution

Fast route: Small n and non normal differences: rank based.

Full working:

- A sign test or permutation test on the signs would also be valid.

Answer: A.

Q9.20 **RECALL** · about 15 s

Pairing **hurts** power, compared with an independent analysis of the same number of observations, when the within pair correlation is

- (A) strongly positive
- (B) irrelevant
- (C) zero or negative
- (D) exactly 0.5

Solution

Fast route: No variance reduction, and fewer degrees of freedom.

Full working:

- $\text{Var}(B) = 2\sigma^2(1 - \rho)$ is not smaller than $2\sigma^2$ when $\rho \leq 0$.

Answer: C.

Q9.21 **RECALL** · about 15 s

Session time is measured on the same 30 users in weeks 1, 2 and 3. A suitable parametric test for any change over time is

- (A) repeated measures ANOVA
- (B) three unadjusted independent t tests
- (C) one way ANOVA treating weeks as independent groups
- (D) a χ^2 test

Solution

Fast route: Same units, three time points: repeated measures.

Full working:

- Its rank based counterpart is the Friedman test.

Answer: A.

Q9.22 **CALC** · 1.5 to 2 min

Two LLMs answer the same 400 prompts. A judge prefers model A on 150, model B on 110, and 140 are ties. Dropping ties, the sign test z statistic is closest to

- (A) 4.00
- (B) 2.00
- (C) 2.48
- (D) 1.00

Solution

Fast route: 260 decisive; mean 130, SD $\sqrt{65} = 8.06$; $\frac{20}{8.06}$.

Full working:

- Under H_0 A's wins $\sim \text{Bin}(260, 0.5)$. $z = 2.48$, $p \approx 0.013$.

Answer: C.

Q9.23 **RECALL** · about 15 s

A paired design has 25 pairs. The degrees of freedom of the paired t test are

- (A) 25
- (B) 50
- (C) 48
- (D) 24

Solution

Fast route: Pairs minus 1.

Full working:

- 25 differences, one mean estimated.

Answer: D.

Q9.24 **REASON** · about 45 s

A paired 95% CI for the mean improvement is (0.4, 2.6) from 16 pairs ($t_{0.025,15} = 2.131$). $\bar{d}$ and s_d are closest to

- (A) 1.5 and 2.06
- (B) 1.5 and 4.13
- (C) 1.5 and 1.10
- (D) 1.1 and 2.06

Solution

Fast route: Midpoint 1.5; half width $1.1 = 2.131 \times \frac{s_d}{4}$.

Full working:

- $s_d = \frac{4.4}{2.131} = 2.06$.

Why the other options fail: (C) takes the half width as the SD.

Answer: A.

Q9.25 **RECALL** · about 15 s

In a crossover study, half the recruiters use tool A then tool B, the other half B then A. Alternating the order mainly serves to

- (A) remove the need for a test
- (B) balance learning and carryover effects
- (C) increase the sample size
- (D) make the data independent

Solution

Fast route: People improve with practice; counterbalancing spreads that equally.

Full working:

- Without it, the second tool would look better simply because it came second.

Answer: B.

References for this chapter

- [Stat Trek: Paired means test](#).
- [Wikipedia: Paired difference test; Wilcoxon signed rank test; McNemar's test](#).
- Dietterich (1998), Approximate statistical tests for comparing supervised classification learning algorithms; Nadeau and Bengio (2003), Inference for the generalization error.

Chapter 10

Linear Algebra I: Matrix Operations, Rank and Null Space

Companion resource for this section: [3Blue1Brown: Essence of Linear Algebra](#) (chapters 1 to 7 cover everything in this chapter visually)

10.1 Vectors: lists of numbers that are also arrows

Start here: the everyday picture

- A candidate described by (years of experience, number of skills) = $(3, 5)$ is a **vector**. So is a word embedding with 300 numbers.
- Picture a 2 number vector as an arrow from the origin to the point $(3, 5)$. Everything in linear algebra is easier with that picture in mind.
- **Adding** vectors: add matching entries; picture: put the arrows head to tail. $(1, 2) + (3, 1) = (4, 3)$.
- **Scaling**: multiply every entry; picture: stretch the arrow. $2 \times (1, 2) = (2, 4)$. A negative number flips it.
- **Length (norm)**: $\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$ (Pythagoras). $\|(3, 4)\| = 5$. The **L1 norm** $\|x\|_1 = \sum |x_i|$ is the “taxi” distance: $\|(3, 4)\|_1 = 7$.
- **Dot product**: $x \cdot y = x^\top y = \sum x_i y_i$. $(1, 2, 3) \cdot (3, 1, 2) = 3 + 2 + 6 = 11$.
- What the dot product means: $x^\top y = \|x\| \|y\| \cos \theta$, where θ is the angle between the arrows. Positive: roughly the same direction. Zero: perpendicular (**orthogonal**). Negative: opposite directions.
- **Cosine similarity** = $\frac{x^\top y}{\|x\| \|y\|}$: direction only, length ignored. $(1, 2, 3)$ and $(3, 1, 2)$: $\frac{11}{\sqrt{14}\sqrt{14}} = 0.786$. This is how embeddings of similar resumes and job descriptions are compared.
- **Projection** of b onto the direction of a : $\frac{a^\top b}{a^\top a} a$, the “shadow” of b on the line through a . Project $(3, 2)$ onto $(1, 1)$: $\frac{5}{2}(1, 1) = (2.5, 2.5)$; the leftover $(0.5, -0.5)$ is perpendicular to $(1, 1)$.

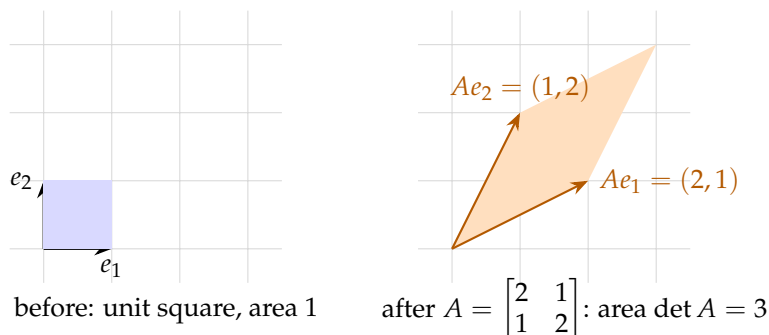
10.2 Matrices: two ways to look at a grid of numbers

- A matrix $A \in \mathbb{R}^{m \times n}$ has m rows and n columns. Entry a_{ij} sits in row i , column j .
- **Data view**: rows are samples, columns are features (the design matrix X , e.g. 10,000 users by 300 features).
- **Machine view**: A is a machine that takes an input vector x with n numbers and outputs Ax with m numbers. It is **linear**: it keeps the origin fixed and keeps grid lines straight, parallel and evenly spaced.
- **The columns tell you everything**: column j is where the j -th basis direction lands. $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ stretches the x direction by 2 and y by 3. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotates by 90° .

- **Matrix times vector** is a weighted mix of the columns, weights taken from the vector:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

The row view gives the same numbers: $17 = (1, 2) \cdot (5, 6)$, $39 = (3, 4) \cdot (5, 6)$.



10.3 Multiplying matrices

- AB means “apply B first, then A ”. The shapes must chain: $(m \times n)(n \times p) = (m \times p)$. The inner numbers must match; the outer numbers give the result’s shape.
- Entry (i, j) of AB is row i of A dotted with column j of B . Example:

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 1 \\ 0 \cdot 0 + 1 \cdot 1 & 0 \cdot 1 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}.$$

- **Not commutative:** usually $AB \neq BA$. Rotating then stretching is not stretching then rotating. In the example, $BA = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}$.
- But it is **associative**, $(AB)C = A(BC)$, and the order of evaluation changes the cost. Multiplying $(m \times n)(n \times p)$ costs mnp multiplications. For $A_{10 \times 100} B_{100 \times 5} C_{5 \times 50}$: $(AB)C$ costs $5000 + 2500 = 7,500$; $A(BC)$ costs $25000 + 50000 = 75,000$.
- **Transpose** $A^\top$: rows become columns. $(AB)^\top = B^\top A^\top$ (order reverses). **Symmetric:** $A = A^\top$ (covariance matrices, $X^\top X$).
- **Identity** I : 1s on the diagonal, 0 elsewhere; $AI = IA = A$.
- **Trace:** sum of diagonal entries. $\text{tr}(AB) = \text{tr}(BA)$, and the trace equals the sum of the eigenvalues.
- **Frobenius norm:** $\|A\|_F = \sqrt{\sum a_{ij}^2}$; for $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ it is $\sqrt{30}$.
- **Outer product** $uv^\top$: a column times a row gives a full matrix whose every column is a multiple of u , so its rank is 1.

10.3.1 NumPy conventions you will be asked about

- $A * B$ is **element wise** (Hadamard) multiplication; $A @ B$ (or `np.dot` for 2D) is the matrix product.
- **Broadcasting:** shapes are compared from the right; a dimension of size 1 is stretched to match. $(3, 1) + (1, 4)$ gives $(3, 4)$. $(3,) + (3, 1)$ gives $(3, 3)$, a common silent bug.
- To solve $Ax = b$ use `np.linalg.solve(A, b)`, not `np.linalg.inv(A) @ b`: faster and numerically safer. For least squares use `np.linalg.lstsq`.

10.4 Determinant: how much a matrix stretches area

- Meaning: the factor by which the matrix scales **area** (in 2D) or **volume** (in 3D). In the picture above, the unit square (area 1) becomes a parallelogram of area 3, so $\det A = 3$. A negative determinant means orientation is flipped (a mirror image).
- 2×2 : $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.
- 3×3 by expanding along the first row (signs $+, -, +$):

$$\det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{bmatrix} = 2 \det \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} - 1 \det \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix} + 0 = 2(11) - 1(4) = 18.$$

- $\det A = 0$ means the matrix squashes space flat (area becomes zero): the columns are dependent and A has no inverse.
- Rules: $\det(AB) = \det A \det B$; $\det A^\top = \det A$; $\det A^{-1} = \frac{1}{\det A}$; for an $n \times n$ matrix $\det(cA) = c^n \det A$ (every one of the n directions is scaled by c); determinant = product of eigenvalues; triangular matrix: product of the diagonal.

10.5 The inverse: undoing a matrix

- A^{-1} undoes A : $A^{-1}A = AA^{-1} = I$. It exists only for a square A with $\det A \neq 0$.
- 2×2 recipe: swap the diagonal, negate the off diagonal, divide by the determinant:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}.$$

- $(AB)^{-1} = B^{-1}A^{-1}$ (to undo "socks then shoes", remove shoes first).
- **Orthogonal matrix** Q : its columns are perpendicular unit vectors, so $Q^\top Q = I$ and $Q^{-1} = Q^\top$. Rotations and reflections; they preserve lengths; $\det Q = \pm 1$. They need not be symmetric.
- For a square matrix, all of these say the same thing: invertible; $\det \neq 0$; rank = n ; columns independent; $Ax = 0$ only for $x = 0$; 0 is not an eigenvalue.

10.6 Independence, rank, column space and null space (a reported InfoEdge interview topic)

- Vectors are **linearly independent** if none of them can be built from the others. $(1, 2)$ and $(2, 4)$ are dependent (the second is twice the first). In $\mathbb{R}^n$ you can have at most n independent vectors.
- **Column space** $C(A)$: every output Ax can produce, i.e. all mixes of the columns. $Ax = b$ has a solution exactly when b is in the column space.
- **Rank**: the number of independent columns = the number of independent rows = the dimension of the column space = the number of non zero singular values. It is the number of truly different directions in the data.
- **Null space** $N(A)$: every input x that the matrix sends to zero, $Ax = 0$. These are the directions the matrix cannot "see".
- **Rank nullity theorem**: for A with n columns, $\text{rank}(A) + \dim N(A) = n$. Each column is either a genuinely new direction (adds to rank) or a combination of others (adds a null space direction).
- Useful facts: $\text{rank}(A) \leq \min(m, n)$; $\text{rank}(AB) \leq \min(\text{rank} A, \text{rank} B)$; $\text{rank}(A^\top A) = \text{rank}(A)$.

10.6.1 How to find the rank and the null space by hand

1. **Row reduce:** use row operations (subtract multiples of one row from another; swap rows) to create zeros below each leading entry.
2. The number of non zero rows left (pivots) is the **rank**.
3. Write the equations from the reduced rows with right hand side 0. Variables without a pivot are **free**; express the others in terms of them. Each free variable gives one null space direction.

• Worked example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$.

- $R_2 - 2R_1 = (0, 0, 0)$. $R_3 - R_1 = (0, -1, -2)$. Two non zero rows remain: $(1, 2, 3)$ and $(0, -1, -2)$. **Rank** = 2.
- Equations: $x + 2y + 3z = 0$ and $-y - 2z = 0$. Free variable z : $y = -2z$, $x = -2y - 3z = 4z - 3z = z$. So $x = z(1, -2, 1)$.
- **Null space** = $\text{span}\{(1, -2, 1)\}$, dimension 1. Check: $1 - 4 + 3 = 0$, $2 - 8 + 6 = 0$, $1 - 2 + 1 = 0$. Rank nullity: $2 + 1 = 3$.
- Meaning: column 1 $-2 \times$ column 2 + column 3 = 0 is the exact dependency among the columns.
- What it means for data: a null space vector of a design matrix is an exact linear relation among features (for example one hot columns that add up to the intercept column, the **dummy variable trap**). Regression coefficients along that direction cannot be identified.

Try it now (answer right below, cover it first)

Find the rank and null space of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

Answer: Row 2 = $2 \times$ row 1: rank 1. $x + 2y = 0$: null space $\text{span}\{(2, -1)\}$. $1 + 1 = 2$ columns.

10.7 Solving $Ax = b$ and least squares

- Three possibilities: exactly one solution (square, full rank); no solution (b outside the column space); infinitely many (a non trivial null space: any solution plus any null space vector is again a solution).
- A 3×5 matrix has rank at most 3, so its null space has dimension at least 2: $Ax = 0$ always has non zero solutions.
- **Least squares:** in regression $y \approx X\beta$ usually has no exact solution. We pick $\hat{\beta}$ so that $X\hat{\beta}$ is the point of the column space closest to y : the projection. The leftover (residual) is perpendicular to every column: $X^\top(y - X\hat{\beta}) = 0$, the **normal equations**

$$X^\top X \hat{\beta} = X^\top y, \quad \hat{\beta} = (X^\top X)^{-1} X^\top y \quad (\text{when } X \text{ has full column rank}).$$

- Worked: points $(0, 1), (1, 2), (2, 4)$ with $X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$. $X^\top X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$ (entries $n, \sum x, \sum x^2$),

$$X^\top y = \begin{bmatrix} 7 \\ 10 \end{bmatrix} \quad (\sum y, \sum xy). \text{ Solve } 3a + 3b = 7, 3a + 5b = 10: b = 1.5, a = 0.833.$$

- The **hat matrix** $H = X(X^\top X)^{-1} X^\top$ maps y to the fitted values. It is a projection: $H^2 = H$ and $H^\top = H$.

10.8 Positive definite matrices and matrix gradients

- A symmetric A is **positive semidefinite** (PSD) if $x^\top Ax \geq 0$ for every x , and **positive definite** if > 0 for every non zero x . Equivalently, all eigenvalues are ≥ 0 (or > 0). Covariance matrices, $X^\top X$ and kernel matrices are PSD.
- Quick test for a symmetric 2×2 : positive definite iff the top left entry is positive and the determinant is positive. $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: yes. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$: determinant -3 , no.
- Gradients you meet in ML: $\nabla_x(a^\top x) = a$; $\nabla_x(x^\top Ax) = (A + A^\top)x$, which is $2Ax$ when A is symmetric; $\nabla_\beta \|y - X\beta\|^2 = -2X^\top(y - X\beta)$.
- A gradient always has the shape of the thing you differentiate with respect to. For a layer $Y = XW$ with X of shape 32×768 and W of shape 768×256 , $\frac{\partial L}{\partial W} = X^\top \frac{\partial L}{\partial Y}$ has shape 768×256 .

Where this shows up in ML and at a company like Naukri

- A dense layer is a matrix product; attention is $\text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$, two matrix products.
- An embedding lookup is a one hot vector times an embedding matrix.
- Duplicate or combined features make $X^\top X$ singular; ridge regression uses $X^\top X + \lambda I$, which is always invertible.
- Low rank structure is everywhere: user by job interaction matrices, LoRA fine tuning ($\Delta W = BA$ with a small inner dimension), compressed layers.

Formula sheet

$$(m \times n)(n \times p) \rightarrow m \times p, \text{cost } mnp \quad (AB)^\top = B^\top A^\top \quad (AB)^{-1} = B^{-1}A^{-1} \quad \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\det(cA) = c^n \det A \quad \det(AB) = \det A \det B \quad \text{rank} + \text{nullity} = n \quad \text{rank}(A^\top A) = \text{rank} A$$

$$\hat{\beta} = (X^\top X)^{-1} X^\top y \quad \text{proj} = \frac{a^\top b}{a^\top a} a \quad \cos \theta = \frac{x^\top y}{\|x\| \|y\|} \quad \nabla(x^\top Ax) = (A + A^\top)x \quad \|A\|_F^2 = \sum a_{ij}^2$$

15 second recall card: memorise every line

- Shapes chain as $(m \times n)(n \times p) = (m \times p)$; $AB \neq BA$ in general; $(AB)^\top = B^\top A^\top$; $(AB)^{-1} = B^{-1}A^{-1}$.
- $\det$ is the area (volume) scale; $\det = 0$ means not invertible; $\det(cA) = c^n \det A$; $\det =$ product of eigenvalues; trace = sum.
- 2×2 inverse: swap, negate, divide by $\det$.
- Rank = independent columns = independent rows = number of non zero singular values. Rank + nullity = number of columns.
- Null space = inputs sent to 0; non trivial null space means infinitely many solutions and unidentifiable coefficients (dummy variable trap).
- $A^\top A$ is symmetric PSD with the same rank as A ; invertible only with full column rank.
- Orthogonal: $Q^{-1} = Q^\top$, lengths preserved, $\det = \pm 1$, not necessarily symmetric.
- $*$ is element wise, $@$ is matrix product; broadcasting $(3, 1) + (1, 4) \rightarrow (3, 4)$; use solve, not inv.
- Outer product $uv^\top$ has rank 1. Hat matrix: $H^2 = H$.

10.9 Question bank: Matrix operations

Q10.1 CALC · 1.5 to 2 min

★ PYQ (INT and OA: rank of a matrix) The rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ is

- (A) 3
- (B) 2
- (C) 1
- (D) 0

Solution

Fast route: Row 2 = $2 \times$ row 1, so rank ≤ 2 ; rows 1 and 3 are clearly not multiples: rank 2.

Full working:

- $R_2 - 2R_1 = (0, 0, 0)$; $R_3 - R_1 = (0, -1, -2)$. Two pivots.
- Cross check: $\det A = 0$, so rank < 3 .

Why the other options fail: (A) would need three independent rows. (C) would need every row to be a multiple of one row.

Answer: B.

Q10.2 CALC · 1.5 to 2 min

★ PYQ (INT and OA: null space) For the same A , a basis of the null space is

- (A) $\{(1, -2, 1)\}$
- (B) $\{(2, -1, 0), (3, 0, -1)\}$
- (C) $\{(1, 2, 1)\}$
- (D) the null space is $\{0\}$

Solution

Fast route: Check each option by multiplying: $(1, -2, 1)$ gives $1 - 4 + 3 = 0$, $2 - 8 + 6 = 0$, $1 - 2 + 1 = 0$.

Full working:

- From the reduced rows: $y = -2z$, $x = z$. Dimension 1 = $3 - \text{rank}$ (rank nullity).

Why the other options fail: (B) has two vectors, but the null space has dimension 1; $(2, -1, 0)$ also fails row 3 ($2 - 1 = 1 \neq 0$). (D) is impossible because rank < 3 .

Answer: A.

Q10.3 RECALL · about 15 s

★ PYQ (OA: matrices (IIT Roorkee paper reported “matrices” questions)) For any real matrix $A \in \mathbb{R}^{m \times n}$, which is always true?

- (A) $\text{rank}(A^T A) = n$
- (B) $A^T A$ is symmetric and positive semidefinite
- (C) $A^T A = A A^T$
- (D) $A^T A$ is invertible

Solution

Fast route: $(A^T A)^T = A^T A$ and $x^T A^T A x = \|Ax\|^2 \geq 0$.

Full working:

- Its rank equals $\text{rank}(D)$, which may be less than n .

Why the other options fail: (D) and (A) need full column rank. (C) the two even have different shapes when $m \neq n$.

Answer: B.

Q10.4 REASON · about 45 s

For $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, the product AB is

- (A) $\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}$
 (B) $\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$
 (C) $\begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}$
 (D) $\begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$

Solution

Fast route: Row 1 of A times columns of B : $(0 + 2, 1 + 2)$.

Full working:

- Row 2: $(0 + 1, 0 + 1)$.

Why the other options fail: (C) is BA . (A) multiplies entry by entry.

Answer: B.

Q10.5 REASON · about 45 s

The inverse of $\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ is

- (A) $\begin{bmatrix} 0.6 & 0.7 \\ 0.2 & 0.4 \end{bmatrix}$
 (B) $\begin{bmatrix} 0.25 & 0.14 \\ 0.5 & 0.17 \end{bmatrix}$
 (C) $\begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$
 (D) $\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$

Solution

Fast route: $\det = 24 - 14 = 10$; swap, negate, divide by 10.

Full working:

- Check: $4(0.6) + 7(-0.2) = 1$.

Why the other options fail: (D) forgets to divide by the determinant. (B) inverts each entry. (A)

forgets the signs.

Answer: C.

Q10.6 CALC · 1.5 to 2 min

$\det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{bmatrix}$ is

- (A) 18
- (B) 26
- (C) 14
- (D) 22

Solution

Fast route: $2(12 - 1) - 1(4 - 0) + 0$.

Full working:

- $22 - 4 = 18$.

Why the other options fail: (D) stops after the first term. (B) adds the second term instead of subtracting.

Answer: A.

Q10.7 RECALL · about 15 s

A is 5×5 with $\det A = 1$. $\det(2A)$ is

- (A) 10
- (B) 2
- (C) 25
- (D) 32

Solution

Fast route: $c^n \det A = 2^5$.

Full working:

- Stretching all 5 directions by 2 multiplies volume by 32.

Answer: D.

Q10.8 RECALL · about 15 s

For a square $n \times n$ matrix, which condition is **not** equivalent to “ A is invertible”?

- (A) $\text{rank}(A) = n$
- (B) $Ax = 0$ has only the solution $x = 0$
- (C) $\det A \neq 0$
- (D) all diagonal entries are non zero

Solution

Fast route: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has non zero diagonal but is singular.

Full working:

- Non zero diagonal is enough only for triangular matrices.

Answer: D.

Q10.9 **RECALL · about 15 s**

A is 5×7 with rank 4. The dimension of its null space is

- (A) 4
- (B) 1
- (C) 3
- (D) 2

Solution

Fast route: Columns minus rank: $7 - 4$.

Full working:

- The column space is a 4 dimensional subspace of $\mathbb{R}^5$.

Why the other options fail: (B) subtracts from the number of rows.

Answer: C.

Q10.10 **RECALL · about 15 s**

For a 3×5 matrix A , which is certainly true?

- (A) A has an inverse
- (B) $Ax = b$ has a solution for every b
- (C) the rank is 3
- (D) $Ax = 0$ has a non zero solution

Solution

Fast route: Rank ≤ 3 , so nullity $\geq 5 - 3 = 2$.

Full working:

- More unknowns than equations always leaves free directions.

Why the other options fail: (B) and (C) hold only if the rank is 3. (A) non square matrices have no inverse.

Answer: D.

Q10.11 **CALC · 1.5 to 2 min**

The least squares line $y = a + bx$ through $(0, 1), (1, 2), (2, 4)$ is

- (A) $y = 0.833 + 1.5x$
- (B) $y = 0.833 + 1.25x$
- (C) $y = 1 + 1.25x$
- (D) $y = 1 + 1.5x$

Solution

Fast route: $X^T X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $X^T y = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$; subtract equations: $2b = 3$.

Full working:

- $b = 1.5; a = \frac{7-4.5}{3} = 0.833$. Check with Chapter 6: $b = \frac{S_{xy}}{S_{xx}} = \frac{3}{2}$.

Answer: A.

Q10.12 **CALC · 1.5 to 2 min**

A is 10×100 , B is 100×5 , C is 5×50 . The numbers of scalar multiplications for $(AB)C$ and $A(BC)$ are

- (A) 5,000 and 25,000
- (B) 50,000 and 50,000
- (C) 7,500 and 75,000
- (D) 75,000 and 7,500

Solution

Fast route: $(AB)C$: $10 \cdot 100 \cdot 5 + 10 \cdot 5 \cdot 50$. $A(BC)$: $100 \cdot 5 \cdot 50 + 10 \cdot 100 \cdot 50$.

Full working:

- $5000 + 2500 = 7500$ vs $25000 + 50000 = 75000$. Same result, ten times the cost.

Answer: C.

Q10.13 **RECALL · about 15 s**

In NumPy, a has shape (3,1) and b has shape (1,4). $(a+b)$.shape is

- (A) (3,4)
- (B) an error
- (C) (1,1)
- (D) (4,3)

Solution

Fast route: Size 1 dimensions stretch: rows from a, columns from b.

Full working:

- Broadcasting compares shapes from the right.

Answer: A.

Q10.14 **RECALL · about 15 s**

A and B are NumPy arrays of shape (2,2). Which is correct?

- (A) $A*B$ and $A@B$ are always equal for square matrices
- (B) $A@B$ is element wise
- (C) $A*B$ is the matrix product
- (D) $A@B$ is the matrix product and $A*B$ is element wise

Solution

Fast route: @ is matmul; * is element wise.

Full working:

- $\text{np.dot}(A,B)$ also gives the matrix product for 2D arrays.

Answer: D.

Q10.15 REASON · about 45 s

The cosine similarity of the embeddings $(1, 2, 3)$ and $(3, 1, 2)$ is closest to

- (A) 0.79
- (B) 1.00
- (C) 0.50
- (D) 0.93

Solution

Fast route: Dot 11; both norms $\sqrt{14}$; $\frac{11}{14}$.

Full working:

- 0.786.

Answer: A.

Q10.16 REASON · about 45 s

The projection of $b = (3, 2)$ onto the direction $a = (1, 1)$ is

- (A) $(1.5, 1)$
- (B) $(3, 3)$
- (C) $(5, 5)$
- (D) $(2.5, 2.5)$

Solution

Fast route: $\frac{a^\top b}{a^\top a} a = \frac{5}{2}(1, 1)$.

Full working:

- The residual $(0.5, -0.5)$ is perpendicular to a .

Why the other options fail: (C) forgets to divide by $a^\top a$.

Answer: D.

Q10.17 RECALL · about 15 s

The Frobenius norm of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is

- (A) 10
- (B) 5
- (C) $\sqrt{30}$
- (D) 30

Solution

Fast route: $\sqrt{1 + 4 + 9 + 16}$.

Full working:

- ≈ 5.48 . (Its trace is 5.)

Answer: C.

Q10.18 RECALL · about 15 s

For invertible square matrices A and B , which identity is correct?

- (A) $(AB)^{-1} = B^{-1}A^{-1}$
 (B) $(AB)^{-1} = A^{-1}B^{-1}$
 (C) $\det(A+B) = \det A + \det B$
 (D) $(AB)^{\top} = A^{\top}B^{\top}$

Solution

Fast route: Undo in reverse order.

Full working:

- $(AB)(B^{-1}A^{-1}) = AIA^{-1} = I.$

Why the other options fail: (D) should be $B^{\top}A^{\top}$. (C) fails for $A = B = I_2$: $4 \neq 2$.

Answer: A.

Q10.19 **CALC · 1.5 to 2 min**

Is $M = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$ invertible?

- (A) No, $\det M = 0$
 (B) Yes, $\det M = -2$
 (C) Yes, $\det M = 4$
 (D) Yes, $\det M = 2$

Solution

Fast route: Expand along row 1: $2(3-2) - 0 + 1(1-3) = 0$.

Full working:

- Column 3 = $\frac{1}{2}(\text{column 1} + \text{column 2})$: $\frac{1}{2}(2+0, 1+3, 1+1) = (1, 2, 1).$

Answer: A.

Q10.20 **RECALL · about 15 s**

A regression uses an intercept plus one hot columns for all 3 cities. The main consequence is

- (A) the R^2 becomes negative
 (B) the design matrix does not have full column rank, so the OLS solution is not unique
 (C) nothing; it is the recommended encoding
 (D) the model becomes non linear

Solution

Fast route: The three dummies add up to the intercept column: an exact dependency.

Full working:

- $X^{\top}X$ is singular. Fix: drop one city (reference category), or use ridge.

Answer: B.

Q10.21 **RECALL · about 15 s**

For a general (not necessarily symmetric) square matrix A , the gradient of $f(x) = x^{\top}Ax$ is

- (A) $(A + A^{\top})x$
 (B) Ax

- (C) $2Ax$
 (D) $A^T x$

Solution

Fast route: Both copies of x contribute; it reduces to $2Ax$ only if A is symmetric.

Full working:

$$\bullet \frac{\partial}{\partial x_k} \sum_{ij} a_{ij} x_i x_j = \sum_j a_{kj} x_j + \sum_i a_{ik} x_i.$$

Answer: A.

Q10.22 RECALL · about 15 s

A layer computes $Y = XW$ with X of shape 32×768 and Y of shape 32×256 . The shape of $\frac{\partial L}{\partial W}$ is

- (A) 32×256
 (B) 256×768
 (C) 768×256
 (D) 32×768

Solution

Fast route: A gradient has the shape of the parameter: W is 768×256 .

Full working:

$$\bullet X^T \frac{\partial L}{\partial Y}: (768 \times 32)(32 \times 256).$$

Answer: C.

Q10.23 REASON · about 45 s

Which matrix is positive definite?

- (A) $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$
 (B) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
 (C) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$
 (D) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Solution

Fast route: Symmetric 2×2 : top left > 0 and $\det > 0$.

Full working:

- $\bullet$ (B): $2 > 0$, $\det = 3 > 0$. Eigenvalues 3 and 1.

Why the other options fail: (C) $\det = -3$ (indefinite). (D) $\det = 0$ (only semidefinite). (A) negative top left entry.

Answer: B.

Q10.24 RECALL · about 15 s

For an orthogonal matrix Q , which statement is **false**?

- (A) $\|Qx\| = \|x\|$ for every x

- (B) $\det Q = \pm 1$
 (C) Q must be symmetric
 (D) $Q^{-1} = Q^T$

Solution

Fast route: A 90° rotation is orthogonal but not symmetric.

Full working:

- $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has transpose $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \neq Q$.

Answer: C.

Q10.25 **RECALL** · about 15 s

For non zero vectors $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$ with $n \geq 2$, the outer product uv^T

- (A) is always symmetric
 (B) has rank 1
 (C) is invertible when $m = n$
 (D) has rank $\min(m, n)$

Solution

Fast route: Every column is a multiple of u .

Full working:

- Its null space is every vector perpendicular to v (dimension $n - 1$). LoRA updates are sums of a few such rank 1 pieces.

Answer: B.

Q10.26 **RECALL** · about 15 s

$\text{rank}(A) = 3$ and $\text{rank}(B) = 5$, with AB defined. The rank of AB is

- (A) at least 5
 (B) exactly 3
 (C) exactly 5
 (D) at most 3

Solution

Fast route: $\text{rank}(AB) \leq \min(\text{rank} A, \text{rank} B)$.

Full working:

- A product can only lose information, never add directions.

Answer: D.

Q10.27 **RECALL** · about 15 s

To solve $Ax = b$ for a large, well conditioned square A in NumPy, the recommended call is

- (A) A / b
 (B) $b @ A$
 (C) $\text{np.linalg.solve}(A, b)$
 (D) $\text{np.linalg.inv}(A) @ b$

Solution

Fast route: Solve directly; never form the inverse.

Full working:

- solve uses an LU factorisation: faster and more accurate. For least squares use `lstsq`.

Answer: C.

Q10.28 RECALL · about 15 s

The hat matrix $H = X(X^\top X)^{-1}X^\top$ of linear regression satisfies

- (A) $\det H = 1$
- (B) $H^2 = 0$
- (C) $H = I$ always
- (D) $H^2 = H$ and $H^\top = H$

Solution

Fast route: It is an orthogonal projection: projecting twice changes nothing.

Full working:

- Its trace equals the number of parameters.

Answer: D.

Q10.29 REASON · about 45 s

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ equals

- (A) (23, 34)
- (B) (17, 39)
- (C) (11, 39)
- (D) (5, 24)

Solution

Fast route: $5 \times \text{column 1} + 6 \times \text{column 2} = (5 + 12, 15 + 24)$.

Full working:

- Row view: $(1, 2) \cdot (5, 6) = 17$, $(3, 4) \cdot (5, 6) = 39$.

Why the other options fail: (A) uses the columns as rows (computes $A^\top x$).

Answer: B.

References for this chapter

- [3Blue1Brown: Essence of linear algebra](#), especially [matrix multiplication as composition](#), the [determinant](#) and [column space](#) and [null space](#).
- [MIT 18.06, Gilbert Strang](#) (lectures 6 to 10 for rank and subspaces).
- [Mathematics for Machine Learning](#) (Deisenroth, Faisal, Ong), free book, chapter 2.

Chapter 11

Linear Algebra II: Eigenvalues and Eigendecomposition

Companion resource for this section: [3Blue1Brown: Eigenvectors and Eigenvalues](#) (Essence of Linear Algebra, chapter 14)

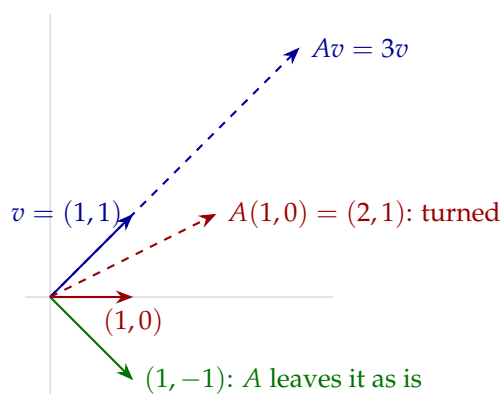
11.1 The idea: directions a matrix does not turn

Start here: the everyday picture

- Spin a globe. Every point on its surface moves, except the points on the axis: they stay on the same line. The axis is the one direction the rotation leaves alone.
- A matrix moves almost every vector off its line. But a few special directions only get **stretched, shrunk or flipped**, never turned. Those directions are the matrix's **eigenvectors**, and the stretch factor is the **eigenvalue**. ("Eigen" is German for "own": the matrix's own directions.)

$$Av = \lambda v, \quad v \neq 0.$$

- Example: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
 - $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$: the direction $(1, 1)$ is only stretched, by 3.
 - $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$: the direction $(1, -1)$ is left unchanged (eigenvalue 1).
 - $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$: this vector is turned, so $(1, 0)$ is not an eigenvector.
- Reading an eigenvalue: $\lambda > 1$ stretch; $0 < \lambda < 1$ shrink; $\lambda < 0$ flip; $\lambda = 0$ crush to zero (that eigenvector is in the null space); a complex λ means rotation, where no real direction survives. A 90° rotation has eigenvalues $\pm i$.
- Eigenvectors are defined only up to length: $(1, 1)$, $(2, 2)$ and $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ are the same eigen direction. Software returns unit length vectors (and the sign is arbitrary).



11.2 How to find them

1. Rewrite $Av = \lambda v$ as $(A - \lambda I)v = 0$. A non zero v can solve this only if $A - \lambda I$ squashes space, i.e. $\det(A - \lambda I) = 0$: the **characteristic equation**.
2. Solve it for λ .
3. For each λ , find the null space of $A - \lambda I$ (Chapter 10): those are its eigenvectors.
 - 2×2 **shortcut**: the equation is $\lambda^2 - (\text{tr} A)\lambda + \det A = 0$. With $m = \frac{\text{tr} A}{2}$ (the mean of the two eigenvalues) and $p = \det A$ (their product): $\lambda = m \pm \sqrt{m^2 - p}$.
 - Example: $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$: $m = 3.5$, $p = 12 - 2 = 10$; $\lambda = 3.5 \pm \sqrt{12.25 - 10} = 3.5 \pm 1.5$, so 5 and 2.
 - For $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$, giving $-x + y = 0$: $v = (1, 1)$.
 - For $\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$, giving $2x + y = 0$: $v = (1, -2)$.
 - **Two checks that are always true** (and solve many MCQs in seconds): the sum of eigenvalues equals the **trace**; the product equals the **determinant**.
 - **Triangular or diagonal** matrices: the eigenvalues are the diagonal entries.
 - Example with a check: a 3×3 matrix with trace 6, determinant 0 and one eigenvalue 2. The product is 0, so another eigenvalue is 0; the sum gives the third: $6 - 2 - 0 = 4$.

Try it now (answer right below, cover it first)

Find the eigenvalues of $\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$.

Answer: trace 7, det 6: $\lambda^2 - 7\lambda + 6 = 0$, so 6 and 1.

11.3 Eigendecomposition: the matrix in its own coordinates

- If an $n \times n$ matrix has n independent eigenvectors, put them as the columns of V and the eigenvalues on the diagonal of Λ . Then $AV = V\Lambda$, so

$$A = V\Lambda V^{-1}.$$

- Meaning: switch to the eigen directions (V^{-1}), stretch each one by its eigenvalue (Λ), switch back (V). In the right coordinates the matrix is just a diagonal stretch.
- **Powers become easy**: $A^k = V\Lambda^k V^{-1}$, because the inner $V^{-1}V$ pairs cancel. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (eigenvalues 3 and 1): $A^{10} = \frac{1}{2} \begin{bmatrix} 3^{10} + 1 & 3^{10} - 1 \\ 3^{10} - 1 & 3^{10} + 1 \end{bmatrix}$, so the top left entry is $\frac{59049+1}{2} = 29525$.
- **Related matrices share eigenvectors**: A^{-1} has eigenvalues $\frac{1}{\lambda}$; $A + cI$ has $\lambda + c$; cA has $c\lambda$; A^k has λ^k .
- **Not every matrix can be diagonalised**. The shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has eigenvalue 1 twice but only one eigen direction, $(1, 0)$: it is **defective**. Distinct eigenvalues guarantee enough eigenvectors.

11.4 Symmetric matrices: the well behaved family

- If $A = A^\top$ (real symmetric), the **spectral theorem** promises: all eigenvalues are **real**; eigenvectors for different eigenvalues are **perpendicular**; and A can always be diagonalised with an orthogonal matrix:

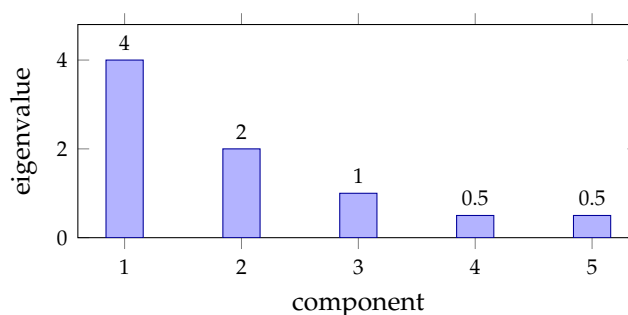
$$A = Q\Lambda Q^\top = \sum_i \lambda_i q_i q_i^\top.$$

- Each $q_i q_i^\top$ is a rank 1 piece. Example: eigenvalues 4 and 1 with unit eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$ rebuild $4 \cdot \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 1 \cdot \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$.
- Covariance matrices, correlation matrices, $X^\top X$, Hessians, graph Laplacians and kernel matrices are all symmetric, which is why eigenvalues are everywhere in ML.
- PSD $\iff$ all $\lambda \geq 0$; positive definite $\iff$ all $\lambda > 0$.
- A rank 1 matrix $uu^\top$: u is an eigenvector with eigenvalue $\|u\|^2$ (because $uu^\top u = u\|u\|^2$), and every vector perpendicular to u has eigenvalue 0. For $u = (1, 2, 2)$: eigenvalues 9, 0, 0.
- Rayleigh quotient**: over all unit vectors, $x^\top A x$ is largest ($= \lambda_{\max}$) at the top eigenvector. This one fact is the mathematics of PCA.

11.5 PCA: what eigenvectors and eigenvalues mean (a reported InfoEdge OA and interview topic)

Start here: the everyday picture

- Imagine a cloud of data points shaped like a cigar lying at an angle. If you had to describe each point with one number, you would measure its position along the length of the cigar, because that is where points differ most. That long axis is the first principal component.
- Centre the data (subtract each feature's mean) and compute the covariance matrix S (symmetric, PSD).
- The variance of the data projected on a unit direction w is $w^\top S w$. Maximising it gives the top eigenvector (Rayleigh quotient).
- Eigenvectors of S** are the **principal directions** (loadings): orthogonal axes of the cloud, ordered by importance.
- Eigenvalues of S** are the **variances of the data along those directions**. The **explained variance ratio** of component k is $\frac{\lambda_k}{\sum_j \lambda_j}$ (and $\sum_j \lambda_j$ is the trace, the total variance).
- Worked: $S = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ has eigenvalues 6 and 1. PC1: $(S - 6I)v = 0$ gives $-x + 2y = 0$, so PC1 $\propto (2, 1)$. It explains $\frac{6}{7} = 85.7\%$ of the variance.
- Scree plot**: eigenvalues in decreasing order. Keep the components before the "elbow", or enough to reach a cumulative threshold such as 90%.
- Standardise first** when features have different units; otherwise the feature with the biggest numbers (salary in rupees) dominates PC1. PCA on standardised data is the eigendecomposition of the **correlation** matrix.



Scree plot for eigenvalues 4, 2, 1, 0.5, 0.5 (total 8): PC1 explains 50%, two components 75%, three 87.5%.

11.6 Other places eigenvalues rule

- **Markov chains:** a long run (stationary) distribution π satisfies $\pi P = \pi$, an eigenvector with eigenvalue 1. Example: active users stay active with 0.9 and go dormant with 0.1; dormant users return with 0.5. Balance: $0.1\pi_a = 0.5\pi_d$, so $\pi_a = \frac{5}{6}$. PageRank is the same idea on the web graph.
- **Power iteration:** repeat $x \leftarrow \frac{Ax}{\|Ax\|}$; it converges to the top eigenvector, at a speed set by $\left| \frac{\lambda_2}{\lambda_1} \right|$.
- **Optimisation:** the eigenvalues of the Hessian (matrix of second derivatives) describe curvature. All positive: a minimum. Mixed signs: a **saddle point** (very common in deep networks). Gradient descent on a quadratic converges only if the learning rate $\eta < \frac{2}{\lambda_{\max}}$, and its speed is limited by the **condition number** $\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$.
- **Ridge regression:** $X^T X + \lambda I$ shifts every eigenvalue up by λ , so the tiny eigenvalues caused by multicollinearity no longer blow up the inverse.
- **RNNs:** repeated multiplication by a weight matrix scales signals by λ^t : $|\lambda| > 1$ explodes, $|\lambda| < 1$ vanishes.
- **Graphs:** the number of zero eigenvalues of the graph Laplacian $L = D - A$ equals the number of connected components (spectral clustering).

Where this shows up in ML and at a company like Naukri

- PCA compresses features, removes correlation between them and visualises embeddings.
- Feature scaling helps gradient descent because it reduces the condition number.
- Covariance eigenvalues near zero reveal redundant features; ridge fixes the resulting instability.

Formula sheet

$Av = \lambda v$ $\det(A - \lambda I) = 0$ $2 \times 2: \lambda = m \pm \sqrt{m^2 - p}$ with $m = \frac{\text{tr}}{2}$, $p = \det$ $\sum \lambda = \text{tr} A$,
 $\prod \lambda = \det A$ $A = V \Lambda V^{-1}$, $A^k = V \Lambda^k V^{-1}$ symmetric: $A = Q \Lambda Q^T$ $A^{-1} \rightarrow \frac{1}{\lambda}$, $A + cI \rightarrow \lambda + c$
 explained variance $\frac{\lambda_k}{\sum \lambda}$ GD stable if $\eta < \frac{2}{\lambda_{\max}}$ $\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$ uu^T : eigenvalue $\|u\|^2$ once, else 0.

15 second recall card: memorise every line

- PCA: eigenvectors of the covariance = orthogonal directions of maximum variance; eigenvalues = variance along each; ratio $\frac{\lambda_k}{\sum \lambda}$.
- Sum of eigenvalues = trace; product = determinant; triangular: the diagonal.
- $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: 3 and 1 with (1, 1) and (1, -1). $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$: 5 and 2.

- Symmetric: real eigenvalues, orthogonal eigenvectors, always diagonalisable. Shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is not diagonalisable.
- 90° rotation: eigenvalues $\pm i$.
- A^{-1} : $\frac{1}{\lambda}$; $A + cI$: $\lambda + c$; A^k : λ^k (same eigenvectors).
- Hessian with mixed signs: saddle. GD needs $\eta < \frac{2}{\lambda_{\max}}$. Ridge adds λ to every eigenvalue.
- Laplacian zero eigenvalues = number of connected components. Stationary distribution: eigenvalue 1.
- Standardise before PCA; standardised PCA uses the correlation matrix.

11.7 Question bank: Eigenvalues and eigendecomposition

Q11.1 RECALL · about 15 s

★ **PYQ (OA: meaning of eigenvectors and eigenvalues in PCA)** In PCA on centred data with covariance matrix S , which statement is correct?

- (A) Eigenvectors of S are the directions of maximum variance, and each eigenvalue is the variance of the data along its eigenvector
- (B) Principal directions are generally not orthogonal
- (C) The eigenvector with the smallest eigenvalue is the most informative direction
- (D) Eigenvalues give the mean of each principal component

Solution

Fast route: Directions = eigenvectors; variances = eigenvalues; orthogonal because S is symmetric.

Full working:

- For a unit eigenvector q : $\text{Var}(Xq) = q^\top S q = \lambda$.

Why the other options fail: (C) the smallest eigenvalue is the direction of least spread, usually noise. (D) the projected data is centred, so every component has mean 0. (B) the spectral theorem makes them orthogonal.

Answer: A.

Q11.2 REASON · about 45 s

★ **PYQ (INT and OA: explained variance and the scree plot)** PCA gives eigenvalues 4, 2, 1, 0.5, 0.5. The variance explained by the first two components, and the number of components needed for at least 85%, are

- (A) 75% and 2
- (B) 60% and 2
- (C) 75% and 3
- (D) 50% and 4

Solution

Fast route: Total 8; $\frac{6}{8} = 75\%$; three give $\frac{7}{8} = 87.5\%$.

Full working:

- Two components reach only 75%, below 85%, so three are needed.

Why the other options fail: (D) gives PC1 alone and overshoots the count.

Answer: C.

Q11.3 REASON · about 45 s

The eigenvalues and eigenvectors of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are

- (A) 3 with $(1, -1)$ and 1 with $(1, 1)$
- (B) 3 with $(1, 1)$ and 1 with $(1, -1)$
- (C) 2 with $(1, 0)$ and 2 with $(0, 1)$
- (D) 4 with $(1, 1)$ and 0 with $(1, -1)$

Solution

Fast route: Trace 4, det 3: 3 and 1. Test $(1, 1)$: $A(1, 1) = (3, 3)$.

Full working:

- The two eigenvectors are perpendicular, as for every symmetric matrix.

Answer: B.

Q11.4 REASON · about 45 s

The eigenvalues of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ are

- (A) 6 and 1
- (B) 5 and 2
- (C) 4 and 3
- (D) 7 and 0

Solution

Fast route: Mean 3.5, product 10: 3.5 ± 1.5 .

Full working:

- Check: $5 + 2 = 7 = \text{trace}$, $5 \times 2 = 10 = \text{det}$.

Why the other options fail: (C) reads the diagonal, which works only for triangular matrices.

Answer: B.

Q11.5 RECALL · about 15 s

The eigenvalues of $\begin{bmatrix} 3 & 7 & -2 \\ 0 & 5 & 1 \\ 0 & 0 & -1 \end{bmatrix}$ are

- (A) 0, 0, 0
- (B) 3, 7, -2
- (C) cannot be found without expanding
- (D) 3, 5, -1

Solution

Fast route: Upper triangular: read the diagonal.

Full working:

- $\det(A - \lambda I) = (3 - \lambda)(5 - \lambda)(-1 - \lambda).$

Answer: D.

Q11.6 REASON · about 45 s

A 3×3 matrix has trace 6 and determinant 0, and one eigenvalue is 2. Its eigenvalues are

- (A) 2, 2, 2
- (B) 0, 0, 6
- (C) 0, 2, 4
- (D) 1, 2, 3

Solution

Fast route: Product 0 forces a 0; sum 6 gives $6 - 2 - 0 = 4$.

Full working:

- The matrix is singular (a zero eigenvalue).

Why the other options fail: (A) and (D) have product 8 and 6, not 0.

Answer: C.

Q11.7 CALC · 1.5 to 2 min

The top left entry of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}^{10}$ is

- (A) 59049
- (B) 29524
- (C) 1024
- (D) 29525

Solution

Fast route: $\frac{3^{10}+1^{10}}{2} = \frac{59050}{2}.$

Full working:

- $A^{10} = 3^{10}q_1q_1^\top + 1^{10}q_2q_2^\top$ with $q_1q_1^\top = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, q_2q_2^\top = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$

Why the other options fail: (B) is the off diagonal entry $\frac{3^{10}-1}{2}.$

Answer: D.

Q11.8 CALC · 1.5 to 2 min

A job seeker is “active” or “dormant” each week. From active: stay active 0.9, become dormant 0.1. From dormant: become active 0.5, stay dormant 0.5. The long run share of active users is

- (A) 0.83
- (B) 0.64
- (C) 0.50
- (D) 0.90

Solution

Fast route: Flow balance: $0.1\pi_a = 0.5\pi_d$, so $\pi_a = 5\pi_d$: $\frac{5}{6}$.

Full working:

- The stationary distribution is the left eigenvector of the transition matrix with eigenvalue 1. The other eigenvalue is $0.9 + 0.5 - 1 = 0.4$, so the chain settles quickly.

Answer: A.

Q11.9 **CALC · 1.5 to 2 min**

A covariance matrix is $S = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$. The first principal direction and its explained variance are

- (A) $\propto (1, -2)$, 14.3%
- (B) $\propto (2, 1)$, 85.7%
- (C) $\propto (1, 2)$, 85.7%
- (D) $\propto (2, 1)$, 71.4%

Solution

Fast route: Eigenvalues 6 and 1; $(S - 6I)$ gives $-x + 2y = 0$.

Full working:

- PC1 $\propto (2, 1)$, explains $\frac{6}{7} = 85.7\%$. PC2 $\propto (1, -2)$ explains 14.3%.

Why the other options fail: (D) uses $\frac{5}{7}$, the share of the first feature's variance, not of the component.

Answer: B.

Q11.10 **RECALL · about 15 s**

The matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

- (A) is not invertible
- (B) has eigenvalues 1 and 0
- (C) is not diagonalisable
- (D) has two independent eigenvectors

Solution

Fast route: Double eigenvalue 1 but only one eigen direction $(1, 0)$: defective.

Full working:

- $A - I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has a one dimensional null space. It is invertible ($\det = 1$).

Answer: C.

Q11.11 **RECALL · about 15 s**

A has eigenvalues 2 and 3. Which statement is correct?

- (A) A^{-1} has eigenvalues $\frac{1}{2}$ and $\frac{1}{3}$ with the same eigenvectors
- (B) $A^\top$ always has the same eigenvectors as A
- (C) A^2 has eigenvalues 4 and 6

(D) $A + 4I$ has eigenvalues 8 and 12

Solution

Fast route: $Av = \lambda v \Rightarrow A^{-1}v = \frac{1}{\lambda}v$.

Full working:

- $A + 4I$ has 6 and 7; A^2 has 4 and 9.

Why the other options fail: (B) $A^\top$ shares the eigenvalues, not in general the eigenvectors.

Answer: A.

Q11.12 REASON · about 45 s

Gradient descent on $f(w) = \frac{1}{2}w^\top Hw$, where H has eigenvalues 10 and 0.1. The largest learning rate that still converges, and the condition number, are

- (A) 20 and 0.01
- (B) 2 and 100
- (C) 0.2 and 100
- (D) 0.2 and 10

Solution

Fast route: $\eta < \frac{2}{\lambda_{\max}} = 0.2$; $\kappa = \frac{10}{0.1}$.

Full working:

- Along eigenvector i each step multiplies by $1 - \eta\lambda_i$; need $|1 - \eta\lambda_i| < 1$ for all i . The flat direction then shrinks only by 0.98 per step: slow, which motivates feature scaling, momentum and Adam.

Answer: C.

Q11.13 REASON · about 45 s

$u = (1, 2, 2)$. The eigenvalues of $M = uu^\top$ are

- (A) 5, 2, 2
- (B) 3, 0, 0
- (C) 1, 2, 2
- (D) 9, 0, 0

Solution

Fast route: $Mu = u(u^\top u) = 9u$; perpendicular vectors give 0.

Full working:

- Check: trace = $1 + 4 + 4 = 9$.

Why the other options fail: (B) uses $\|u\|$ instead of $\|u\|^2$.

Answer: D.

Q11.14 RECALL · about 15 s

For every real symmetric matrix, which property is guaranteed?

- (A) Its eigenvalues are its diagonal entries
- (B) All eigenvalues are real

- (C) All eigenvalues are positive
(D) It is invertible

Solution

Fast route: Spectral theorem: real eigenvalues, orthogonal eigenvectors.

Full working:

- $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is symmetric with eigenvalue -1 , so (C) fails; $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is symmetric and singular, so (D) fails.

Answer: B.

Q11.15 **RECALL** · about 15 s

The 90° rotation $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues

- (A) 0 and 0
(B) 1 and 1
(C) i and $-i$
(D) 1 and -1

Solution

Fast route: Trace 0, det 1: $\lambda^2 + 1 = 0$.

Full working:

- No real direction survives a rotation.

Answer: C.

Q11.16 **RECALL** · about 15 s

$X^T X$ has eigenvalues 50, 5 and 0.001. After adding a ridge penalty with $\lambda = 1$, the eigenvalues of $X^T X + \lambda I$ are

- (A) 49, 4, 0
(B) 50, 5, 0.001
(C) 50, 5, 1
(D) 51, 6, 1.001

Solution

Fast route: Adding λI adds λ to every eigenvalue.

Full working:

- The near zero eigenvalue (from near collinearity) becomes about 1, so the inverse no longer explodes: the condition number drops from 50,000 to about 51.

Answer: D.

Q11.17 **RECALL** · about 15 s

The Hessian at a critical point has eigenvalues 3, 0.5 and -2 . The point is

- (A) a saddle point
(B) a local maximum

- (C) cannot be determined
(D) a local minimum

Solution

Fast route: Mixed signs: curves up in some directions, down in another.

Full working:

- Saddles vastly outnumber local minima in high dimensional deep learning losses.

Answer: A.

Q11.18 RECALL · about 15 s

A graph Laplacian $L = D - A$ has eigenvalues 0, 0, 0, 1.2, 3.5, 4.1. The graph has

- (A) a cycle
(B) 1 connected component
(C) 6 connected components
(D) 3 connected components

Solution

Fast route: Number of zero eigenvalues = number of connected components.

Full working:

- Each component's indicator vector lies in the null space of L .

Answer: D.

Q11.19 RECALL · about 15 s

Power iteration $x \leftarrow \frac{Ax}{\|Ax\|}$ on a symmetric matrix with eigenvalues 5, 4, 1 converges to

- (A) the eigenvector of eigenvalue 5, at a rate of about $(4/5)^t$
(B) the eigenvector of eigenvalue 5, at a rate of about $(1/5)^t$
(C) it never converges
(D) the eigenvector of eigenvalue 1

Solution

Fast route: Top eigenvector; the ratio $\frac{\lambda_2}{\lambda_1}$ sets the speed.

Full working:

- Assumes the start vector has some component along the top eigenvector. This is how PageRank is computed at scale.

Answer: A.

Q11.20 RECALL · about 15 s

PCA is run on unstandardised features: salary in rupees (variance about 10^{10}) and age in years (variance about 100). PC1 will be

- (A) a 50/50 mix
(B) undefined
(C) almost exactly the salary axis
(D) almost exactly the age axis

Solution

Fast route: PCA chases raw variance; salary's scale dominates.

Full working:

- Standardise first (correlation matrix PCA) unless the units are comparable.

Answer: C.

Q11.21 **CALC · 1.5 to 2 min**

A symmetric 2×2 matrix has eigenvalues 4 and 1 with unit eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$. The matrix is

- (A) $\begin{bmatrix} 2.5 & -1.5 \\ -1.5 & 2.5 \end{bmatrix}$
 (B) $\begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$
 (C) $\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$
 (D) $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$

Solution

Fast route: $4 \cdot \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 1 \cdot \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$.

Full working:

- Check: trace $5 = 4 + 1$; det $6.25 - 2.25 = 4 = 4 \times 1$.

Why the other options fail: (A) attaches the eigenvalue 4 to $(1, -1)$. (C) would have eigenvectors along the axes.

Answer: B.

Q11.22 **RECALL · about 15 s**

An RNN's hidden to hidden matrix has spectral radius (largest $|\lambda|$) 1.3. Ignoring the non linearity, over long sequences the gradients tend to

- (A) become zero immediately
 (B) explode
 (C) vanish
 (D) stay constant

Solution

Fast route: 1.3^t grows: $1.3^{20} \approx 190$.

Full working:

- Remedies: gradient clipping, orthogonal initialisation, LSTMs or GRUs.

Answer: B.

Q11.23 **RECALL · about 15 s**

The first principal component of standardised data is the eigenvector of

- (A) the correlation matrix with the largest eigenvalue
- (B) the covariance matrix with the smallest eigenvalue
- (C) XX^T with the smallest eigenvalue
- (D) the identity matrix

Solution

Fast route: Standardised covariance = correlation matrix; PC1 = top eigenvector.

Full working:

- Largest eigenvalue = most variance.

Answer: A.

Q11.24 **CALC** · 1.5 to 2 min

Three standardised features have all pairwise correlations equal to 0.5. The share of variance explained by PC1 is

- (A) 66.7%
- (B) 50%
- (C) 83.3%
- (D) 33.3%

Solution

Fast route: $R = 0.5I + 0.5\mathbf{1}\mathbf{1}^T$: eigenvalues $0.5 + 1.5 = 2$, then 0.5, 0.5, $\frac{2}{3}$.

Full working:

- $\mathbf{1}\mathbf{1}^T$ has eigenvalues 3, 0, 0 (a rank 1 matrix), scaled by 0.5 and shifted by 0.5.
- PC1 $\propto (1, 1, 1)$: essentially the average of the three features. Total variance = 3 (the trace).

Answer: A.

Q11.25 **RECALL** · about 15 s

Which is **not** a valid reason to use PCA before training a model?

- (A) Visualise high dimensional data in 2D
- (B) Remove correlation between features
- (C) Reduce dimensionality and noise
- (D) Guarantee better accuracy for any downstream model

Solution

Fast route: PCA ignores the label; discarded low variance directions might carry the signal.

Full working:

- Supervised alternatives (LDA, PLS) use the label.

Answer: D.

References for this chapter

- [3Blue1Brown: Eigenvectors and eigenvalues](#); a quick trick for computing eigenvalues; change of basis.
- [StatQuest: PCA step by step](#).

- [Mathematics for Machine Learning](#), chapters 4 and 10.

Chapter 12

Linear Algebra III: Singular Value Decomposition and PCA

Companion resource for this section: [Steve Brunton: SVD overview](#) (the whole short series on SVD is excellent)

12.1 Why we need one more decomposition

Start here: the everyday picture

- Eigenvectors are wonderful, but they need a **square** matrix and sometimes do not exist (the shear in Chapter 11). Real data is a **rectangle**: 10,000 users by 300 features, or users by jobs.
- The Singular Value Decomposition (SVD) works for **every** matrix, of any shape. It says: every linear map, however complicated, is just **rotate, then stretch along perpendicular axes, then rotate again**.

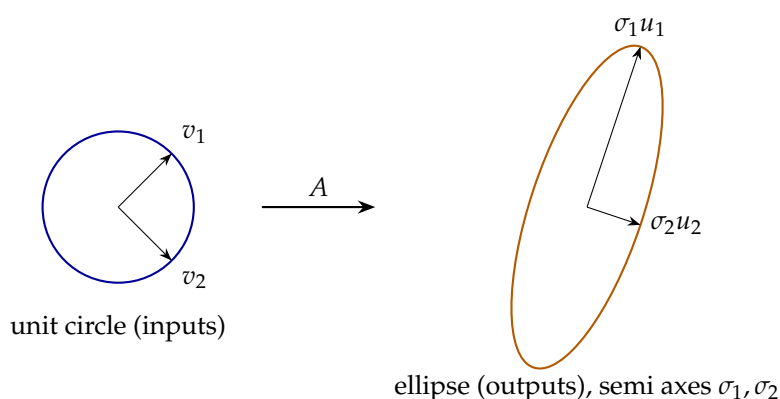
$$A = U\Sigma V^\top$$

- A is $m \times n$. V ($n \times n$) and U ($m \times m$) are **orthogonal** (pure rotations or reflections). Σ ($m \times n$) is zero except on its diagonal, which holds the **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.
- Read it from right to left: $V^\top$ rotates the input, Σ stretches axis i by σ_i , U rotates the result.
- Columns of V : **right singular vectors** (special input directions). Columns of U : **left singular vectors** (the output directions). The key relation is

$$Av_i = \sigma_i u_i :$$

input direction v_i comes out in direction u_i , stretched by σ_i .

- The picture: the unit circle always turns into an **ellipse**. Its semi axes point along u_1, u_2 and have lengths σ_1, σ_2 .



$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ (drawn at one third scale): $\sigma_1 = \sqrt{45} \approx 6.7$ along $u_1 \propto (1, 3)$ and $\sigma_2 = \sqrt{5} \approx 2.2$ along $u_2 \propto (3, -1)$.

12.2 Connecting SVD to eigenvalues, and computing it by hand

- Multiply A by its transpose: $A^T A = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$. So:
 - the v_i are the eigenvectors of $A^T A$;
 - the squared singular values σ_i^2 are its eigenvalues;
 - similarly the u_i are eigenvectors of $A A^T$ (same non zero eigenvalues).
- **Recipe:** (1) eigen decompose $A^T A$ to get $\sigma_i = \sqrt{\lambda_i}$ and v_i ; (2) compute $u_i = \frac{A v_i}{\sigma_i}$.
- **Worked example:** $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.
 - $A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$. For a matrix $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ the eigenvalues are $a \pm b$: here 45 and 5, with eigenvectors $\frac{1}{\sqrt{2}}(1, 1)$ and $\frac{1}{\sqrt{2}}(1, -1)$.
 - $\sigma_1 = \sqrt{45} = 6.708$, $\sigma_2 = \sqrt{5} = 2.236$.
 - $u_1 = \frac{A v_1}{\sigma_1} = \frac{(3, 4)/\sqrt{2}}{\sqrt{45}} = \frac{(1, 3)}{\sqrt{10}}$; $u_2 = \frac{(3, -4)}{\sqrt{10}}$.
 - Checks: $\sigma_1 \sigma_2 = 15 = |\det A|$; $\sigma_1^2 + \sigma_2^2 = 50 = 9 + 16 + 25 = \|A\|_F^2$.
- **Singular values vs eigenvalues:** they coincide for symmetric PSD matrices. For a symmetric matrix with negative eigenvalues, $\sigma_i = |\lambda_i|$: $\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$ has eigenvalues 2, -3 but singular values 3, 2. For non symmetric matrices they differ completely: the shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has eigenvalues 1, 1 but singular values 1.618 and 0.618.

12.2.1 What the singular values tell you instantly

- **Rank** = number of non zero singular values. (Numerically: those above a small tolerance.)
 $\begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ has row 2 = 2 × row 1, so rank 1: singular values $\sqrt{125} = 11.18$ and 0.
- **Spectral norm** $\|A\|_2 = \sigma_1$: the biggest stretch any vector can get.
- **Frobenius norm** $\|A\|_F = \sqrt{\sum \sigma_i^2}$.
- **Determinant** (square A): $|\det A| = \prod \sigma_i$.
- **Condition number** $\kappa = \frac{\sigma_{\max}}{\sigma_{\min}}$: how much relative errors can be amplified when solving $Ax = b$.
 $\kappa = 1000$ means you may lose about 3 digits of accuracy.

12.3 Shapes, thin SVD, and building a matrix from rank 1 layers

- Full SVD of a 100×20 matrix: U is 100×100 , Σ is 100×20 , V is 20×20 . The **thin** (economy) SVD keeps only 20 columns of U : $U_{100 \times 20} \Sigma_{20 \times 20} V_{20 \times 20}^T$. In NumPy: `np.linalg.svd(A, full_matrices=False)`.
- The SVD writes A as a sum of rank 1 layers, most important first:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T.$$

- **Eckart Young theorem:** the best rank k approximation of A is simply the first k layers, A_k . No other rank k matrix is closer. Its errors:

$$\|A - A_k\|_2 = \sigma_{k+1}, \quad \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \sigma_{k+2}^2 + \cdots}$$

- **Energy** captured by rank k : $\frac{\sigma_1^2 + \dots + \sigma_k^2}{\sum \sigma_i^2}$. Always square the singular values. With $\sigma = 10, 3, 1$: rank 1 keeps $\frac{100}{110} = 90.9\%$.
- **Storage**: rank k of an $m \times n$ matrix needs $k(m + n + 1)$ numbers instead of mn . A 512×512 image at rank 20: $20 \times 1025 = 20,500$ numbers, 7.8% of 262,144.

Try it now (answer right below, cover it first)

Singular values are 20, 10, 5, 1, 0.5, 0.5. What is the smallest k that keeps at least 90% of the energy?

Answer: squares 400, 100, 25, 1, 0.25, 0.25 (total 526.5). $k = 1$: 76%. $k = 2$: 95%. So $k = 2$. (Using σ instead of σ^2 would wrongly give $k = 3$.)

12.4 PCA via SVD (a reported InfoEdge interview question)

1. Centre the data matrix X (n rows, d features): subtract each column's mean (and scale if units differ).
 2. Compute the thin SVD $X = U\Sigma V^\top$.
 3. **Principal directions** (loadings) = columns of V .
 4. **Variance along PC i** $= \lambda_i = \frac{\sigma_i^2}{n-1}$, because the covariance is $S = \frac{X^\top X}{n-1} = V \frac{\Sigma^2}{n-1} V^\top$.
 5. **Scores** (the projected data) $= XV = U\Sigma$. Keep the first k columns to reduce dimension.
 6. Explained variance ratio $= \frac{\sigma_i^2}{\sum \sigma_j^2}$ (the $n-1$ cancels); the scree plot shows these in decreasing order.
- Example: $n = 101$ rows with singular values 50 and 20. Variances $\frac{2500}{100} = 25$ and $\frac{400}{100} = 4$. PC1 explains $\frac{25}{29} = 86.2\%$.
 - **Why SVD rather than eigen of the covariance?** Forming $X^\top X$ squares the condition number and loses precision for small components; SVD works on X directly and gives scores for free. Randomised truncated SVD computes only the top k components for huge data. This is what scikit-learn's PCA does.
 - The sign of each singular vector is arbitrary: $\sigma uv^\top = \sigma(-u)(-v)^\top$. Two runs may give PC1 as $(0.6, 0.8)$ or $(-0.6, -0.8)$; both are correct.
 - Reconstruction: $\hat{X} = Z_k V_k^\top$ (plus the mean). A point's reconstruction error is the part of it outside the kept subspace; large errors flag anomalies.

12.5 Pseudoinverse and least squares

- **Moore Penrose pseudoinverse**: $A^+ = V\Sigma^+U^\top$, where Σ^+ inverts each non zero σ_i and leaves zeros as zeros. It exists for every matrix and equals A^{-1} when A is invertible.
- $x = A^+b$ is the least squares solution, and, when many least squares solutions exist (a non trivial null space), the one with the **smallest length**.
- Tiny singular values become huge $\frac{1}{\sigma_i}$ and amplify noise. **Truncated SVD** drops them; **ridge regression** replaces $\frac{1}{\sigma_i}$ by the tamer $\frac{\sigma_i}{\sigma_i^2 + \lambda}$.

Where this shows up in ML and at a company like Naukri

- **Recommenders**: the user by job interaction matrix is roughly low rank (a few latent tastes). Because most entries are missing, practical systems fit a factorisation $R \approx PQ^\top$ on observed entries only (ALS or SGD); plain SVD needs a full matrix.

- **Latent semantic analysis:** truncated SVD of a term by document (TF IDF) matrix finds topics, so resumes using synonyms end up close together.
- **LoRA:** fine tuning a 4096×4096 weight with $\Delta W = BA$ (B : 4096×8 , A : 8×4096) trains 65,536 numbers instead of 16.8 million, about 0.4%.
- **Spectral normalisation:** dividing a weight matrix by σ_1 limits how much a layer can amplify its input (stable GAN training).
- **Compression:** replace a big layer by two thin layers from its truncated SVD.

Formula sheet

$$A = U\Sigma V^\top \quad \sigma_i = \sqrt{\lambda_i(A^\top A)} \quad u_i = \frac{Av_i}{\sigma_i} \quad \text{rank} = \#\{\sigma_i > 0\} \quad \|A\|_2 = \sigma_1 \quad \|A\|_F = \sqrt{\sum \sigma_i^2}$$

$$|\det A| = \prod \sigma_i \quad \kappa = \frac{\sigma_1}{\sigma_r} \quad \|A - A_k\|_2 = \sigma_{k+1} \quad \|A - A_k\|_F = \sqrt{\sum_{i>k} \sigma_i^2} \quad \text{storage } k(m+n+1)$$

PCA: directions V , variances $\frac{\sigma_i^2}{n-1}$, scores $U\Sigma$ $A^+ = V\Sigma^+U^\top$.

15 second recall card: memorise every line

- Every matrix has an SVD: rotate, stretch, rotate. Circle becomes an ellipse with semi axes σ_i .
- σ_i^2 are the eigenvalues of $A^\top A$; V holds its eigenvectors; U holds those of $AA^\top$.
- Rank = number of non zero σ . $\|A\|_2 = \sigma_1$. $\|A\|_F^2 = \sum \sigma^2$. $|\det| = \prod \sigma$. $\kappa = \frac{\sigma_{\max}}{\sigma_{\min}}$.
- Full SVD of $m \times n$: U $m \times m$, Σ $m \times n$, V $n \times n$.
- Best rank k : truncate (Eckart Young). Error σ_{k+1} (spectral). Energy uses σ^2 . Storage $k(m+n+1)$.
- PCA via SVD: centre; directions V ; variances $\frac{\sigma_i^2}{n-1}$; scores $U\Sigma$. Signs are arbitrary.
- Pseudoinverse: minimum norm least squares. Ridge: $\frac{\sigma}{\sigma^2 + \lambda}$.

12.6 Question bank: SVD

Q12.1 RECALL · about 15 s

★ **PYQ (INT and OA: PCA via SVD, scree plot, explained variance)** Centred data X ($n \times d$) has thin SVD $X = U\Sigma V^\top$. Which statement is correct?

- (A) The variance along PC i is σ_i
 (B) The scores (projected data) are $V\Sigma$
 (C) The principal directions are the columns of V , and the variance along PC i is $\frac{\sigma_i^2}{n-1}$
 (D) The principal directions are the columns of U

Solution

Fast route: $X^\top X = V\Sigma^2V^\top$, so V diagonalises the covariance.

Full working:

- $S = \frac{X^\top X}{n-1}$ has eigenvectors V and eigenvalues $\frac{\sigma_i^2}{n-1}$. Scores $XV = U\Sigma$.
- A scree plot shows σ_i^2 (or the ratios $\frac{\sigma_i^2}{\sum \sigma_j^2}$) in decreasing order.

Why the other options fail: (D) U lives in sample space (n dimensional). (A) forgets to square and divide. (B) the scores are $U\Sigma$.

Answer: C.

Q12.2 CALC · 1.5 to 2 min

The singular values of $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ are

- (A) $\sqrt{45}$ and $\sqrt{5}$
- (B) 3 and 5
- (C) 45 and 5
- (D) 7 and 1

Solution

Fast route: $A^\top A = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$: eigenvalues 25 ± 20 ; take square roots.

Full working:

- Checks: product $\sqrt{225} = 15 = |\det A|$; sum of squares $50 = \|A\|_F^2$.

Why the other options fail: (B) reads the eigenvalues of the triangular A . (C) forgets the square root.

Answer: A.

Q12.3 RECALL · about 15 s

A is 100×20 . In the full SVD, the shapes of U , Σ and V are

- (A) 100×100 , 100×20 , 20×20
- (B) 20×20 , 20×100 , 100×100
- (C) 100×20 , 20×20 , 20×20
- (D) 100×100 , 20×20 , 20×100

Solution

Fast route: U : $m \times m$; Σ : $m \times n$; V : $n \times n$.

Full working:

- (C) is the thin SVD.

Answer: A.

Q12.4 RECALL · about 15 s

A matrix has singular values 3 and 4. Its Frobenius norm and spectral norm are

- (A) 7 and 4
- (B) 5 and 4
- (C) 5 and 3
- (D) 25 and 4

Solution

Fast route: $\sqrt{9 + 16}$; the largest singular value.

Full working:

- $\|A\|_F = 5$, $\|A\|_2 = 4$.

Answer: B.

Q12.5 REASON · about 45 s

Singular values are 10, 3 and 1. For the best rank 1 approximation, the spectral norm error and the Frobenius norm error are

- (A) 3 and 4
- (B) 3 and $\sqrt{10}$
- (C) $\sqrt{10}$ and 3
- (D) 1 and 1

Solution

Fast route: Spectral: next singular value. Frobenius: root of the sum of discarded squares.

Full working:

- $\sigma_2 = 3; \sqrt{9+1} = 3.16.$

Why the other options fail: (A) adds the discarded values without squaring.

Answer: B.

Q12.6 REASON · about 45 s

Singular values are 10, 3 and 1. The fraction of energy captured by the best rank 1 approximation is closest to

- (A) 77%
- (B) 71%
- (C) 100%
- (D) 91%

Solution

Fast route: $\frac{100}{100+9+1}.$

Full working:

- 90.9%.

Why the other options fail: (B) uses σ instead of σ^2 : $\frac{10}{14}.$

Answer: D.

Q12.7 CALC · 1.5 to 2 min

Singular values are 20, 10, 5, 1, 0.5, 0.5. The smallest k whose truncated SVD keeps at least 90% of the energy is

- (A) 4
- (B) 3
- (C) 1
- (D) 2

Solution

Fast route: Squares 400, 100, 25, ... (total 526.5): $k = 2$ gives $\frac{500}{526.5} = 95\%.$

Full working:

- $k = 1$ gives only 76%.

Why the other options fail: (B) comes from using σ instead of σ^2 ($\frac{30}{37} = 81\%$ at $k = 2$).

Answer: D.

Q12.8 REASON · about 45 s

A 512×512 grayscale image is stored with a rank 20 SVD. The fraction of the original storage used is closest to

- (A) 15.6%
- (B) 7.8%
- (C) 3.9%
- (D) 20%

Solution

Fast route: $\frac{20(512+512+1)}{512^2} = \frac{20500}{262144}$.

Full working:

- 20 columns of U , 20 of V , 20 singular values.

Why the other options fail: (C) stores only U or only V .

Answer: B.

Q12.9 REASON · about 45 s

Centred data with $n = 101$ rows has singular values 50 and 20. The variances of PC1 and PC2, and PC1's explained variance, are

- (A) 2500, 400 and 86.2%
- (B) 50, 20 and 71.4%
- (C) 25, 4 and 86.2%
- (D) 0.5, 0.2 and 71.4%

Solution

Fast route: $\frac{\sigma^2}{n-1}$: $\frac{2500}{100}$ and $\frac{400}{100}$; ratio $\frac{25}{29}$.

Full working:

- The $n - 1$ cancels in the ratio.

Why the other options fail: (B) uses σ as the variance. (A) forgets to divide by $n - 1$.

Answer: C.

Q12.10 RECALL · about 15 s

Which relationship between SVD and eigendecomposition is correct?

- (A) The left singular vectors are the eigenvectors of $A^\top A$
- (B) SVD exists only for square matrices
- (C) The singular values of A are the square roots of the eigenvalues of $A^\top A$
- (D) The singular values equal the eigenvalues of A for every square A

Solution

Fast route: $A^\top A = V\Sigma^2V^\top$.

Full working:

- The *right* singular vectors are the eigenvectors of $A^\top A$; the left ones those of $AA^\top$.

Why the other options fail: (D) the shear has eigenvalues 1, 1 but singular values 1.618 and 0.618.

Answer: C.

Q12.11 RECALL · about 15 s

A matrix has singular values 7.2, 3.1, 0, 0. Its rank is

- (A) 2
- (B) 0
- (C) 4
- (D) 3

Solution

Fast route: Count the non zero singular values.

Full working:

- Numerically, count those above a small tolerance.

Answer: A.

Q12.12 CALC · 1.5 to 2 min

The singular values of $\begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ are closest to

- (A) 11 and 0
- (B) 3 and 8
- (C) 5 and 10
- (D) 11.18 and 0

Solution

Fast route: Rank 1 (row 2 = 2 × row 1), so $\sigma_2 = 0$ and $\sigma_1 = \|A\|_F = \sqrt{125}$.

Full working:

- $A = uv^\top$ with $u = (1, 2)$, $v = (3, 4)$: $\sigma_1 = \|u\| \|v\| = \sqrt{5} \times 5 = 11.18$.

Answer: D.

Q12.13 RECALL · about 15 s

A matrix has largest singular value 100 and smallest 0.1. Its condition number is

- (A) 100
- (B) 0.001
- (C) 10
- (D) 1000

Solution

Fast route: $\frac{100}{0.1}$.

Full working:

- Relative errors in b can be amplified up to 1000 times when solving $Ax = b$.

Answer: D.

Q12.14 RECALL · about 15 s

The Moore Penrose pseudoinverse $A^+ = V\Sigma^+U^\top$

- (A) exists only for invertible matrices

- (B) is always equal to $A^\top$
- (C) gives the minimum norm least squares solution $x = A^+b$
- (D) inverts the zero singular values too

Solution

Fast route: It exists for every matrix and picks the shortest least squares solution.

Full working:

- It equals A^{-1} when A is invertible and $(A^\top A)^{-1}A^\top$ when A has full column rank.

Why the other options fail: (D) zero singular values stay zero.

Answer: C.

Q12.15 RECALL · about 15 s

The main numerical reason scikit-learn computes PCA with an SVD of the centred data rather than the eigenvectors of $X^\top X$ is that

- (A) SVD gives different principal directions
- (B) forming $X^\top X$ squares the condition number and loses precision
- (C) eigen decomposition cannot handle symmetric matrices
- (D) SVD does not need centring

Solution

Fast route: Squaring the matrix squares κ .

Full working:

- Mathematically both routes give the same directions (up to sign); SVD also yields the scores directly.

Answer: B.

Q12.16 RECALL · about 15 s

Running PCA twice on the same data gives PC1 as $(0.6, 0.8)$ once and $(-0.6, -0.8)$ the other time. This means

- (A) the data changed
- (B) the sign of singular vectors is arbitrary; both are correct
- (C) a bug
- (D) PC1 is undefined

Solution

Fast route: $\sigma uv^\top = \sigma(-u)(-v)^\top$.

Full working:

- Fix a sign convention if you need to compare runs.

Answer: B.

Q12.17 RECALL · about 15 s

A user by job rating matrix ($10^6 \times 10^5$) has 99.9% of entries missing. Which statement is correct?

- (A) Low rank models cannot predict missing ratings
- (B) Matrix factorisation fitted only on observed entries (ALS or SGD) is the practical approach

- (C) Plain SVD can be applied directly to the missing entries
 (D) A rank of 10^5 is needed for good recommendations

Solution

Fast route: SVD needs a complete matrix; factorisation methods fit the observed cells.

Full working:

- The low rank assumption says preferences depend on a few latent factors; typical ranks are tens to hundreds.

Answer: B.

Q12.18 REASON · about 45 s

LoRA fine tunes a 4096×4096 weight using $\Delta W = BA$, with B of shape 4096×8 and A of shape 8×4096 . The trainable parameters, as a fraction of full fine tuning, are about

- (A) 0.4%
 (B) 100%
 (C) 4%
 (D) 50%

Solution

Fast route: $\frac{2 \times 4096 \times 8}{4096^2} = \frac{16}{4096}$.

Full working:

- 65,536 vs 16,777,216: 0.39%. The update has rank at most 8.

Answer: A.

Q12.19 RECALL · about 15 s

In terms of the SVD of X , ridge regression replaces each factor $\frac{1}{\sigma_i}$ of the least squares solution by

- (A) $\sigma_i + \lambda$
 (B) $\frac{1}{\sigma_i + \lambda}$
 (C) $\frac{\lambda}{\sigma_i}$
 (D) $\frac{\sigma_i}{\sigma_i^2 + \lambda}$

Solution

Fast route: Small σ_i no longer explode: $\frac{\sigma_i}{\sigma_i^2 + \lambda} \rightarrow 0$ as $\sigma_i \rightarrow 0$.

Full working:

- For large σ_i it is close to $\frac{1}{\sigma_i}$, so well determined directions are barely changed.

Answer: D.

Q12.20 RECALL · about 15 s

Geometrically, applying $A = U\Sigma V^\top$ to the unit circle produces

- (A) an ellipse with semi axes σ_1, σ_2 along u_1, u_2
 (B) a square
 (C) always a line segment
 (D) a rotated circle

Solution

Fast route: Rotate (circle stays a circle), stretch axes, rotate.

Full working:

- It collapses to a segment only when $\sigma_2 = 0$.

Answer: A.

Q12.21 RECALL · about 15 s

Latent semantic analysis applies truncated SVD to a term by document TF IDF matrix. A key benefit is that

- (A) it always improves classification accuracy
- (B) it requires dense word embeddings as input
- (C) it removes the need for tokenisation
- (D) documents that use synonyms can end up close together in the reduced space

Solution

Fast route: Shared latent topics link different words with similar meaning.

Full working:

- scikit-learn's TruncatedSVD works on sparse matrices without centring.

Answer: D.

Q12.22 REASON · about 45 s

For $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$, the eigenvalues and the singular values are

- (A) 3, 2 and 2, -3
- (B) 2, 3 and 4, 9
- (C) 2, -3 and 3, 2
- (D) 2, -3 and 2, -3

Solution

Fast route: Eigenvalues from the diagonal; singular values are non negative and sorted.

Full working:

- The sign moves into U or V .

Why the other options fail: (B) squares them.

Answer: C.

Q12.23 RECALL · about 15 s

The spectral norm $\|W\|_2 = \sigma_1$ of a layer's weight matrix bounds

- (A) how much the layer can stretch any input: $\|Wx\| \leq \sigma_1 \|x\|$
- (B) the bias
- (C) the rank of W
- (D) the number of neurons

Solution**Fast route:** $\sigma_1 = \max_{\|x\|=1} \|Wx\|$.**Full working:**

- Spectral normalisation divides by σ_1 to keep layers stable.

Answer: A.**Q12.24 RECALL · about 15 s**

A PCA based anomaly detector keeps k components and flags points with large reconstruction error. The reconstruction error of a point is

- (A) the length of its component perpendicular to the kept k dimensional subspace
- (B) the largest singular value
- (C) always zero for anomalies
- (D) its projection on PC1

Solution**Fast route:** What the kept components cannot explain.**Full working:**

- Normal points lie close to the subspace; anomalies often break the usual correlation pattern.

Answer: A.**Q12.25 REASON · about 45 s**

A 1000×500 matrix is approximated with rank 20. The number of stored values, and the fraction of the original, are

- (A) 20,000 and 4%
- (B) 30,020 and about 6%
- (C) 500,000 and 100%
- (D) 30,020 and about 3%

Solution**Fast route:** $20(1000 + 500 + 1) = 30020$; $\frac{30020}{500000}$.**Full working:**

- About 6%.

Answer: B.**Q12.26 RECALL · about 15 s**

A square matrix has singular values 4, 2 and 0.5. $|\det A|$ is

- (A) 16
- (B) 6.5
- (C) 4
- (D) cannot be determined

Solution

Fast route: Product of singular values: $4 \times 2 \times 0.5$.

Full working:

- $|\det U| = |\det V| = 1$ for orthogonal matrices.

Why the other options fail: (B) adds them.

Answer: C.

References for this chapter

- [Steve Brunton: SVD overview; mathematical overview](#).
- [StatQuest: PCA step by step](#).
- [MIT 18.06 \(lecture 29: SVD\)](#).
- [Mathematics for Machine Learning, chapter 4](#).

Timed Mock Papers: 40 Questions in 40 Minutes

Companion resource for this section: Re read the OA intelligence and the 40 in 40 time plan at the front of this book before each mock

How to use these mocks

- Each mock is a list of 40 questions taken from the chapter banks (no question repeats across the three mocks), mixed across the whole syllabus the way the real paper mixes topics. No new questions are added, so the book still contains exactly 352.
- Each paper has about 18 recall, 10 quick reasoning and 12 calculation questions, which by the time tags adds up to roughly 34 minutes: the same budget as the plan at the front of the book, with a few minutes of buffer.
- Set a 40 minute timer. Answer in the order listed, using the two pass strategy: answer recall questions immediately, mark calculations you cannot see through at once, come back to them.
- Score +1 per correct answer and -0.25 per wrong answer (a common scheme; adjust if your paper states otherwise). Leave blanks as 0.
- Afterwards: for every wrong or slow answer, reread its solution and the recall card of that chapter. A recall question that took more than 20 seconds counts as “slow”.
- Targets: 32 or more correct in 40 minutes means you are ready. Under 25: redo the recall cards before the next mock.
- Maths only sprint: if you expect around 13 maths questions in the real paper, also try the first 13 questions of any mock in 13 minutes.

Mock paper 1

Mix: 18 recall, 10 quick, 12 calculation questions; estimated 34 minutes at target speed. ★ marks a PYQ.

#	Question	Topic	Time	#	Question	Topic	Time
1	Q5.15	CLT	45 s	21	Q3.34	Distributions	15 s
2	Q10.18	Matrices	15 s	22	Q5.18	CLT	15 s
3	Q12.1 ★	SVD	15 s	23	Q2.2 ★	Bayes	15 s
4	Q6.4	Correlation	1.5 to 2 min	24	Q5.3	CLT	45 s
5	Q10.24	Matrices	15 s	25	Q3.4 ★	Distributions	45 s
6	Q6.13	Correlation	1.5 to 2 min	26	Q9.11	Paired tests	1.5 to 2 min
7	Q8.1 ★	Significance	15 s	27	Q4.24	Sampling	15 s
8	Q7.1 ★	Hypothesis tests	15 s	28	Q3.11	Distributions	45 s
9	Q3.1 ★	Distributions	45 s	29	Q9.8	Paired tests	15 s
10	Q7.7	Hypothesis tests	1.5 to 2 min	30	Q2.18	Bayes	45 s
11	Q10.6	Matrices	1.5 to 2 min	31	Q11.11	Eigen	15 s
12	Q11.1 ★	Eigen	15 s	32	Q1.4 ★	Probability	45 s
13	Q3.7	Distributions	45 s	33	Q1.1 ★	Probability	1.5 to 2 min
14	Q4.6	Sampling	1.5 to 2 min	34	Q7.4	Hypothesis tests	15 s
15	Q7.10	Hypothesis tests	1.5 to 2 min	35	Q10.1 ★	Matrices	1.5 to 2 min
16	Q6.1 ★	Correlation	15 s	36	Q1.6	Probability	45 s
17	Q11.19	Eigen	15 s	37	Q8.16	Significance	1.5 to 2 min
18	Q4.12	Sampling	15 s	38	Q1.24	Probability	45 s
19	Q12.2	SVD	1.5 to 2 min	39	Q1.41	Probability	1.5 to 2 min
20	Q8.20	Significance	15 s	40	Q2.1 ★	Bayes	15 s

Answer key (mock 1): 1 B, 2 A, 3 C, 4 B, 5 C, 6 D, 7 A, 8 A, 9 C, 10 C, 11 A, 12 A, 13 B, 14 B, 15 C, 16 A, 17 A, 18 A, 19 A, 20 B, 21 C, 22 C, 23 B, 24 A, 25 A, 26 D, 27 D, 28 A, 29 B, 30 D, 31 A, 32 C, 33 A, 34 B, 35 B, 36 D, 37 A, 38 A, 39 A, 40 B.

Mock paper 2

Mix: 18 recall, 10 quick, 12 calculation questions; estimated 34 minutes at target speed. ★ marks a PYQ.

#	Question	Topic	Time	#	Question	Topic	Time
1	Q2.22	Bayes	45 s	21	Q1.18	Probability	45 s
2	Q7.3 ★	Hypothesis tests	15 s	22	Q4.10	Sampling	15 s
3	Q1.12	Probability	1.5 to 2 min	23	Q8.2 ★	Significance	15 s
4	Q11.2 ★	Eigen	45 s	24	Q7.27	Hypothesis tests	15 s
5	Q7.28	Hypothesis tests	15 s	25	Q10.3 ★	Matrices	15 s
6	Q5.22	CLT	45 s	26	Q6.15	Correlation	45 s
7	Q2.12	Bayes	45 s	27	Q6.19	Correlation	1.5 to 2 min
8	Q11.16	Eigen	15 s	28	Q1.8	Probability	45 s
9	Q6.2 ★	Correlation	15 s	29	Q10.21	Matrices	15 s
10	Q8.17	Significance	1.5 to 2 min	30	Q9.17	Paired tests	15 s
11	Q10.2 ★	Matrices	1.5 to 2 min	31	Q2.23	Bayes	15 s
12	Q7.2 ★	Hypothesis tests	1.5 to 2 min	32	Q8.8	Significance	45 s
13	Q3.22	Distributions	45 s	33	Q12.16	SVD	15 s
14	Q5.7	CLT	1.5 to 2 min	34	Q4.8	Sampling	15 s
15	Q5.25	CLT	1.5 to 2 min	35	Q10.7	Matrices	15 s
16	Q3.5 ★	Distributions	1.5 to 2 min	36	Q3.39	Distributions	45 s
17	Q3.2 ★	Distributions	1.5 to 2 min	37	Q9.1	Paired tests	1.5 to 2 min
18	Q1.28	Probability	15 s	38	Q11.21	Eigen	1.5 to 2 min
19	Q3.33	Distributions	15 s	39	Q4.14	Sampling	15 s
20	Q12.13	SVD	15 s	40	Q1.2 ★	Probability	1.5 to 2 min

Answer key (mock 2): 1 D, 2 C, 3 C, 4 C, 5 B, 6 B, 7 A, 8 D, 9 B, 10 D, 11 A, 12 A, 13 B, 14 C, 15 D, 16 D,

17 B, 18 C, 19 B, 20 D, 21 B, 22 A, 23 D, 24 B, 25 B, 26 D, 27 C, 28 C, 29 A, 30 D, 31 B, 32 D, 33 B, 34 C, 35 D, 36 D, 37 C, 38 B, 39 B, 40 C.

Mock paper 3

Mix: 19 recall, 9 quick, 12 calculation questions; estimated 33 minutes at target speed. ★ marks a PYQ.

#	Question	Topic	Time	#	Question	Topic	Time
1	Q11.7	Eigen	1.5 to 2 min	21	Q1.3 ★	Probability	15 s
2	Q7.5	Hypothesis tests	1.5 to 2 min	22	Q10.12	Matrices	1.5 to 2 min
3	Q9.23	Paired tests	15 s	23	Q8.3 ★	Significance	15 s
4	Q3.32	Distributions	45 s	24	Q12.24	SVD	15 s
5	Q5.19	CLT	15 s	25	Q11.13	Eigen	45 s
6	Q1.11	Probability	15 s	26	Q1.10	Probability	1.5 to 2 min
7	Q8.9	Significance	1.5 to 2 min	27	Q3.28	Distributions	15 s
8	Q2.7	Bayes	1.5 to 2 min	28	Q9.24	Paired tests	45 s
9	Q3.3 ★	Distributions	15 s	29	Q11.15	Eigen	15 s
10	Q6.20	Correlation	1.5 to 2 min	30	Q1.22	Probability	45 s
11	Q12.19	SVD	15 s	31	Q2.19	Bayes	45 s
12	Q5.11	CLT	15 s	32	Q2.17	Bayes	1.5 to 2 min
13	Q7.20	Hypothesis tests	15 s	33	Q4.2	Sampling	15 s
14	Q6.3 ★	Correlation	45 s	34	Q4.23	Sampling	15 s
15	Q1.46	Probability	45 s	35	Q10.15	Matrices	45 s
16	Q6.6	Correlation	15 s	36	Q10.26	Matrices	15 s
17	Q7.17	Hypothesis tests	45 s	37	Q8.25	Significance	15 s
18	Q10.13	Matrices	15 s	38	Q7.8	Hypothesis tests	1.5 to 2 min
19	Q5.14	CLT	1.5 to 2 min	39	Q3.6 ★	Distributions	15 s
20	Q3.10	Distributions	1.5 to 2 min	40	Q4.5	Sampling	1.5 to 2 min

Answer key (mock 3): 1 D, 2 A, 3 D, 4 B, 5 D, 6 C, 7 B, 8 A, 9 D, 10 C, 11 D, 12 D, 13 A, 14 D, 15 D, 16 D, 17 C, 18 A, 19 D, 20 B, 21 D, 22 C, 23 C, 24 A, 25 D, 26 A, 27 D, 28 A, 29 C, 30 A, 31 A, 32 D, 33 C, 34 D, 35 A, 36 D, 37 A, 38 D, 39 A, 40 D.

A 7 day revision plan

- **Day 1:** Chapter 0 (skim), Chapter 1 teaching half, then Q1.1 to Q1.25 with a timer.
- **Day 2:** Q1.26 to the end; Chapter 2 fully (Bayes appears in both the OA and the interviews).
- **Day 3:** Chapter 3. Make one line flashcards: story, mean, variance, one ML use.
- **Day 4:** Chapters 4 and 5 (sampling, CLT). Short, but they feed every test in Chapters 7 to 9.
- **Day 5:** Chapters 6 and 7. Practise naming the right test in 10 seconds. Evening: Mock paper 1.
- **Day 6:** Chapters 8 and 9; review every mistake from Mock 1. Evening: Mock paper 2.
- **Day 7:** Chapters 10 to 12 (rank, null space, eigen, SVD, PCA). Evening: Mock paper 3. Night before the test: read only the twelve recall cards and the quick reference tables.